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Road Design Manual

Volume 6: Traffic Control Facilities & Communication Systems Design

Part 3: Traffic Signals and Communication System

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Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

The Traffic Signals Manual has been prepared to assist traffic engineers in designing and installing traffic signals at the road junctions. It details the warrants required to justify installation of traffic signals. The data required to be collected for designing a signal are elaborated. This includes data on traffic volume, pedestrians, road crashes and anticipated future developments that may contribute to traffic at the junction.

Traffic signal is intricately connected with the geometric layout of the junction. Turning lane arrangements, traffic islands, kerbs, median island, strips connected to installation of traffic signals are explained. Signals are connected with various equipment, including signal poles, lantern, mast arm, equipment associated with lanterns, power distribution, controller equipment, detection equipment. The equipment are described, along with facilities required for pedestrians, bus priority and remote monitoring. Signal face and display for arrows, vehicles, bicycles, pedestrians are explained, along with timing of signals for all the phases. Traffic detection and signal controller are explained along with the electrical connections, maintenance requirements. The various tools available for traffic design and coordination of signal timing, to set up signal system are detailed.

Eng. Joseph M. Mbugua, CBS

Principal Secretary, State Department for Roads

Document Management

Document Status

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document

Copies of the document can be obtained from:

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While all care and consideration has been applied in the compilation of this document, the Ministry accepts no responsibility for failure in any way related to the application of this manual or any reference documents cited in it.

Requests for edits and corrections can be freely sent to the following address:

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Type of request: General – G; Editorial – E; Technical - T

Amendments to Date:

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The technical work was undertaken under the guidance of a Technical Task Force, chaired by Eng. David Maganda, with the following gazetted members: Francis Gichaga (Prof.) (Eng.), Andrew Gitonga (Eng.), Timothy Nyomboi (Dr.) (Eng.), Rosemary Kungu (Eng.), Charles Obuon (Eng.), Sylvester Abuodha (Prof.) (Eng.), Samuel Kathindai (Eng.), Nicholas Musuni (Eng.), Charles Muriuki (Eng.), Tom Opiyo (Eng.), John Maina (Eng.), Fidelis Sakwa (Eng.), Daniel Cherono (Eng.), Maurice Ndeda (Eng.), Theo Uwamba (Eng.).

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Abbreviations

AM	Ante Meridiem
ATC	Area Wide Traffic Control
BRT	Bus Rapid Transit
CT	Cycle Time
DIDO	Dial In Dial Out
ECO	Early Cut Off
FT	Fixed Time
Hz	Hertz
kmph	Kilometres per Hour
LRT	Light Rail Transit
LT	Left Turn
LTGA	Left Turn Green Arrow
LTRA	Left Turn Red Arrow
MOVA	Microprocessor Optimised Vehicle Actuation
MVIG	Maximum Variable Initial Green
NMT	Non-motorised Traffic
OMU	Outstation Monitoring Unit
PBU	Push Button Unit
PCU	Passenger Car Unit
PL	Primary Lantern
PM	Post Meridiem
PP	Primary Pole
PPH	Passengers per Hour
PSTN	Public Switched Telephone Network
QH	Quartz Halogen
RT	Right Turn
RTGA	Right Turn Green Arrow
RTRA	Right Turn Red Arrow
SA	Straight Ahead
SDO	Single Diamond Overlap
SP	Secondary Pole
SVD	Selective Vehicle Detection
TSCT	Traffic Signal Countdown Timer
V	Volts
VA	Vehicle Actuated
VIG	Vehicle Initial Green
VPH	Vehicles per Hour

Glossary of Terms

Acceleration Lane	A paved auxiliary lane, including tapered areas, allowing vehicles to accelerate when entering the through-traffic lane of the roadway
Active Warning Device	A device which changes state to display a warning of a hazard.
Actuated Control	See Traffic-Actuated Control
Actuation	The electrical action produced by a vehicle (on a vehicle detector) or pedestrian (on a push button switch) to enable the controller to recognise its presence.
Adaptive Engineering	See Controller Programming
Advance Warning Signals	An active warning device consisting of a warning sign with Alternating Flashing Amber displays to warn approaching drivers of their imminent arrival at traffic signal installation.
Algorithm	See Controller Algorithm
All-Red Interval	A period of time for the clearance of conflicting movements within the controlled area, during which only Red aspects are illuminated for conflicting movements.
Approach	That section of road, consisting of one or more lanes, used by vehicles approaching an intersection or mid-block site.
Approach Speed	See Speed
Area Traffic Control System	See Wide Area Control System
Arrow Aspect	A masked vehicle aspect that displays an arrow shape when illuminated.
Arrow-Controlled Turn	A left-turn or right-turn movement at a signalised intersection that is made with a Green arrow display, and therefore is unopposed. Also see Unopposed Turn , Full Control , Partial Control .
Arrow Mask	A mask placed over a vehicle aspect so that an arrow shape will be displayed when the aspect is illuminated. Also see Symbolic Mask .
Arterial Demand	A feature of traffic signal controllers whereby a permanent demand is placed for a particular Phase or signal group. Also see Recall Feature .
Arterial Road	A road that predominantly carries through traffic from one region to another, forming principal avenues of travel for traffic movements.
Aspect	A single optical system (circular, arrow, or symbolic) on a signal face capable of being illuminated at a given time. Red, Amber, Green and white aspects are used for vehicle movements. Also see Pedestrian Aspects , Bicycle Aspect , Special Vehicle Aspects , Lantern .
Availability	A measure of the probability in signal maintenance that the equipment will be available for use, determined as $MTBF/(MTBF + MTTR)$ where MTBF = Mean Time Between Failures and MTTR = Mean Time in order to Repair.
Background Intensity	The luminous intensity of the signal background.

Glossary of Terms *(continued)*

Bandwidth	The amount of Green time common to all signals along the route in a coordinated system (Green Band). This can be determined as the time between the first and the last vehicle, travelling at the design speed, which can pass through a coordinated system of signals receiving a Green signal on arrival at each stop line.
Bicycle Aspect	A masked aspect that displays a bicycle symbol when the aspect is illuminated. Also see Bicycle Signal Face .
Bicycle Signal Face	A signal face for the control of bicycle movements that consists of two bicycle aspects (Red, Green) or three bicycle aspects (Red, Amber, Green). Also see Bicycle Aspect .
Cable-Less Linking	A mode of signal coordination in which linking is achieved by reference to an accurate clock in each signal controller. The clocks are initially set to exactly the same time and maintained in synchronism by reference to the mains supply frequency. The clock initiates the operation of a certain plan at a certain time of day and day of week, according to a predetermined schedule.
Cable Linking	See Local Interlinking
Capacity	The maximum sustainable flow rate at which vehicles or persons reasonably can be expected to traverse a point or uniform segment of a lane or roadway during a specified time period under given roadway, geometric, traffic, environmental, and control conditions; usually expressed as vehicles per hour, passenger cars per hour, or persons per hour.
Channelling	A system of controlling traffic by the introduction of a traffic island or median, or markings on a carriageway to direct traffic into predetermined paths, usually at an intersection or junction.
Clearance Time	Time given to allow a terminating movement of vehicles or pedestrians to vacate the controlled area, before the beginning of the next movement of traffic. Also see Pedestrian Clearance Period .
Column (of Aspects)	A combination of signal aspects arranged in a, vertical assembly.
Conflicting Movements	The vehicle or pedestrian traffic streams at an intersection, whose paths cross or merge when moving simultaneously.
Controlled Area	That portion of a carriageway or intersection, the entry into which is controlled by traffic signals.
Controller	The equipment (including the housing) that switches power to signal lanterns and controls the duration and sequence of signal displays.
Controller Algorithm	Programmed logic sequence internal to the controller, which transforms operator input and traffic demands into traffic control signal sequences.
Controller Programming	The task of configuring a controller to the specific requirements of a particular site. Also called Adaptive Engineering.
Controller Settings	User-defined control parameters within a traffic controller that determine signal timings (together with current traffic demands in the case of traffic-actuated controller).
Coordination	The operation of a traffic signal system where the control of individual installations is interrelated by means of a Wide Area Control System, Local Interlinking or Cable-Less Linking.
Cowl	see Visor

Glossary of Terms *(continued)*

Critical Intersection	The intersection in a coordinated signal system that operates with the highest overall degree of saturation during a given period.
Critical Lane	The lane in a lane group or approach that has the highest degree of saturation and places the highest demand on Green time.
Critical Movements	The set of movements that determine the capacity and timing requirements of a signalised intersection.
Crosswalk Lines	Continuous or broken lines marked transversely across the road to define the limits of a signalised crossing. Also see Marked Foot Crossing , Signalised Crossing .
Cycle	A complete sequence of signal phases.
Cycle Length (Cycle Time)	Time required for one complete sequence of signal displays (sum of Phase Green and inter-green times). For a given movement, cycle time is the sum of the durations of Red, Amber and Green signal displays, or sum of Effective Green and Red Times.
Degree of Saturation	The ratio of arrival (demand) flow rate to capacity during a given flow period.
Delay	The additional travel time experienced by a vehicle or pedestrian with reference to a base travel time (e.g. the free flow travel time).
Demand	The registration of the presence of vehicle or pedestrian traffic waiting for the right of way.
Demand Flow (Demand Volume)	The number of vehicles or pedestrians arriving during a given period as measured at the back of queue (as distinct from departure flows measured in front of the queue). Also see Flow Rate , Traffic Count , Traffic Volume .
Density	The number of vehicles per unit distance along the road as measured at an instant in time.
Design Life	The number of years into the future while the intersection operates satisfactorily considering increases in traffic demand volumes.
Detector	A device by which vehicle or pedestrian traffic registers its presence. The most common detectors are the inductive loop detectors for vehicles and the pushbutton detectors for pedestrians. Other detector types include microwave/radar, infrared, sonic, video image processing, magnetic and pressure detectors
Detector Loop	One or more loops of wire embedded in the road surface and connected to the controller, energised by a low voltage current. The inductance of the circuit changes when a vehicle passes over the loop, which is detected by a unit in the controller.
Diamond Overlap Phasing	Phasing that allows right turns from opposing directions to operate either simultaneously, or independently with the through movement on the same approach, depending on demand for the right turns and conflicting through movements.
Display	A signal aspect that is illuminated.
Display Sequence	The order in which traffic signal displays occur. Also see Phase Sequence .
Divided Road	A road with a median that separates the opposing directions of travel.
Dividing Line	A road marking formed by a line, or two parallel lines, whether broken or continuous, designed to indicate the parts of the road to be used by vehicles travelling in opposite directions.
Dividing Strip	An area or structure that divides a road lengthways, but does not include a nature strip, bicycle path, walkway or shared path.

Glossary of Terms *(continued)*

Downstream	In the direction of the movement of traffic.
Dual Primary Signal Face	The signal face mounted on a post either on the median at or near the right of the stop line, or if there is no median or median is too narrow, to the right and near the projection of the stop line. Also see Primary Signal Face .
Dual Secondary Signal Face	The signal face mounted on a post on the downstream side to the right of that approach, in addition to the secondary signal face located on the median.
Early Cut-Off (ECO)	The Phase interval used at the end of a Phase for allowing the termination of some signal groups earlier than others.
Effective Green and Red Times	The movement Green and Red times for capacity and performance analysis purposes, which are determined by adjusting the displayed Green and Red times for Start Loss and End Gain effects. Also see Green Time , Red Time .
Eighty-fifth Percentile Speed	See Speed End Gain . Duration of the interval between the end of displayed Green period and the end of effective Green period for a movement. This is used in signal timing and performance analysis to allow for additional departures after the end of Green period. Early Cut-Off time can be treated as an end gain for the movement that is terminated at the end of the Early Cut-Off interval. Also see Start Loss .
Exclusive Pedestrian Phase	The Phase at an intersection during which all pedestrian displays are Green and all vehicle displays are Red, allowing all pedestrian movements to operate simultaneously while all vehicle movements are stopped. Also see Scramble- Crossing Phase .
Exclusive Lane	A lane (or length of lane) allocated for use only by a particular movement or a type of vehicle, e.g. left-turn lane, through lane, right-turn lane, bus lane. Also see Shared Lane .
Extension Period	The interval of variable length during the Green period that begins after a demand for another Phase is registered following the minimum Green interval, and extends according to vehicle detector actuations subject to a maximum Green extension setting in the controller
Face	See Signal Face
Filter Turn	A turning movement that must give way to and find safe gaps in conflicting (opposing) vehicle or pedestrian traffic before proceeding, e.g. filter right-turn, slip-lane left turn, left turn on Red. Also called Opposed Turn .
Finishing-Offset Coordination	Coordination of the end times of Green periods using a signal offset that equals the average uninterrupted travel time in the travel direction so as to minimise delay along the route. Also see Offset .
Fixed-Time Control	A signal control method that allows for only a fixed sequence and fixed duration of displays.
Fixed-Time Plan Selection	A signal coordination method that uses predetermined signal timing plans introduced according to a weekly schedule or timetable.
Flexilink	The cable-less linking mode of operation in the SCATS control system, used to maintain a level of signal coordination in the event of failure of the regional computer or parts of the communication system.
Flow Rate	Number of vehicles or pedestrians per unit time passing (arriving or departing) a given reference point. Also see Demand Flow (Demand Volume) , Traffic Count , Traffic Volume .
Flow Ratio	The ratio of arrival (demand) flow rate to saturation flow rate during a given flow period.

Glossary of Terms *(continued)*

Full Control	Control of a turning movement using three-aspect (Red, Amber, Green) turn arrows on a six-aspect signal face, where the Green arrow indicates that the vehicle can turn unopposed (with no opposing vehicle or pedestrian traffic) and the Red arrow indicates that the vehicle is not permitted to turn (filter turns not permitted).
Fully-Actuated Control	See Traffic-Actuated Control
Fully-Adaptive Control	A signal coordination method that generates appropriate signal timing plans on-line in a continuously variable fashion using extensive data provided by vehicle detectors.
Gap Setting	A controller setting equivalent to a predetermined space time measured between successive vehicles at the given (approach) speed, detection zone length and vehicle length values. Also see Space Time .
Give-Way Line	A broken line (double continuous lines in New Zealand, known as limit line) marked across all or part of a road, behind which vehicles should slow down and give way to opposing traffic. Also see Stop Line .
Green Band	See Bandwidth
Green Time	Duration of the Green display for a Phase or a movement. Also see Effective Green and Red Times .
Guide Sign	A sign which is erected to inform and advise road users of directions, distances, destinations, routes and location of services for road users, and points of interest.
Headway	The time between passage of the front ends of two successive vehicles (corresponds to Spacing). Not to be confused with the Headway Setting in a signal controller.
Headway Setting	A controller setting equivalent to a predetermined space time measured between successive vehicles at the given (saturation, or queue discharge) speed, detection zone length and vehicle length values. Also see Space Time .
HGV	Heavy Goods Vehicle
Intensity	See Luminous Intensity
Intersection	A place at which two or more roads meet or cross.
Inter-green Time	Duration of the clearance part of the Phase corresponding to the period between the Phase change point (the end of running intervals) and the beginning of the Green display for the next Phase (end of Phase). Normally, it comprises Amber Time and All-Red Time. The Early Cut-off Green interval that follows the Phase change point is also considered to be part of the Inter-green Time.
Irradiation	Blurring of the edges of a bright object viewed adjacent to a dark background.
Junction	See Intersection
Lagging Right Turn	An arrow-controlled right-turn movement that is started and terminated in the Phase that immediately follows the Phase in which the opposing through movement runs.
Lane	A portion of the carriageway allocated for the use of a single line of vehicles.
Lane-Control Signal	A signal face mounted above a lane in an overhead lane control scheme, which is used to control the direction of vehicle flow in the lane during a particular time.
Lane Group	A set of lanes allocated to a particular movement in exclusive lanes, or to several movements with common shared lanes.

Glossary of Terms *(continued)*

Lantern	A signal assembly of optical components (one or more aspects), together with the means of connecting them to power supply and facilities for mounting the complete assembly. Also see Aspect .
Late Start	The Phase interval used at the start of a Phase for delaying the start of some movements.
Leading Right Turn	An arrow-controlled right-turn movement that is started and terminated in the Phase that immediately precedes the Phase in which the opposing through movement turns.
Lead-Lag Right Turn Phasing	A Phase sequence that commences with a fully-controlled leading right turn from one approach, followed by a through Phase, and terminates with a lagging right turn from the opposing approach.
Level of Service	An index of the operational performance of traffic on a given traffic lane, carriageway, road or intersection, based on service measures such as speed, travel time, delay and degree of saturation during a given flow period.
Limit Line	See Give-Way Line , Stop Line
Linking	See Coordination
Local Interlinking	A signal coordination system comprising a small number of closely-spaced signals, interconnected by a cable, usually with one of the signal controllers assuming the role of master. Also see Master Controller .
Louvres	An assembly of mechanical baffles mounted within the visor to reduce sun phantom (horizontal louvres) or to restrict the angular coverage of a signal (vertical louvres).
Luminance	Luminance at a point of a surface and in a given direction is the luminous intensity per unit projected area of a light emitting or reflecting surface.
Luminous Intensity	Luminous intensity in a given direction is the luminous flux emitted by a light source in an infinitesimal cone containing the given direction divided by the solid angle of that cone.
Marked Foot Crossing	A transverse strip of carriageway marked for the use of pedestrians crossing the road (mid-block or at an intersection) controlled by vehicular and pedestrian signals. The term Signalised Crossing used in this document includes Marked Foot Crossing.
Masterlink-Median	The fully-adaptive mode of operation of the SCATS control system.
Master Controller	A dividing strip that separates vehicles travelling in opposite directions.
Movement	A traffic signal controller controlling a system of secondary controllers.
MTBF	A stream of vehicles that enters from the same approach and departs from the same exit (i.e. with the same origin and destination).
MTTR	Mean time between failures (a signal maintenance term).
Negotiation Speed	Mean time to repair (a signal maintenance term).
Ninety-Fifth Percentile Queue Length	See Speed
	Queue length expected to be exceeded in 5 per cent of signal cycles only, used for designing adequate queue storage length.

Glossary of Terms *(continued)*

Occupancy Time	The time that starts when the front of a vehicle enters the detection zone and finishes when the back of the vehicle exits the detection zone. Thus, it is the duration of the period when the detection zone is occupied by a vehicle.
Off-Peak Period	The periods that have low demand volumes of traffic during the day. Also see Peak Period .
Offset	The difference between the start or end times of Green periods at adjacent (upstream and downstream) signals.
Opposed Turn	See Filter Turn
Opposing Movement	A movement that conflicts with, and has priority over, another (opposed) movement. A movement which approaches from the opposite direction as another on the same road regardless of destination.
Outreach (of Mast-Arm)	The horizontal distance from the centre line of the vertical member to the centre line of the overhead lantern assembly.
Overhang (of Mast-Arm)	The horizontal distance from the kerb alignment to the centre line of the overhead lantern assembly.
Overhead Signal Face	The signal face mounted above the roadway.
Overlap Movement	A movement that runs in consecutive phases without stopping during the associated inter-green period(s). Also see Overlapping Signal Group .
Overlapping Signal Group	A signal group that displays Green in consecutive phases and during the associated Phase transition(s). Also see Overlap Movement .
Painted Island	An area of a road, surrounded by a line or lines (whether broken or continuous), on which there are stripes marked on the road surface in white or another colour contrasting with the colour of the road.
Parallel Pedestrian Movement	A signalised pedestrian movement that runs at the same time as the parallel vehicle movement(s) that are controlled by circular Green displays.
Partial Control	Control of a turning movement by a Green arrow display in one Phase (as an unopposed movement) and by a Green circle display in another Phase (as a filter, or opposed movement).
Passage Detector	A detector which produces a short output (pulse) of relatively constant duration, independent of the mass or type of a moving vehicle within the detection zone.
PBU	Push Button Unit for pedestrian traffic detection
Peak Period	The period that has the highest demand volume of traffic during the day (peak hour, peak half hour, etc). Also see Off-Peak Period .
Pedestrian-Actuated Control	See Traffic-Actuated Control
Pedestrian Aspects	Signal aspects for pedestrians, consisting of the Walk aspect (a Green walking human figure) and the Don't Walk aspect (Red standing human figure). Also see Pedestrian Signal Face .

Glossary of Terms *(continued)*

Pedestrian Clearance Period	The Flashing Don't Walk period that immediately follows the termination of pedestrian Walk display to enable pedestrians, who have just stepped off the kerb at the commencement of this period, to complete their crossing to the nearest kerb or refuge. Pedestrian Clearance Time is the duration of the Pedestrian Clearance Period.
Pedestrian Crossing	A transverse strip of carriageway marked for the use of pedestrians crossing the road (mid-block or at intersections) at a place with a pedestrian crossing sign, and with or without alternating flashing twin Amber lights.
Pedestrian Green Time	The duration of the Green Walk display.
Pedestrian Indicator	An indicator mounted on the pedestrian push-button assembly that is illuminated to acknowledge when a demand has been recorded.
Pedestrian Push Button Assembly	Device to enable pedestrians to register a demand for right of way. It includes the pedestrian indicator. Push button devices are also used for bicycles and emergency vehicles.
Pedestrian Refuge	A place, usually in a carriageway, set aside for exclusive use of pedestrians.
Pedestrian Signal Face	A two-aspect signal face for the control of pedestrian movements that consists of Green Walk and Red Don't Walk aspects. Also see Pedestrian Aspects .
Perception and Reaction Time	The time between the commencement of a stimulus, e.g. change in signal condition, and the taking of appropriate action, e.g. application of vehicle brakes.
Phase	That part of a signal cycle during which one or more movements receive right of way subject to resolution of any vehicle or pedestrian conflicts by priority rules. A Phase is identified by at least one movement gaining right of way at the start of it and at least one movement losing right of way at the end of it.
Phase Sequence	The order of phases in a signal cycle. Also see Display Sequence , Signal Phasing .
Phase Split	Duration of each Phase (Green Time and Inter-green Time) within a signal cycle. It is normally expressed as a percentage of cycle length.
Phasing	See Signal Phasing
Plan	See Timing Plan
Platoon	A group of vehicles or pedestrians travelling together because of signal control, geometric conditions or other factors.
Pole / Post	Vertical tubular support for traffic signal lanterns and associated signs.
Presence Detector	A detector that produces a continuous output while a moving or stationary vehicle is present within the detection zone.
Primary Signal Face	The signal face mounted on a post at or near the left of the stop line of the approach. Also see Dual Primary Signal Face , Secondary Signal Face , and Tertiary Signal Face .
Priority Rule	A traffic regulation which assigns priority to one stream of traffic. Also see Conflicting Movements , Filter Turn , Opposing Movement .
Priority Traffic	That traffic which is allocated priority service at an intersection by the operation of signals. It may include emergency vehicles or public transport vehicles. Not to be confused with the term Priority Road (Major Road) for un-signalised intersections (a road on which traffic has right of way over all entering or crossing traffic).

Glossary of Terms *(continued)*

Progression	Progression is a time-relationship, between adjacent traffic signals, which allows vehicle platoons to be given a Green signal as they pass through the sequence of intersections.
PwD	Persons with Disabilities
Queue	A line of vehicles or pedestrians waiting to proceed through an intersection. Slowly moving vehicles or pedestrians joining the back of the queue are usually considered part of the queue. The internal queue dynamics can involve starts and stops. A faster-moving line of vehicles is often referred to as a moving queue or a platoon.
Reaction Time	See Perception and Reaction Time
Recall Feature	A feature of traffic signal controllers whereby a demand is placed for a particular Phase when no other demands are present and all approach timers have expired. Also see Arterial Demand .
Red Arrow Drop Out	A form of partial control that uses three-aspect (Red, Amber, Green) right-turn arrows on a six-aspect signal face and extinguishes the Red arrow for the adjacent Green circle display to permit filter turns after a few seconds in the through Phase.
Red Time	Duration of the Red signal display for a Phase or a movement. Also see Effective Green and Red Times .
Repeat Right Turn Phasing	Phasing that introduces the arrow-controlled right turn twice in the same cycle.
Reversible Lanes	See Tidal Flow Scheme
Right-Turn Trap	A situation where a driver executing a filter right-turn manoeuvre at the start of Amber interval thinks that the signals change to Amber for the opposing traffic at the same time, and therefore proceeds and runs into an opposing through vehicle for which the signal display would still be Green. Also known as 'lagging right turn problem' or 'Amber trap'. Also see Lagging Right Turn .
Road Marking	A word, figure, symbol, mark, line, raised marker or stud, or something else, on the surface of a road to direct or warn traffic, but does not include a Painted Island.
Scramble-Crossing Phase (All-Round Pedestrian Movement)	An Exclusive Pedestrian Phase at an intersection where pedestrians are allowed to cross in any direction including diagonally within the limits of the crosswalk lines. Also see Exclusive Pedestrian Phase .
Secondary Signal Face	The signal face mounted on a post on the downstream side to the right of the approach. Also see Primary Signal Face , Tertiary Signal Face .
Semi-Actuated Control	See Traffic-Actuated Control
Sequence	See Display Sequence
Shared Lane	A lane allocated for use by two or more movements, e.g. shared through and right-turn lane. Also see Exclusive Lane .
Sight Distance	The distance over which a road user needs to have unobstructed sight to respond to a visual cue, or to safely avoid a conflict.
Sign	See Traffic Sign
Signal	See Traffic Signal

Glossary of Terms *(continued)*

Signal Aspect	See Aspect
Signal Controller	See Controller
Signal Coordination	See Coordination
Signal Cycle	See Cycle
Signal Display	See Display
Signal Face	A set of signal aspects in a common assembly, generally in one or two columns placed together with a target board to improve signal visibility, facing traffic from one direction.
Signal Group	A signal group is a set of lanterns with common electrical switching such that the aspects illuminated in each lantern are always identical.
Signal Group Overlap	See Overlapping Signal Group
Signal Intensity	The luminous intensity of the signal aspects. Also see Luminance and Luminous Intensity .
Signal Lantern	See Lantern
Signal Linking	See Coordination
Signal Offset	See Offset
Signal Phase	See Phase
Signal Phasing	Sequential arrangement of separately controlled groups of vehicle and pedestrian movements within a signal cycle to allow all vehicle and pedestrian movements to proceed.
Signal Range (Visual Range)	The distance over which the signal is intended to be clearly visible to approaching traffic.
Signal Sequence	see Display Sequence
Signal Timing	The process of determining durations of successive intervals of Green, Amber and Red displays, actuated signal control settings, as well as offsets for coordinated signals.
Signalled Crossing	An area of the road used by pedestrians when crossing the road with the guidance of pedestrian signals at a mid-block or intersection location, and can be used by cyclists if bicycle signals are provided.
Simultaneous-Offset Coordination	Signal coordination where Green periods in the travel direction start at the same time (zero offset). This is useful when the back of queue at the downstream intersection interrupts progression at closely spaced intersections. Also see Offset .
Slip Lane	A left-turn lane separated from an adjacent lane by a triangular island.
Space Length (Gap Distance)	The following distance between two successive vehicles as measured between the rear end of one vehicle and the front end of the next vehicle in the same traffic lane (spacing less vehicle length).
Space Occupancy Ratio	The proportion of a road section (distance) occupied by vehicles at an instant in time.

Glossary of Terms *(continued)*

Space Time	The time between the detection of two consecutive vehicles when the presence detection zone is not occupied. It is equivalent to gap time less the time taken to travel the effective detection zone length.
Spacing	The distance between the front ends of two successive vehicles in the same traffic lane.
Special Vehicle Aspects	Signal aspects for special vehicles (bus, tram, emergency vehicle) consisting of Red, Amber and white B, T and E symbols. Also see Special Vehicle Signal Face .
Special Vehicle Signal Face	A single-column signal face for the control of special vehicle (bus, tram, emergency vehicle) movements. Also see Special Vehicle Aspects .
Speed-	Distance travelled per unit time. In a time – distance diagram, the slope of the trace of a vehicle is its speed. Approach Speed is the uninterrupted (mid-block) cruise speed of vehicles before being affected by traffic signals. This can be represented by the speed limit. Negotiation Speed is the safe speed of a vehicle moving through the controlled area of the intersection. For turning vehicles, this can be determined as a function of the negotiation radius. Eighty-fifth Percentile Speed is the speed at or below which 85 per cent of the traffic travels.
Split-Approach Phasing	The signal phasing arrangement that allocates separate phases to opposing approaches, allowing the through and all turning movements from each approach to operate simultaneously.
Staged Signalled Crossing	A system by which a long signalised crossing is divided or 'staged' into several time-separated sections, each being a separate group controlled by individual signals.
Start Loss	Duration of the interval between the start of displayed Green period and the start of effective Green period for a movement. This is used in signal timing and performance analysis to allow for queue discharge time losses at the start of Green period due to vehicles accelerating to saturation speed, or due to giving way to opposing vehicle or pedestrian movements. Late Start time can be treated as a start loss for the movement that starts after the Late Start interval. Also see End Gain .
Starting Display	See Starting Signal
Starting-Offset Coordination	Coordination of the start times of Green periods using a signal offset that equals the average uninterrupted travel time in the travel direction so as to minimise stops along the route. Also see Offset .
Starting Signal	A signal which is located so that the drivers who have stopped at the stop line can see the signal display.
Stop Line	A single continuous line (double continuous lines in New Zealand, known as limit line) marked across all or part of a road, behind which vehicles should stand when required to stop by traffic light signals or regulatory signs. Also see Give-Way Line .
Stopping Display	See Stopping Signal
Stopping Signal	A signal which is used to enable the approaching driver to make a decision either to stop safely in front of the stop line or proceed into the intersection.
Stopping Sight Distance	The sight distance which is necessary to enable a driver to stop safely in response to a Red signal.

Sun Phantom	The internal reflection of light from external sources (especially the sun) onto the lens of a signal lantern which makes it appear to be internally illuminated.
Symbolic Mask	A mask used to create a symbol (e.g. arrow, pedestrian or bicycle) or letter (e.g. B or T) to dedicate the aspect to a particular vehicle movement (e.g. turning traffic) or a special vehicle movement (e.g. bus, tram). Also see Arrow Mask .
Synchronous Linking	See Cable-less Linking
T-Intersection	An intersection where two roads meet (whether or not at right angles) and one of the roads ends.
Target Board	The panel attached to a signal face to improve its visibility by reducing background luminance.
Tertiary Signal Face	The signal face mounted on a post on the downstream side to the left of the approach. Also see Primary Signal Face , Secondary Signal Face .
Through Phasing	Phasing that allows through and left-turn movements and filter right turns from opposing approaches to operate in the same Phase using the three-aspect circular (Red, Amber, Green) signal faces.
Tidal Flow Scheme	A traffic management method for increasing capacity under conditions of high traffic demand and marked directional split by means of reversible lanes or carriageways.
Time-Distance Diagram	A graphical representation of the movement of a vehicle or traffic stream in terms of its time and distance coordinates, e.g. used to show signal coordination along a route or for showing vehicles queuing at traffic signals.
Time Occupancy Ratio	The proportion of time in a given period when the passage or occupancy detector at a point along the road is occupied by vehicles.
Timing Plan-	A plan that defines the cycle time, Green splits and offsets for each intersection in a coordinated signal system.
Traffic-	Movement of vehicles and pedestrians (people and goods) along a route.
Traffic-Actuated Control	A control method that allows a variable sequence and variable duration of signal displays depending on vehicle and pedestrian traffic demands. All vehicle movements (phases) are actuated in Fully-Actuated Control in contrast with Semi-Actuated Control where only minor vehicle movements (e.g. side road traffic) are actuated.
Traffic Composition	The proportions (usually expressed as a percentage) of the different vehicle classes and turning vehicles within the total traffic flow.
Traffic Control Device	Any traffic sign, road marking, traffic signal, or other device, placed or erected under public authority for the purpose of regulating, directing, warning or guiding traffic.
Traffic Count-	The process of determining the number of vehicles or pedestrians passing a given point or points during a specified period of time. Also see Demand Flow (Demand Volume) , Flow Rate , Traffic Volume .
Traffic Demand	See Demand
Traffic Flow	See Traffic Volume
Traffic Island	A structure on a road to direct traffic, but does not include a Road Marking or Painted Island.

Traffic-Responsive Plan Selection	A signal coordination method that introduces predetermined plans by means of algorithms that respond to changing traffic conditions using data collected from detectors.
Traffic Sign	A board, plate, screen, or another device, whether or not illuminated, displaying words, figures, symbols or anything else to regulate, direct or warn road users, and includes a children crossing flag, a hand-held stop sign, a parking control sign and a variable illuminated message sign, but does not include traffic signals.
Traffic Signal	A signal that controls vehicle and pedestrian traffic at an intersection or on a road by means of Red, Amber, Green or white light displays, and includes circular and arrow signals, pedestrian signals, bicycle crossing signals, B (bus) and T (tram) signals, overhead lane control signals, and twin Red or Amber signals.
Traffic Signal Countdown Timer	It is a timer device displayed along with traffic signal to show how many seconds are remaining.
Traffic Volume	The number of vehicles or pedestrians passing a given point on a lane or carriageway during a specified period of time. Also see Demand Flow (Demand Volume) , Flow Rate , Traffic Count .
Trailing Right Turn	See Lagging Right Turn
Transponder	A device fitted to a bus, tram or emergency vehicle that allows the signal controller to selectively identify the presence of that vehicle in a stream of mixed traffic.
Undivided Road	A road without a median. Also see Divided Road .
Unopposed Turn	A left-turn or right-turn movement at a signalised intersection that is made with no opposing or conflicting vehicular or pedestrian flow allowed.
Upstream	In the direction opposite to the movement of traffic.
Urban Traffic Control System	See Wide Area Control System
Vehicle-Actuated Control	See Traffic-Actuated Control
Vehicle Passage Time	The time between the passage of the front and back ends of a vehicle from a given point along the road.
Visor	An attachment to the face of a signal aspect to minimise the sun-phantom effect and/or to reduce the possibility of a signal being seen by traffic for which it is not intended. Also called Cowl .
Volume of Traffic	See Traffic Volume
Walk Time	Duration of the Walk display (steady Green person) for pedestrians.
Walkway	That portion of a road reserved exclusively for pedestrians. Often referred to as footpath.
Wide Area Control System	A signal coordination system comprising one or more centrally or regionally located computers controlling relatively large numbers of signals, with all signals connected to the traffic control computers, usually by leased data lines or, in some cases, by dedicated cable systems.

Contents

	Foreword	iii
	Preface	v
	Document Management	vi
	Acknowledgements	vii
	Abbreviations	viii
	Glossary of Terms	ix
	Contents	xxii
	List of Tables	xxvii
	List of Figures	xxviii
1	Introduction	1
	1.1 General	1
	1.2 Scope	3
	1.3 Organisation of the Manual	3
	1.4 Design Process	4
	1.5 History	4
2	Design data for Signal Sites	5
	2.1 Justification for Traffic Signal Sites	5
	2.1.1 Vehicle and Pedestrian Counts for Justification	5
	2.1.2 Road Safety	5
	2.1.3 Other Justifications	5
	2.2 Warrants for Traffic Signals	6
	2.2.1 Traffic volume criteria	6
	2.2.2 Warrant 4 - Crash experience	7
	2.2.3 Warrant 5 – Combination of warrants	7
3	Design Data for Traffic Signals	9
	3.1 Introduction to Geometric Layouts	9
	3.2 Physical Layout of Signal Sites	9
	3.3 Traffic Count Data	10
	3.3.1 Detailed Traffic Counting Guidelines	10
	3.3.2 Analysis of Traffic Count Data	11
	3.3.3 Motor Vehicle Counts	12
	3.3.4 Pedestrians and Cyclist Counts	12
	3.3.5 Road Crash Data	12
	3.3.6 Traffic Impact - Future Developments	12
4	Traffic Signal Control Equipment	13
	4.1 Introduction to Signal Equipment	13
	4.2 Signal System and Detection	13
	4.3 Components	15
	4.4 Signal Pole and Lanterns - Equipment Layout	15
	4.5 Signal Lantern Assembly	17
	4.6 Lantern Visibility Factors	17
	4.6.1 Lantern Visibility Distances	17
	4.7 Poles and Mast-arms	18
	4.8 Associated Lantern Equipment	19
	4.8.1 Visors	19
	4.8.2 Louvres	19
	4.8.3 Target Boards	19
	4.8.4 Symbolic Masks	20
	4.8.5 Lantern Supports	20
	4.9 Power Distribution to Signals Displays	20
	4.10 The Traffic Signal Controller	20
	4.11 Traffic Detection	20

5	4.12	Other Signal Control Equipment	21
	4.12.1	<i>Audio Tactile Facilities</i>	21
	4.12.2	<i>Disability Facilities - Ramps</i>	21
	4.12.3	<i>Pedestrian Guardrails</i>	21
	4.12.4	<i>Pedestrian Crossings</i>	21
	4.12.5	<i>Street Lighting</i>	22
	4.12.6	<i>Bus or BRT Priority at Traffic Signals</i>	22
	4.12.7	<i>Dual Ring Operation</i>	22
	4.12.8	<i>Microprocessor Optimised Vehicle Actuation (MOVA)</i>	22
	4.12.9	<i>Remote Monitoring (Dial-Up)</i>	22
6		Geometric Design	23
	5.1	Lane Arrangements	23
	5.1.1	<i>Right-Turn (RT) Lanes</i>	23
	5.1.2	<i>Left-Turn (LT) Lanes</i>	23
	5.2	Guidelines for Geometric Designs at Signal Sites	23
	5.3	Design Considerations for Special Junction Characteristics	24
	5.4	Traffic Islands and Corner Kerb Radius	24
	5.5	Median Islands and Median Strips	25
	5.5.1	<i>Divided Roads</i>	25
	5.5.2	<i>Undivided Roads with Small Median 'Strips' at the Junction</i>	25
	5.5.3	<i>Painted Medians and Islands</i>	25
	5.6	Parking	25
	5.7	Bus Zones	26
	5.8	Signalised Roundabout	26
	5.8.1	<i>Types of Signal control</i>	26
	5.8.2	<i>Method of Signal control</i>	26
	5.8.3	<i>Duration of Signal Control</i>	28
	5.8.4	<i>Geometric Modifications</i>	28
		Signal Face Layouts and Display Sequences	29
	6.1	Introduction to Signal Displays	29
	6.2	Signal Aspects	29
	6.3	Size of Aspects	29
	6.4	Types of Aspects	30
	6.4.1	<i>Full Roundel (Circular) Aspects</i>	30
	6.4.2	<i>Arrow Aspects</i>	30
	6.4.3	<i>Pedestrian Aspects</i>	30
	6.4.4	<i>Cycle Aspects</i>	31
	6.4.5	<i>Special Vehicle Aspects</i>	31
	6.4.6	<i>Combination of Aspects (Full Roundel and Arrow)</i>	31
	6.5	Basic Signal Face Layouts	31
	6.5.1	<i>Multi-Column Signal Face Layouts</i>	32
	6.5.2	<i>Four-Aspect Single-Column Signal Face Layouts</i>	32
	6.6	Signal Face Layouts with RT Arrow Aspects	33
	6.6.1	<i>Six-Aspect Signal Face Layouts with RT Arrow Aspects</i>	33
	6.6.2	<i>Four-Aspect Signal Face Layouts with LT & RT Arrow Aspect</i>	33
	6.7	Face Layouts with Left-Turn Arrow Aspects	33
	6.7.1	<i>Six-Aspect Signal Face Layouts with LT Arrow Aspects</i>	33
	6.8	Allowable Lantern Aspect Face Combinations	34
	6.9	Types of Aspects	34
	6.9.1	<i>Full Roundel (Circular) Aspects</i>	34
	6.9.2	<i>Arrow Aspects</i>	35
	6.9.3	<i>Pedestrian Aspects</i>	36
	6.9.4	<i>Cycle Aspects</i>	37
	6.10	Pedestrian Signals	38
	6.11	Cycle Signals	38

6	6.12	Special Vehicle Signals	38
	6.13	Summary of Rules for Signal Display Sequences	38
	6.14	Basic Signal Face Layouts	39
	6.15	Basic Signal Face Layouts	39
7		Determination of Signal Phasing (or Staging)	41
	7.1	Introduction and Scope	41
	7.2	Fundamental Rules of Signal Control	41
	7.3	Determination of Phase Sequences	41
	7.4	Signal Groups	43
	7.5	Two Phase Sequences	43
	7.6	Three Phase Sequence	45
	7.7	Multi-Phase Signal Sites	47
	7.8	Evaluation of Phase Design	47
	7.9	Road Crash Risk Factors	47
	7.10	Phasing Design	47
	7.11	Through Phase with Filter RTs	50
	7.12	Leading RT Phasing	51
	7.13	Lagging RT Phasing	51
	7.14	Split-Approach Phasing	51
	7.15	Diamond Overlap Phase	52
	7.16	Lead-Lag RT Phase	52
	7.17	Repeat RT Phase	52
	7.18	LT Movements	53
	7.19	Pedestrian Movements	54
	7.19.1	<i>General</i>	54
	7.19.2	<i>Pedestrian Movement Operation</i>	55
	7.19.3	<i>Protected Pedestrian Movements and Conditional Movements</i>	57
	7.19.4	<i>Pedestrian Movements at Intersections</i>	59
	7.19.5	<i>Mid-block Pedestrian Signal Crossings</i>	59
	7.19.6	<i>Pelican Crossing Signals</i>	59
	7.19.7	<i>Crossing Width</i>	60
8		Layout of Traffic Signal Equipment	61
	8.1	Introduction to Pole Placement and Lantern Alignment	61
	8.2	Arrangement of Signal Lanterns (Faces)	61
	8.2.1	<i>Primary Signal Poles</i>	61
	8.2.2	<i>Secondary Signal Poles</i>	61
	8.2.3	<i>Tertiary Signal Face</i>	61
	8.2.4	<i>Dual Primary Signal Lanterns</i>	61
	8.2.5	<i>Dual Secondary Signal Lantern</i>	61
	8.2.6	<i>Overhead Signal Lanterns</i>	62
	8.3	Signal Lantern Functions	63
	8.4	Signal Displays - Site Requirements	65
	8.4.1	<i>Recommended Minimum Number of Signal Displays</i>	65
	8.4.2	<i>Dual Primary Signal Displays</i>	65
	8.4.3	<i>Mast-Arm or Overhead Signal Warrants</i>	65
	8.4.4	<i>Overhead Signal Displays Placement</i>	65
	8.4.5	<i>Pedestrian Signal Displays</i>	66
	8.5	Positioning of Poles, Lanterns and Signal Lantern Displays	67
	8.6	Equipment Placement for Four-Way Intersections	67
	8.7	Equipment Placement at T-Junction (Three-Way)	67
	8.8	Equipment Placement at Mid-Block Signal Crossings	67
	8.9	Lantern Mounting Heights	69
	8.10	Lantern Aiming	70
	8.11	Visor Types	70
	8.12	Louvre Types and Placement	71

9	Traffic Detection	73
	9.1 Introduction	73
	9.2 Vehicle Detection Systems	73
	9.3 Characteristics of Inductive Loop Detectors	73
	9.4 Traffic Detection During the Signal Cycle	74
	9.5 Vehicle Actuation – Detector Demands	75
	9.6 Vehicle Detection Modes	75
	9.7 Inductive Loop Vehicle Detection	76
	9.7.1 <i>Detector Types</i>	76
	9.7.2 <i>Other Detection Devices and Methods</i>	76
	9.7.3 <i>Demands for a Phase</i>	76
	9.7.4 <i>Green Extension</i>	76
	9.7.5 <i>Detector Functions in an ATC System</i>	76
	9.7.6 <i>Inductive Loop Shape and Size</i>	76
	9.7.7 <i>Placement of Detectors</i>	79
	9.7.8 <i>Stop-Line Loop Detection</i>	79
	9.7.9 <i>Advance or 'Passage' Detection</i>	80
	9.7.10 <i>Queue Detection</i>	81
	9.7.11 <i>Operational Characteristics of Loop Detectors</i>	81
	9.8 Pedestrian Traffic Demands	82
	9.8.1 <i>Pedestrian Detection</i>	82
	9.8.2 <i>Automatic Introduction of Pedestrian Movements</i>	82
	9.8.3 <i>Pedestrian Push-Button Placement</i>	82
	9.8.4 <i>Cycle Detection</i>	83
	9.9 Special Traffic Detection	84
	9.9.1 <i>Emergency Vehicle Detection</i>	84
	9.9.2 <i>Buses</i>	84
	9.9.3 <i>Trams / LRT</i>	84
	9.9.4 <i>Tram / LRT Operation at Signal Sites</i>	84
	9.9.5 <i>Railway Traffic</i>	85
	9.9.6 <i>Railway Linking Requirements</i>	85
10	Traffic Signal Controller	87
	10.1 Introduction to Traffic Signal Control	87
	10.2 Local Controller Operational Modes	87
	10.2.1 <i>Fixed-Time Control</i>	88
	10.2.2 <i>Vehicle Actuated (VA) Control (Isolated Mode)</i>	88
	10.2.3 <i>Vehicle Semi-Actuated Control</i>	88
	10.2.4 <i>Local Link Control</i>	89
	10.2.5 <i>Central Computer Control</i>	89
	10.2.6 <i>Fallback Mode</i>	89
	10.2.7 <i>Maintenance Mode</i>	89
	10.3 Controller Programming	89
	10.4 Preventing Conflicting (Hazardous) Displays	89
	10.5 Controller Timing Intervals and Movements Strategies	90
	10.5.1 <i>Vehicle Phase Intervals</i>	91
	10.5.2 <i>Pedestrian Phase Intervals</i>	92
	10.5.3 <i>Vehicle Actuation Timers</i>	94
	10.5.4 <i>Control of Movements</i>	95
	10.5.5 <i>Detector Logic</i>	95
	10.5.6 <i>Conditional Demands</i>	95
	10.5.7 <i>Conditional Extension</i>	95
	10.5.8 <i>Conditions for Phase Termination</i>	95

11	Electrical Design	97
	11.1 Introduction to Electrical Design	97
	11.2 Cable Connection Design	97
	11.3 Cable Colours and Numbers	97
	11.4 Cable Installation	98
	11.5 Vehicle Loop Detector Cables	98
12	Installation Checks and Maintenance	101
	12.1 Commissioning Check List	101
	12.2 Monitoring and Evaluation of Operation	101
	12.3 Monitoring Check List	101
	12.4 Maintenance	101
	12.5 Maintenance Management	101
	12.6 Maintenance Records	102
	12.7 Urgent Maintenance	102
	12.8 Accident Damage	102
	12.9 Cleaning and Painting	102
13	Traffic Control Strategies and Tools	103
	13.1 Overview of Signal Systems	103
	13.2 Signal Coordination Methodology	103
	13.3 Local Signal Linking Systems	103
	13.4 Area Wide Traffic Control (ATC) Systems	103
	13.4.1 Types of ATC Systems	103
	13.5 Traffic System Design Tools	104
	13.6 Time Distance Diagrams	104
	13.6.1 Site Data Collection for Coordination Plans	105
	13.6.2 Site Data Analysis for Coordination Plans	105
	13.6.3 Construction of One Directional Diagrams	106
	13.6.4 Construction of Two Directional Diagrams	107
	13.6.5 Compensation for Queue Formation	107
	13.6.6 Offset Strategies	108
	13.7 Traffic Signal Coordination	109
	13.7.1 Introduction to Coordination	109
	13.7.2 The Case for Coordination	109
	13.7.3 Local Linking by Cable	109
	13.7.4 Local Linking by Synchronous Clock (Cable-Less)	109
	13.7.5 Area Wide Traffic Control Systems	109
	13.8 Coordination Requirements Affecting the Design of Signal Sites	110
14	Traffic Signal Timings	111
	14.1 Introduction to Timings	111
	14.2 Signal Timing Methods	111
	14.2.1 Green Times and Cycle Time (CT)	111
	14.2.2 Maximum Tolerable Delay	111
	14.2.3 Time Transfer and Stealing	111
	14.2.4 Vehicle Phase Intervals	112
	14.2.5 Inter-Green	113
	14.2.6 Late Start and ECO	113
	14.2.7 Minimum Green	114
	14.2.8 Variable Initial Green Interval	114
	14.2.9 Rest and Extension Green Intervals	115
	14.2.10 Gap Timer	115
	14.3 Pedestrian Intervals	115
	14.4 Actuated Controller Settings	116

14	14.5	Vehicle Settings	116
	14.5.1	Late Start Setting	117
	14.5.2	Minimum Green Setting	117
	14.5.3	Basic Minimum Green Setting	117
	14.5.4	Increment and Maximum Variable Initial Green (VIG)	117
	14.5.5	Maximum Extension Green Setting (Max Green)	118
	14.5.6	Gap Setting	119
	14.5.7	Early Cut-Off Green and Early Cut-Off Amber Settings (ECO)	119
	14.5.8	Vehicle Clearance (Inter-Green) Settings	120
	14.5.9	Amber Time	120
	14.5.10	All-Red Time	121
	14.6	Other Vehicle Settings	122
	14.6.1	Timed Presence Time-Setting	122
	14.6.2	Headway and Waste Time Settings	122
	14.7	Pedestrian Movement Settings	122
	14.7.1	Pedestrian Green 'Man' (Walk) Time	122
	14.7.2	Pedestrian Clearance Time	123
	14.7.3	Pedestrian Delay Setting	123
	14.8	Cycle Settings	123
15		Bibliography	125
16		Appendices	126
	Appendix A	Loop Detectors	126
	Appendix B	Monitoring and Evaluation Checklist	127

List of Tables

Table 1.1	Road Design Manual (RDM) Coding Structure	2
Table 2.1	Warrant 1- Minimum Traffic Volume	3
Table 2.2	Warrant 2- Interruption of continuous traffic	3
Table 2.3	Passenger Car Units	7
Table 4.1	Traffic Signal System - Components and Purpose	15
Table 4.2	Visibility Distances	17
Table 8.1	Signal Face Functions	64
Table 8.2	Visual Coverage of Lanterns	70
Table 8.3	Recommended Aiming Distances	70
Table 8.4	Cut-Off Angles for Visors	71
Table 9.1	Lateral Dimensions of Symmetric pole	79
Table 10.1	Interval Description	91
Table 10.2	Interval Range	91
Table 14.1	Summary of Signal Controller Time-Settings	112
Table 14.2	Typical Signal Controller Settings Used with Stop-Line Detectors	116
Table 14.3	Amber Time Values (deceleration 3.0 m / sec)	121

List of Figures

Figure 3.1	Intersection Showing Classified Traffic Count Data	10
Figure 4.1	The Traffic Signal System	14
Figure 4.2	Simple Junction Signal Layout	16
Figure 4.3	Traffic Signal Components Signal Poles and Lanterns	18
Figure 5.1	Intersection Geometric Layout	24
Figure 5.2	Signalised Roundabout with Partial, Direct Signal Control	27
Figure 5.3	Signalised Roundabout with Indirect Signal Control	27
Figure 6.1	Recommended Signal Aspects	30
Figure 6.2	Signal Aspects Orientation	30
Figure 6.3	Basic Three-Aspect Signal Face Layout	31
Figure 6.4	Six-Aspect Multi-Column Signal Face Layouts	32
Figure 6.5	Four-Aspect Single-Column Signal Face Layouts	32
Figure 6.6	Four-Aspect Signal Face Layouts Single RT Green Arrow Aspect	33
Figure 6.7	Four-Aspect Signal Face Layout Single LT Green Arrow Aspect	33
Figure 6.8	Allowable Signal Face Layouts	34
Figure 6.9	Basic Display Sequence Three Aspect Signals	34
Figure 6.10	Fully-Controlled RT Display Sequences Six-Aspect Arrangements	31
Figure 6.11	Display sequence to Initiate an Arrow - Controlled RT Movement	36
Figure 6.12	Display Sequence to Terminate an Arrow - Controlled RT Movement	36
Figure 6.13	Display Sequence to Initiate an Arrow – Controlled LT movement	37
Figure 6.14	Display Sequence to Terminate an Arrow – Controlled LT movement	37
Figure 6.15	Sequence for LT Amber	39
Figure 7.1	Simple Junction Signal Layout	42
Figure 7.2	Right Turn 'B' Phase Arrows Application of Late Start Control	44
Figure 7.3	Operation of a Late Start Interval for LT movement	45
Figure 7.4	Alternative Overlap Operation (Single Diamond Overlap – S.D.O)	46
Figure 7.5	Basic Elements of Signal Phasing	48
Figure 7.6	Two Phase Control – Filter RTs and Parallel Pedestrians	49
Figure 7.7	Conflicts in Two Phase Control	49
Figure 7.8	Three-Phase System	50
Figure 7.9	Three-Phase System	51
Figure 7.10	Complementary LT and RT arrow-Control	53
Figure 7.11	Complementary LT and RT Arrow-Control at T-Junctions	53
Figure 7.12	Pedestrian Movement Operation	55
Figure 7.13	'WALK for Green' Operation	56
Figure 7.14	Fully Protected Pedestrian Movement	57
Figure 7.15	Partially and Fully Protected WALK	58
Figure 8.1	Signal Pole & Lantern Arrangements Four-Way Intersections	62
Figure 8.2	Signal Lantern Arrangements for T-Intersections & Mid-Block Signal Crossings	63
Figure 8.3	Symbols Used to Describe Signal Displays and PBUs	64
Figure 8.4	Pedestrian Signal Faces: Pedestrian Arrow Orientations Shown In (a) and (b)	66
Figure 8.5	Signal Lantern Visibility Templates	68
Figure 8.6	Signal Lantern Placement Channelled LT Control	68
Figure 8.7	T Intersection signal Location	69
Figure 8.8	Midblock Crossing Arrangement	69

List of Figures

Figure 9.1	Presence Vehicle Loop Detector System Relevant Measurements to Traffic Parameters	74
Figure 9.2	Detector and Demand Operation	77
Figure 9.3	Typical Loop Shapes	78
Figure 9.4	Stop-Line Loop Configuration and Layout	80
Figure 9.5	Advance Loop	81
Figure 9.6	Queue Loop for Detectors Demanding a RT Phase	81
Figure 9.7	PBUs Location and Orientation	83
Figure 10.1	Typical Interfacing Requirements for Traffic Signal Controllers	87
Figure 10.2	Safety of Signal Displays	90
Figure 10.3	Vehicle Timing Intervals	92
Figure 10.4	Pedestrian WALK and Red Arrow Protection	93
Figure 10.5	Headway and Waste Time Operation	94
Figure 10.6	Conditions for Phase Termination	96
Figure 11.1	Typical Detector Location Layout	99
Figure 13.1	PC Based ATC System Components	104
Figure 13.2	Time Distance Diagram –Base Graph	106
Figure 13.3	Time distance Diagram – Signal Phases	106
Figure 13.4	Time Distance Diagram – Progression Line	107
Figure 13.5	Time Distance Diagrams –Through Band	107
Figure 13.6	Time Distance Diagrams –Queue Formation	108
Figure 14.1	Transfer of Unused Maximum Time	113
Figure 14.2	Maximum Time ‘Stealing’	114
Figure 14.3	Phase (or Signal Group) Vehicle Intervals	114
Figure 14.4	Definition of Pedestrian Movement (Phase) Intervals - Parallel Vehicle Movement Intervals	115
Figure 14.5	Pedestrian & Vehicle Phase Intervals - Pelican Crossing	116
Figure 14.6	ECO Green and ECO Amber Operation	119
Figure 14.7	Typical Use of Early Cut-Off Periods	120
Figure 14.8	FALL-Red Time as a Function of Speed (V_d) and Clearance Distance (L_c)	121

1 Introduction

1.1 General

This Manual was prepared by the Ministry of Roads and Transport as part of a series of manuals that cover the entire project cycle. The series incorporate best practices, climate change considerations, and recent technologies thereby enabling the provision of road infrastructure that is safe, secure, and resilient.

The Kenya road manual series is as follows:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audits	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

This table continues onto the next page...

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges & Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

This Road Design Manual, Volume 6, Part 3 – Traffic Signals is part of the Roads Design Manual made up of a series of volumes and shown in [Table 1.1](#).

Table 1.1 Road Design Manual (RDM) Coding Structure

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 – Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation & Design	RDM 4.1
		Part 2 – Bridge & Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Reinforced Fill Structures Design	RDM 4.4
		Part 5 – Bridges and Structures Condition Survey	RDM 4.5
		Part 6 – Bridge Maintenance Design	RDM 4.6
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This manual must be applied sensibly and flexibly in conjunction with the skill and judgement of the designer. Compliance with the guidance given in the manual does not relieve designers of the responsibility for establishing that their design is suitable, appropriate, safe, and adequate for the purpose stated in the project requirements.

1.2 Scope

The scope of the manual includes design and installation of traffic signal at road junctions. It covers the warrants required to justify installation of signals, the data required for designing signals, the geometric layout needed for installation of signals. It details the equipment, including lanterns, poles, mast heads, along with the pedestrian crossing and bus priority facility. Signal face layout, display sequence, time for each phase, layout of the equipment are presented. Traffic detection mechanism, controller equipment, electrical design and maintenance requirements are elaborated. The various tools available for signal design are mentioned along with the strategies for setting up signal systems.

1.3 Organisation of the Manual

This Manual gives guidance on traffic signals design and associated equipment and presents detailed information and guidelines on the following topics:

1. Collection of Intersection and Pedestrian Crossing Signals Design Data.
2. Geometric Elements at Signal Sites.
3. Signal Systems and Components.
4. Signal Face Layouts and Signal Display Sequences.
5. Signal Phasing Designs.
6. Placement of Signal Equipment.
7. Traffic Detection Methods and Detection Devices.
8. Traffic Signal Controllers.
9. Electrical Design of Traffic Signals and Detection Systems.
10. Coordination of Traffic Signal Areas.
11. Traffic Signal Installation and Maintenance.

Information is also provided on Other Special Applications including:

- A Glossary of Traffic Signal Engineering Terms
- Traffic Signal Control of Emergency Vehicles
- Public Transport Signal Priority
- Bicycle Facilities at Signal sites
- Special Intersection Treatments
- Determination of Traffic Signal Timings

This manual is to be read in conjunction with RDM Vol 1, Part 2: Traffic Surveys, RDM Vol 1, Part 3: Geometric Design for Highways, Rural, and Urban Roads, RDM Vol 6, Part 1: Road Marking and RDM Vol 6, Part 2: Traffic Signs.

Refer RDM 1.3 for geometric design of intersections, roundabouts, capacity at intersections and signal time design methodology.

1.4 Design Process

The process of designing and installing traffic signal is:

1. Collect data on traffic volume, pedestrian volume, turning movement, crashes history.
2. Check warrants to see if signal installation is justified.
3. Analyse the signal phasing and timing and detection mechanism.
4. Consider requirements for bus priority, pedestrian / cycle requirement, any special requirement. like signal progression on corridors.
5. Design traffic signal, using tools available.
6. Prepare drawings and BOQ for geometric improvements at the junction and for layout of signal equipment.
7. Procure necessary equipment, controller, computer.

design process. **Chapter 2** gives a background of signs, hierarchy of roads, classification of signs, siting, orientation, material, post, mounting, maintenance and also on testing of signs. **Chapter 3** is about regulatory signs- mandatory, prohibitive and priority. **Chapter 4** describes the warning signs. Informatory signs are given in **Chapter 5** and other Informatory signs like facilities signs are given in **Chapter 6**. **Chapter 5** also details design of letters, block, stroke width, border to design sign faces. Signs for persons with disabilities are given in **Chapter 7**. **Chapter 8** is on Expressway signs and guidelines for urban roads are detailed in **Chapter 9**. **Chapter 10** elaborates signs for roadworks.

This manual is to be read in conjunction with RDM Vol 1, Part 3: Geometric Design for Highways and Rural and Urban Roads, RDM Vol 6, Part 1: Road Marking and RDM Vol 6, Part 3: Traffic Signals and Communication Systems.

1.5 History

With the growth of cities and increase in the number of vehicles on the roads, the need for separation of traffic streams and movement have become essential for safety and efficiency. The use of hand signal control by police officers was supplanted by a gas operated signalling device in 1868. It was installed opposite the Houses of Parliament in London. Signalling systems of various kinds were subsequently developed and the now familiar three coloured light signals were installed in New York in 1918. Similar signals were installed in England in 1925. As the installation of traffic signals progressed, refinements in equipment began to appear.

The manual operation of early signals was replaced by automatic operation controlled by a simple electromechanical timer. The limitations of fixed time operation were quickly recognised. A system was devised for automatically altering the signal timing according to the volume and flow of traffic. This form of operation has come to be known as 'Vehicle actuated' (VA) or 'traffic actuated' operation.

Most modern cities now employ a traffic control room with central computer control of traffic signal sites. Detectors at these sites can send information on traffic volumes, thus enabling the central computer to vary timings for individual sites and to ensure coordination between signal sites.

2 Design Data for Signal Sites

2.1. Justification for Traffic Signal Sites

The reasons for installing traffic signals are:

1. Increasing safety at a road junction.
2. Assisting movement of traffic in and out of side roads.
3. Augmenting a large traffic management scheme.
4. Providing enhanced facilities for pedestrians, cyclists, or non-motorised traffic (NMT).
5. Controlling movement of buses, trams / trains / LRT/ BRT, emergency vehicles.
6. Increase road capacity.
7. Reduce stops and delays.

A decrease in delay for side roads and minor roads may result in an increase in delays on the major route, once the signals have been installed. The overall delay on the major route can be decreased by using coordinated signal linking techniques.

The design engineer must be clear on the objectives of the traffic signal scheme and use the flexibility of the controller and signal displays to achieve safety and efficiency. If alternate or additional criteria for traffic signal design is present at local level, those should be applied.

2.1.1 Vehicle and Pedestrian Counts for Justification

Traffic counts are required at two stages in the process of providing traffic signals.

1. To determine if the provision of signal is really justified.
2. During the detailed design process, in order to establish the best phasing and timing arrangements.

Traffic counts for signal design are carried out over peak and off-peak periods and classified according to different vehicle types and their turning movements. For different classes of vehicles and their turning movements, refer to RDM Vol 1, Part 2: Traffic Surveys.

2.1.2 Road Safety

Traffic signals are often used to improve safety at a junction. It is important to consider the crash rate at the junction and compare it to similar locations in the area.

2.1.3 Other Justifications

New signal installations may also be justified as part of a Area-wide Traffic Control (ATC) scheme. New signal installations are also considered where the old or dilapidated signals need to be upgraded and replaced. Modern signal heads usually consist of LED (Light Emitting Diode) type aspects that provide clearer visibility to the road users. When such replacements are called for, it is usual to renew the poles and cabling at the same time.

In such cases, the controller may be replaced and then connected to a centralised computer. Road signs and marking must be upgraded in a consistent manner to that of the traffic signals layout and operation.

When designing traffic signals for installation at an existing intersection, it is important to note any shortcomings in the geometric layout and current signal operations.

2.2 Warrants for Traffic Signals

In general, design of signals first considers the movement of vehicles. Pedestrian and NMT movements are incorporated next, especially where such movement is high. Signals may be considered at an intersection if one of the following warrants is met.

2.2.1 Traffic Volume Criteria

Turning movement classified traffic volume counts are conducted over a 24-hour period on an average day. Counts are usually done on a Tuesday or Thursday, during school terms. The highest hourly traffic for the four directions are used.

A signal is justified when the following criteria are met:

Warrant 1 – minimum vehicle volumes. This is to be used when the main reason for providing signal is the volume of intersecting traffic. The warrant is satisfied when for any 8 hours on an average day, the traffic volume in passenger car units (pcu) exceeds the value given in [Table 2.1](#)

Table 2.1 Warrant 1 - Minimum Traffic Volume

Number of lanes in each approach		Volume (in pcu) per hour on the major road (Both approaches)	Volume (in pcu) per hour on higher volume minor road approach (one direction only)
Major road	Minor road		
1	1	500	150
2 or more	1	600	150
2 or more	2 or more	600	200
1	2 or more	500	200

The pcu values for various classes of vehicles are given in RDM 1.3 Part 3, Geometric Design Manual. The values are repeated in [Table 2.3](#) for easy reference.

Warrant 2 - Interruption of continuous traffic. This is applied when traffic volume on the major road is so heavy that the traffic on the minor road suffers excessive delay or hazard in trying to enter the junction. The warrant is satisfied when, for any 8 hour period on an average day, the traffic volume in pcu exceeds the values given in [Table 2.2](#)

Table 2.2 Warrant 2- Interruption of Continuous Traffic

Number of lanes in each approach		Volume (in pcu) per hour on the major road (Both approaches)	Volume (in pcu) per hour on higher volume minor road approach (one direction only)
Major road	Minor road		
1	1	750	75
2 or more	1	900	75
2 or more	2 or more	900	100
1	2 or more	750	100

Warrant 3 - pedestrian volume. This is used when traffic volume on the major road is so heavy that pedestrians experience excessive delay in crossing the road. This is applicable if, on the major road, 600 or more pcu per hour enter the intersection (both approaches put together). This is also applicable if there is a raised median of 1.5m or more width for pedestrians, and 1000 or more pcu per hour enter the intersection (both approaches put together). If 150 or more pedestrians cross the major road per hour, this warrant is applicable.

Note: When the 85th percentile speed on the major road is more than 60 kmph or when the intersection is in a built-up area of an isolated community of population less than 150,000, the values for pedestrian volume warrant can be 70% of requirement given. This is in consideration of the difference in the nature and operational characteristics of traffic in urban and rural environments.

2.2.2 Warrant 4 – Crash Experience

This warrant is intended for application when the severity and frequency of crashes are principal reasons for installing a traffic signal. Signal can be installed if alternatives were tried but were not effective in reducing the crash frequency. If a study shows that 5 or more crashes happened in the previous 12 month period, each with injury and property damage, which could have been prevented by installing a traffic signal, then installation is warranted.

2.2.3 Warrant 5 – Combination of Warrants

If no single factor provides the warrant for traffic signals, then two or more of the warrants may be considered by the designer, if they meet 80 per cent or more of the stated criteria. Adequate trial or other remedial measures which may cause less delay and inconvenience to traffic should be tried before installing a signal.

Table 2.3 Passenger Car Units

Vehicle class	Rural Standard	Urban Standard	Roundabout Design	Traffic Signal Design
Pedal Cycle (PC)	0.3	0.3	0.3	0.2
Animal/ hand cart	0.7	0.7	0.7	0.7
Motorcycle (MC)	0.5	0.5	0.5	0.33
Three wheelers and Tuk-tuk	0.7	0.7	0.7	0.5
Cars, Jeeps, SUV, Pick-up (C)	1.0	1.0	1.0	1.0
Microbus (MCB)	1.5	1.5	1.5	1.2
Minibus (MB)	2.0	2.0	2.0	2.0
Bus (B)	2.5	2.5	2.3	2.0
Omnibus (OB)	3.0	3.0	2.8	2.25
Light Goods Vehicle (LGV)	1.5	1.5	1.5	1.2
Medium Goods Vehicle (MGV)	2.5	2.5	2.8	1.75
Heavy Goods Vehicle (HGV)	3.0	3.0	2.8	2.25
Articulated Heavy Goods Vehicle (AHGV)	3.5	3.5	3.5	3.0

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Design Data For Signal Sites

3 Design Data For Traffic Signals

3.1 Introduction to Geometric Layouts

Design data is used in order to assess the required geometric layouts, the placement of signal equipment, the signal controller programmes and the time-settings. This data should be recent and accurate and are of five categories:

1. Physical layout – existing and planned.
2. Traffic counts – classified turning movement counts.
3. Pedestrian and other road-user counts.
4. Crash data.
5. Traffic impacts of future developments.

3.2 Physical Layout of Signal Sites

The following information should be shown on drawings prepared from topographic survey of the signal site:

1. Relevant property boundaries and building lines.
2. Type and location of kerbs and channels.
3. Layout and condition of pavements, medians and traffic islands.
4. Approach gradients.
5. Parking restrictions, car-bays, bus bays, taxi zones, fire stations, etc.
6. Location of all poles and aerial cables - (whether high voltage, low voltage or telecommunication lines).
7. Position, type and size of existing road signs, (directional, informatory, regulatory, etc).
8. Existing road-markings.
9. Existing advertising signs.
10. Location and nature of street lighting.
11. Position of any overhead obstructions (horizontal and vertical clearances, to verandas, signs, power lines, shop awnings).
12. Location of trees and any other foliar growth.
13. Type and location of surrounding development (residential, suburban shopping, industrial, central business district, railway stations, schools, etc).
14. Access to properties, gates, driveways, etc.
15. Location of drains, manholes, pits, ducting.
16. Location of telecommunication, sewer, gas, water and any other underground services.
17. Work proposed by other authorities and constraints on the locations of:
18.
 - a) Controller, power supply and communication links.
 - b) Signal Poles and Mast-Arms or Gantries.
 - c) Other signal sites and pedestrian crossings.
 - d) Cable ducts, junction pits (for ducting).
 - e) Any special traffic which uses the intersection, e.g. over- dimensional Vehicles, special Pedestrian needs, and strategic location within the road network.
 - f) Placement of detection devices.

3.3 Traffic Count Data

Traffic volumes (classified traffic count data) are essential for the design of traffic signal installations. The design process needs to consider serving the transport needs of people, goods as well as passenger vehicles.

Traffic counts for signal designs and traffic impact analysis are usually taken on average days (usually Tuesdays and Thursday at peak hours) when all known factors e.g. schools, shopping centres, stadiums, etc, are deemed to be operating.

Traffic signals should operate efficiently over a wide variety of traffic conditions. It is usual to base counts on peak traffic conditions for the purposes of setting controller parameters for phase movements and time-settings and for designing detector layouts. Efficient operation throughout the day can be managed by proper understanding of traffic engineering, local conditions, and traffic controller and detection capabilities.

3.3.1 Detailed Traffic Counting Guidelines

If recent traffic counts at the intersection are not available, a detailed count should be carried out to establish:

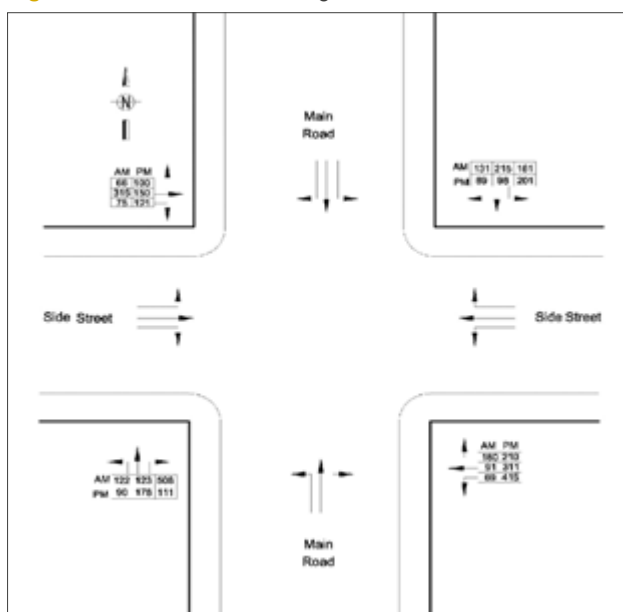
1. AM and PM peak vehicle volumes.
2. Turning movement vehicle volumes.
3. Volume and percentage of Heavy Goods Vehicles (HGVs).
4. Pedestrian / cyclists volume and directional flows.

Factors which affect the relevancy of the count are:

1. Traffic generators, e.g. existing or future shopping centres, industrial and commercial peaks, stadiums, local car-parks, etc.
2. Seasonal and/or recreational peaks.
3. Traffic management proposals.
4. Modal shifts – public transport options (e.g. New BRT or LRT).

Classified turning movement counts and pedestrian counts are carried out at the proposed signal site prior to installation. The count data is used to assist the traffic engineer in providing time for pedestrians and for various vehicular movements in the overall signal time.

Figure 3.1 Intersection Showing Classified Traffic Count Data



Refer to RDM Vol 1 Part 2 Traffic Surveys for guidance on conducting traffic surveys.

High right turn (RT) movement volumes may dictate that a controlled RT Green Arrow (GA) movement is required. Also, the volume data will assist in the setting of green times for each approach at various times throughout the day.

The count data may indicate that there is a capacity constraint at the intersection, that may not be solved by the signal installation alone. In that case, other measures may be required in the form of adding more lanes, islands or channelisation.

There are various software packages available to assist in the design of traffic signal.

Figure 3.1 shows an example of Classified Turning Movements for AM and PM peaks.

3.3.2 Analysis of Traffic Count Data

The traffic count data should be analysed in order to determine:

1. Optimum staging and phase sequences.
2. Optimum controller signal timing.
3. Lane configuration and associated signs and markings.
4. Storage requirements.
5. Pedestrian / cyclist needs.
6. Signal equipment requirements.
7. Inter linking of junctions.

A detailed capacity analysis in order to determine the above can be carried out by using a suitable software package. There are multiple software available in the market. The software should be properly evaluated in terms of its capability, need for data and suitability in the jurisdiction before they are used. For methodology to estimate capacity of an intersection, refer to RDM 1.3.

Software packages are used to determine the signal timing. The input for the packages are:

1. Classified turning movement volume counts.
2. Pedestrian data.
3. Non motorised traffic volume and composition.
4. Geometry data – number of lanes, exclude turning lanes, width of lanes, width of intersection, gradient, type of pedestrian crossing.
5. Available signal time data.
6. Crash data.
7. Threshold values for signal timing and delay for vehicles and pedestrians.
8. Type of signal operation preferred.
9. Any other special conditions.

The package calculates the time required to clear each movement, based on the preference of movements given and intersection geometry. It calculates the overall delay and provides the minimum time required for each movement.

The output of the packages are signal timing required for various movements including pedestrians.

Depending on the software programme settings, HGVs may need to be converted to PCUs or treated as a percentage of total counts. Some software packages are not good at providing time for all movements, or providing all pedestrian phases. Judgement and experience are required on the part of the Traffic Engineer to obtain sensible outputs from the programmes, and to interpret the results in a meaningful manner.

3.3.3 Motor Vehicle Counts

Traffic volumes for the relevant peak periods are expressed in vehicles per hour (vph). The volume data should also include information about traffic composition in each turning movement.

Usually 15-minute segments are used for the count data so that the peak flow rates can be identified. Traffic counts may vary by day of the week, or because of weather, school or public holidays or other abnormal conditions.

The designer should consider the circumstances of the count and adjust accordingly, before determining the volumes for the design. For considering seasonal variation of traffic counts refer to RDM 1.2 Traffic Surveys Manual.

3.3.4 Pedestrians and Cyclist Counts

Counts should identify the pedestrian volumes crossing each vehicle approach. The proportion of children, aged or other pedestrians with special needs (women with children, pregnant women, persons with disabilities, persons on wheel chairs) should be included in these volumes. These are required in order to enable specific features to be designed where necessary.

Cyclist volumes should also be counted both along and across each vehicle approach to identify the need for special features in the design. For details on pedestrian and cycle counts, refer to RDM 1.2 Traffic Surveys Manual.

3.3.5 Road Crash Data

Crash data for the location should be obtained for the last three years from the Kenya Traffic Police Department. The data should be used to identify hazards in the existing geometry, or any special problems. Crash data are best represented in form of collision diagrams.

For the various type of crash data, refer to 'Road Accidents and Safety Statistics' published by the Department for Transport, UK

3.3.6 Traffic Impact - Future Developments

During the design of traffic signals, future potential changes to traffic demand patterns should be researched. For example, the building of a large hotel or conference centre or new transport link may have impacts on the signal. In some cases, the actual provision of hotels, housing developments, shopping malls, etc, may give reasons for the provision of traffic signals.

Projections need to be made for the design life of the signals and associated site. The likely traffic at the intersection in the future years need to be estimated. This will determine the adequacy of the geometric layout and phasing. It will also highlight the future needs for additional road lanes and additional traffic signal equipment. In addition, the queue storage capacity of right turn (RT) and left turn (LT) lanes need to be considered from the projected turning movement counts.

Through lanes across an intersection should be aligned in order to achieve a clear vehicle path. When it is required to store vehicles in separate turning lanes adjacent to through traffic lanes, adequate storage length needs to be provided

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Design Data For Traffic Signals

4 Traffic Signal Control Equipment

4.1 Introduction to Signal Equipment

A traffic signal installation consists of an electronic unit (traffic controller) that is connected by cables to signal displays with lanterns on poles at various approaches to the site or junction. In addition, there may be detection devices connected to the traffic controller. In such cases, signal timings vary with the volume of detected vehicles and pedestrians.

In wide area traffic control schemes (ATC schemes), it is usual for the traffic controller to be connected to a central computer so that signal coordination may be achieved. If a central control system is not warranted or not available, modern local controllers can be coordinated with each other. This system is commonly referred to as 'Cable-Less Linking' (in U.K.) or 'Flexilink' (in Australia).

The signal lamps comprise of a three colour arrangement in a single housing known as a signal head or lantern. Each lamp or display (green, amber, or red) is commonly known as a signal aspect and each 'roundel' is usually 200 mm in diameter. Red is the top-most aspect (lamp). By the virtue of its position, there is no ambiguity to the road users with colour blindness regarding recognition of the red light.

There may be many lanterns and aspects at a junction depending on the complexity of the junction. Lanterns are mounted on metal poles or mast-arms, or gantries. In addition to vehicle lanterns, there may be lanterns for pedestrians, cyclists, emergency vehicles, buses, trams / LRT / BRT.

In addition to controllers, poles and lanterns, there may also be detection devices in order to record the presence of vehicles, pedestrians, cyclists, or trams /LRT/ BRT. Vehicles, cycles and trams are usually detected by sub-surface inductive loops. Pedestrians / cyclists may be detected by their own action of the pressing of a push-button usually situated on a unit attached to a pole (PBU – Push-Button Unit).

Mast-arms with signals may be provided where there is a visibility problem, on wide roads or hills, or where the average vehicular speeds are high. On mast arm, aspect roundels are usually 300 mm in diameter.

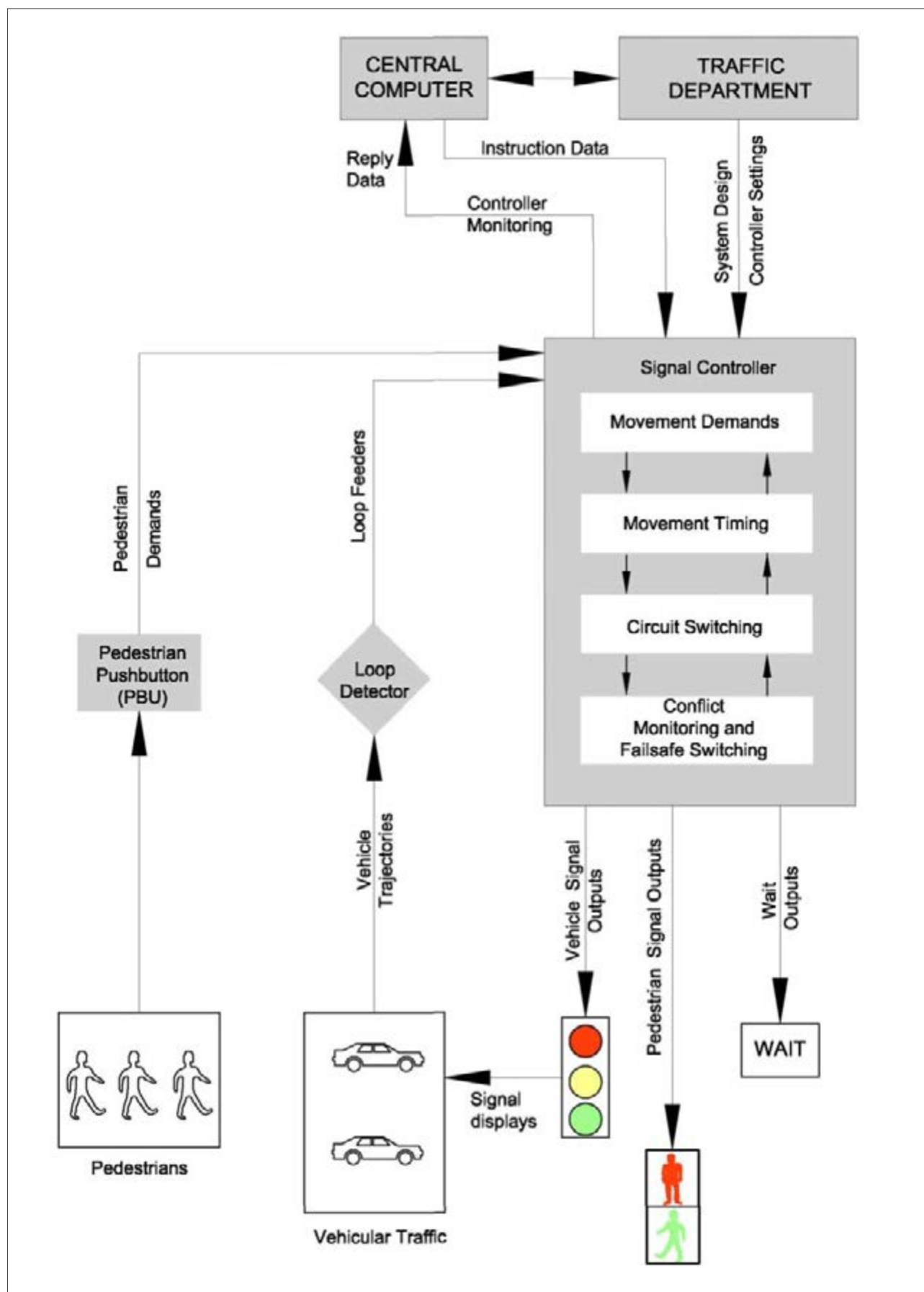
Mast-arms may also be employed throughout a wide-area traffic scheme in order to achieve consistency of design and uniformity of signal displays to all road-users.

Mast-arm poles are cantilevered and stretch out over the approach lanes of an intersection. If there are many lanes, then signal gantries may be employed in order to span all lanes.

4.2 Signal System and Detection

Demands from pedestrian and vehicles are relayed to the controller. The signal system transforms the demands into a sequence of signal displays. The controller sets limits for phase transitions, conflicting displays and safety timings that cannot be overridden by the master computer. The controller also reports activities at the junction to the central computer, such as failed lamps, failed detectors, and other events, such as interference by Police manual control or loss of power. A schematic diagram of traffic signal system is shown in [Figure 4.1](#).

Figure 4.1 The Traffic Signal System



Source: Guide to Traffic Management, Part 9, Austroads

4.3 Components

The components of the signal system are listed in Table 4.1 along with the purpose of each component.

Table 4.1 Traffic Signal System - Components and Purpose

Component	Purpose
Signal lanterns	Provide signal displays
Signal visors	Restrict visibility (one approach) and sun phantom
Signal louvres	Restrict visibility (near traffic) and sun phantom
Target boards	Provide good visual background for lanterns
Traffic signal lamps	Light source for the lanterns
Traffic signal poles	Support signal lanterns mounted beside the roadway
Lantern straps	Attach signal lantern to poles
Pole brackets	Attach signal lantern to poles
Signal mast-arms	Support signal lanterns mounted above the roadway
Mast arms and poles	Support lanterns and other utilities
Multi-core traffic signal cables	Carry power from controller to lanterns and demands from PBUs to the controller
Ducts, pits, conduits	Provide for underground installation of cable and access for maintenance
Terminal assemblies	Connect lanterns, PBUs, etc to the multi-core cables
Traffic signal controller	Determines display sequence and timings and provides switching of power to the signal lanterns, communicates with central computer
Loop Vehicle detectors	Detects Vehicles
Loop feeder cable	Completes detector sensor circuits enabling the controller to 'sense' Vehicles
Pedestrian push-button unit (PBU)	Allows Pedestrians to register their presence
Traffic signs	Regulate, inform, and guide traffic
Finial cap (on pole top)	Prevents inadvertent contact with live terminals and to protect the terminal assembly from the weather
Arrow masks	Permit the display of special Arrow symbols that control specific turn movements

4.4 Signal Pole and Lanterns - Equipment Layout

The most important traffic pole is situated on the left of each vehicle approach at the stop-line and is referred to as the Primary Pole (PP). This PP along with the Primary Lantern (PL) or signal head, is the basic element of any traffic signal installation or scheme. The PP should be placed 1 m ahead of the stop line and 1 m from any pedestrian crossing line.

Note: In the case of Pelican Crossings, the distance is increased to 2m.

Figure 4.2(a), (b) & (c) shows a simple junction layout with movements.

A Secondary Pole (SP) and signal aspect should be provided on the pole of the opposite approach.

If push button units (PBU) are provided for pedestrians, those should be placed such that the pedestrian faces the vehicle stream at right-angle, when approaching the crossing.

Poles carrying pedestrian aspects are usually placed in diagonally opposing positions and may also be used as the PP / SP. On one way streets, PP are placed on both sides of the approach. Secondary signals are also required.

In Figure 4 2(a) there are four approaches to the junction – from the north, south, east, and west. The signal aspects are labelled V1 – from the south, V2 from the north, V3 from the west and V4 from the east. Primary poles are shown as being 1m behind the respective vehicle stop-lines. Each pole has

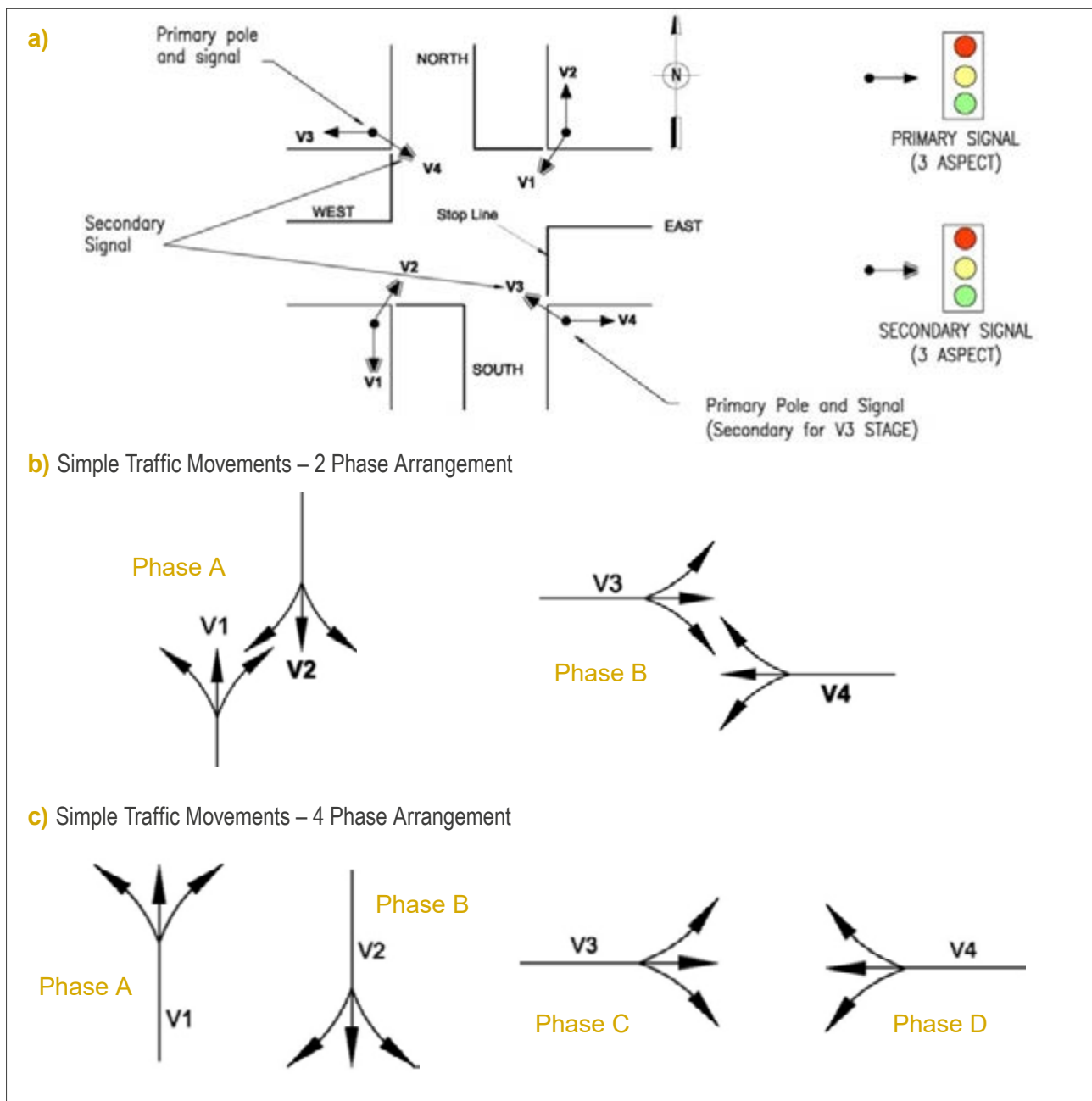
a dual function being a PP and a SP in the diagonally opposite side of the junction. For example, the PP of approach lantern V1 is the SP of approach lantern V2.

If mast-arms are provided then the associated signal aspects should be designed and placed from the perspective of the primary position. However, as mast-arms usually require robust footings and foundations, depending on their size, weight, and wind-loadings, it may not always be possible to place them in ideal positions. This could be because of the presence of trees, drains, overhead cables, etc.

Where traffic islands carry a single pole with 3-aspect signal head, the island should be minimum 1.2 m wide, preferably 1.4 m wide. Where the lantern carries a box sign or GA unit on the side, this minimum width required is 1.7m. Where the island forms part of a pedestrian crossing it should be as wide as possible in order to accommodate a pedestrian pushing a pram.

Where there is a 'staggered' pedestrian facility, the island should be a minimum of 3 m wide. In the case of a reverse staggered facility, where pedestrians walk with their backs to the traffic, the minimum width should be 4m in order to accommodate the signal poles within the pedestrian area.

Figure 4.2 Simple Junction Signal Layout



4.5 Signal Lantern Assembly

The signal lantern is an assembly of one or more signal aspects, and cable connection to the controller. In addition, the lantern has facilities for mounting the complete assembly to a pole. The signal aspect is a single optical system on a signal face capable of being illuminated at a given time.

Modern lanterns use LED (Light-Emitting Diode) type assemblies with low power consumption.

The lanterns consist of 200 mm diameter green, amber, red and white aspects. White aspect is for special vehicles and trams/LRT/BRT. 300 mm aspects are for overhead use, or when approaches are particularly wide. Each aspect may have Arrow Masks, or other qualifying symbols that may be rotated to any angle within the aspect, depending on traffic flow direction and site geometrics.

Pedestrian lanterns are provided for the control of pedestrians. Red 'Man' and Green 'Man' symbols are used in order to convey the meaning of 'WALK' or 'DON'T WALK'. The 'Man' symbols may be replaced with cycle symbols. Cycle lanterns have symbol of cycle on it.

4.6 Lantern Visibility Factors

The factors that affect visibility of lanterns are:

1. The size of the lantern (diameter in mm).
2. Approach speed on the road.
3. Width of the intersection.

For roads where the approach speed is 70 km/h or less, 200 mm lantern can be used. For approach speed more than 70 km/h, 300 mm lantern should be used. 300 mm lanterns should also be used on mast arms.

At intersections, the lantern should be visible at least 40 m ahead of the junction. For standard intersections of 2-lane road with 2-lane or 4-lane road, 200 mm lantern should be used. For intersections of more than 4-lane roads in each direction, 300 mm lantern should be used. The lanterns can be placed on 4 m poles. Lantern on mast arm is preferable for wider intersections for better visibility.

Where geometry and other special features allow it, 3 (but never less than 2) lanterns may be provided. The lantern should be aimed in the right direction. For this, visors and louvres may be used.

4.6.1 Lantern Visibility Distances

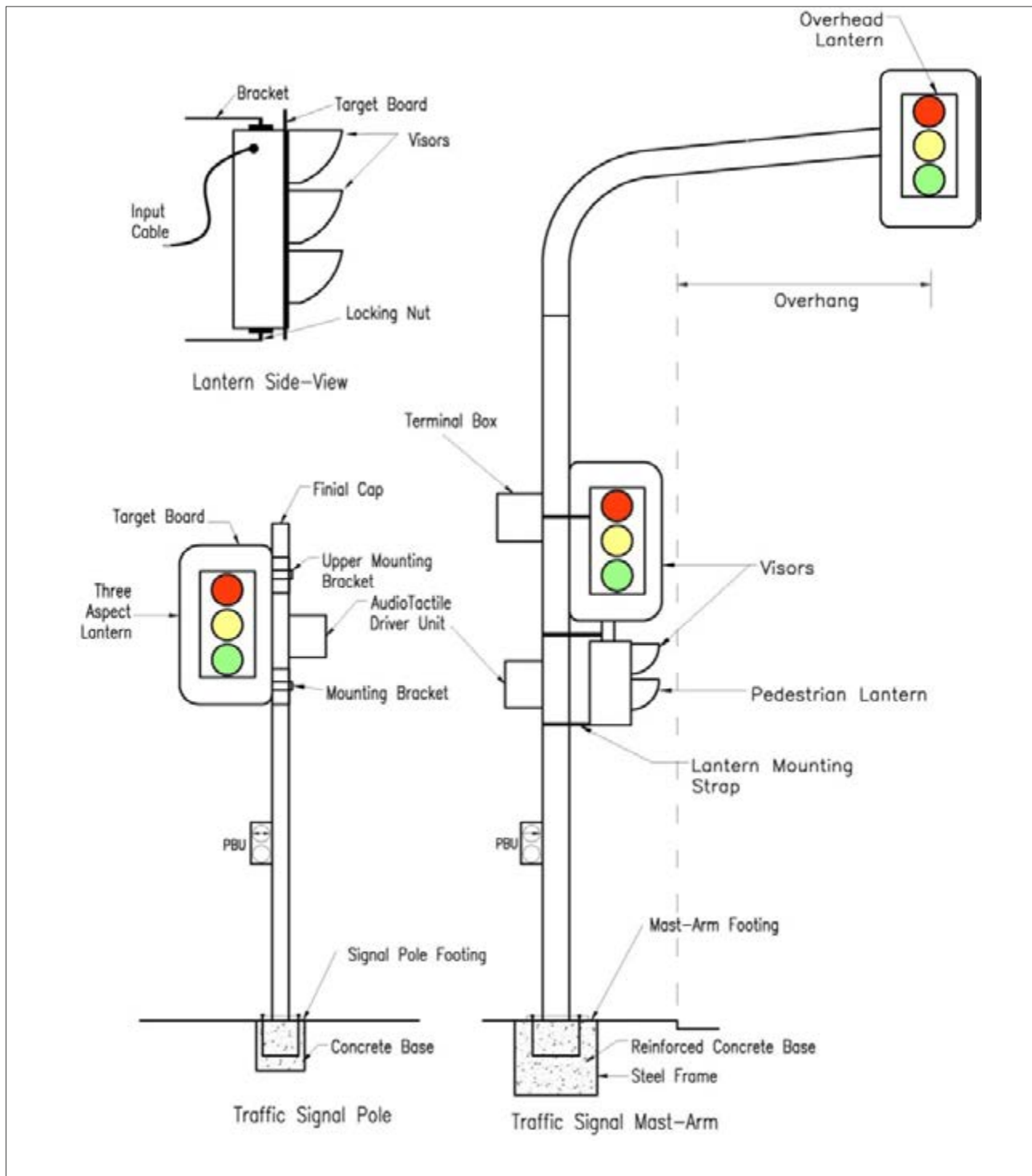
Recommended visibility distance to the junction with respect to speed is given in [Table 4.2](#).

Signal controlled junctions are not recommended where the 85th percentile approach speed exceeds 100 km/h.

Table 4.2 Visibility Distances

85 Percentile Approach Speed (km/h)	Visibility Distance (m)
32	22
40	31
48	40
56	51
64	80

Source: *Traffic Signs Manual, Chapter 6, Department for Transport, UK*

Figure 4.3 Traffic Signal Components Signal Poles and Lanterns

4.7 Poles and Mast-arms

Traffic signal poles are normally 4 m in height above the walkway level. Where used exclusively for pedestrian lanterns, the height is usually 3.2 m above walkway level. Special heights may be prescribed where necessary, to avoid awnings, tree branches, foliage, or where a centre island is already high above road level. A higher pole at 4.6 m may also be used where placement of a mast-arm is not feasible.

Special type poles may be used where directional signs and signal lanterns are placed on the same pole.

Short poles may be used for the mounting of PBUs, or

1. Where the pedestrian lanterns are mounted on other poles.
2. When the distance to walk to the pedestrian pole is inconvenient to the main stream of pedestrians.
3. Where too many persons crowd around one pole.
4. Where cyclists cannot conveniently access the PBU.

Mast-arms should be installed above road surface, provided there is sufficient clearance from overhead power lines. Mast arm are provided:

1. At approaches where stopping sight distance is not available with a 4 m pole.
2. At or near the stop-line of such approaches.
3. On the departure side of the intersection if it cannot be installed at the approach side.
4. On any three lane approach if no median pole can be installed.
5. On any approach with more than four lanes with a median pole.

Long outreach mast-arms should normally be used to satisfy the above criteria.

Short outreach mast-arms should only be used if the long outreach type is not warranted e.g. where restriction of sight by utility poles in a single or two-lane approach is the only reason for installing a mast-arm.

Where trees are present at an approach where mast-arm is required, the long outreach type should always be used.

4.8 Associated Lantern Equipment

4.8.1 Visors

Visors are used to:

1. Provide visibility of signals to certain traffic movements only, so that vehicle drivers on other approaches cannot see those signals.
2. Limit sun phantom effects (direct sunlight on a lantern face reduces its clarity).

4.8.2 Louvres

Louvres are used to:

1. Limit the possibility of a signal being seen by 'far back' traffic on the same approach to different stop-lines.
2. Limit sun phantom effects when oriented horizontally.

The use of louvres should be restricted to locations where traffic might otherwise be confused by distant signals. The louvre directs visibility of signal light towards the approaching traffic.

4.8.3 Target Boards

Target boards form a border to lantern assemblies so that the visibility of displays is enhanced, particularly when the sun shines from the same direction as the signal displays. Black target boards provide the best background.

Multi-column signal faces should be provided with a common target board. Target boards should not be fitted to pedestrian lanterns.

4.8.4 Symbolic Masks

A symbolic mask can be used in order to create an arrow symbol or letter that will control a particular type of movement or vehicle, e.g. Bus 'B', or Tram 'T'.

4.8.5 Lantern Supports

Poles, brackets and straps are used to attach signal lanterns to the required position, angle and height.

Lanterns may also be fixed to existing power-poles and street lighting columns in order to reduce pole clutter.

4.9 Power Distribution to Signals Displays

The supply of power to the traffic lanterns at the signal site is made by traffic signal cables, which are laid in underground ducts. The ducts are accessible by means of junction pits. Terminal assemblies are used to connect cables to the signal controller and lanterns.

The voltage level is usually 230 Volts at a frequency of 50 Hz. In recent times, with the advances in LED lantern technology, some lanterns can now be powered at only 42 V.

4.10 The Traffic Signal Controller

The traffic signal controller sets up the sequence and duration of signal displays and monitors their operation.

The main components of a traffic signal controller are:

1. Power switching for the signal lanterns.
2. Power switching for intersection signals.
3. Failsafe systems in order to prevent conflicting signal displays.
4. Signal dimming at night-time.
5. Vehicle detection system.
6. Provision for inputs from pedestrian PBUs or any other device.
7. A logic unit to control the sequence and timing of movements or phases, including minimum green time and safe clearance periods between conflicting movements (inter-green time).
8. Facilities for traffic personnel to monitor and alter signal operations.
9. Facilities to allow data communications between the controller and a master control computer.

4.11 Traffic Detection

The most common form of vehicle detection is by inductive loop detector. Inductive loop consists of several turns of insulated wire that is placed in a slot cut in the roadway. The cut is typically 6 mm deep. Once the wire is placed, the slot is sealed with a bituminous compound or epoxy resin.

The wire forms part of a tuned circuit and is connected by a feed cable to a detector sensor unit mounted in the controller cabinet or on a signal pole. Presence or movement of a vehicle over a loop reduces the loop inductance and causes a detector output or logical '1' to be registered in the controller memory as a vehicle demand.

Pedestrian demands are usually recorded by means of a PBU. The PBU is operated once by a pedestrian and the demand or logical '1' is stored in the controller logic until such time as the green 'Man' appears.

Other pedestrian sensors such as overhead infrared or microwave sensors, and pressure pads in the walkways, may also be used.

Inductive loops can be used for the detection of cycles and motorcycles. Loops should be design such these vehicles can be detected. Special detector systems may also be required for Emergency or Priority Vehicles.

Alternative forms of detection such as microwave, infrared or video may be used where appropriate.

4.12 Other Signal Control Equipment

4.12.1 Audio Tactile Facilities

Audible signals can be to indicate that the Green 'Man' period has commenced, for the benefit of the visually impaired. The Traffic Engineer should make sure that there is no danger of the unit being heard at adjacent signalled pedestrian crossings. Similarly, tactile (active vibration sensor during the Green 'Man') can be provided for deaf persons. Audio and vibration units are usually provided by signal suppliers as a single unit and are triggered by the Green 'Man' display.

Where it is not practical to install audible signals it may be desirable to provide tactile only signals (a knurled knob or pad mounted beneath the push button box which vibrates when the steady Green 'Man' is shown)

4.12.2 Disability Facilities - Ramps

Sloping ramps are essential where there is a need for wheelchair users to cross the carriageway. Tactile surfaces behind the kerb are of benefit to the visually impaired.

The needs of blind pedestrians and wheel-chair users sometimes conflict. The kerb, which is useful for blind persons, becomes a barrier for wheelchair users. Sloping ramps associated with a textured surface are useful for wheelchair users.

The immediate approaches to the pedestrian crossings should be indicated by contrasting coloured textured surfaces. The design, texture and recommendations for the production and use of textured pavement are detailed in most Disability Unit Publications.

4.12.3 Pedestrian Guardrails

Guard rails should be provided if it is required to restrict the crossing of pedestrians at certain locations. It is advisable to provide guard railing 50 m on either side of a crossing to encourage the use of the crossing facilities provided.

Care should be taken to ensure that guard rail does not impair visibility, particularly where children cross or wait in the pedestrian refuge island. Installation of guard rail should not reduce walkway widths and should not restrict pedestrian movement.

The effectiveness of guard rail is reduced if gaps have to be left for access of vehicles and for loading/unloading of goods.

4.12.4 Pedestrian Crossings

Pedestrian crossings are marked by pedestrian lines and alternate black and white stripes. Flashing beacons may also be provided.

These are not suitable for roads with more than 50 kmph speed limits or where there is more than one lane in each direction.

4.12.5 Street Lighting

Adequate street lighting should be provided so that the crossing can be seen against the background of other lights and signs.

Lighting at pedestrian crossing sites should be installed as per the requirements of BS 5489 and BS EN 13201.

4.12.6 Bus or BRT Priority at Traffic Signals

Selective Vehicle Detection (SVD) is a means of electronically recognising and giving priority to certain vehicles.

Bus/BRT priority is a technique whereby buses are detected on an approach to an intersection. The Green signal is either held, or recalled, in order to enable the bus to pass through. The location of the bus detector has to be at a position sufficiently upstream to allow time for the signals to change.

Buses in the system are fitted with transponders. When a demand is inserted for a Priority Phase, VA demands for other non-running Phases are allowed, enforced or excluded, to serve the Priority demand. The transponders are mounted underneath the buses and the bus-lane or shared lane is fitted with subsurface inductive loops. The transponders require a connection to the buses' electrical system to obtain power.

The bus priority system may employ Controller Dual Ring programming so that bus Phases may be serviced more than once during a complete cycle.

4.12.7 Dual Ring Operation

'Dual Ring' system utilises Parallel Phase streaming which allows various non-conflicting movements to occur in the junction at the same time. The controller is designed to give a more direct response to traffic demand and may operate in a non-cyclic fashion.

4.12.8 Microprocessor Optimised Vehicle Actuation (MOVA)

MOVA is isolated adaptive control system based on minimising stops and delays which maximises capacity at a single controlled junction. It is capable of rapidly responding to all conditions and to varying traffic patterns, particularly rapidly changing traffic flows. It is particularly well suited to sites with high traffic flow, particularly where these are seasonal or intermittent, such as diversion routes or holiday routes.

4.12.9 Remote Monitoring (Dial-Up)

Out-Station Monitoring Unit (OMU) or Dial-in Dial-out (DIDO), is used for the remote monitoring system and microprocessor controllers. It allows a variation of ATC using the Public Switched Telephone Network (PSTN) for groups of signals.

Remote monitoring of signals can check working signal lamps, controller timings, vehicle detection.

The components of Remote Monitoring System are:

1. Operator Terminal.
2. In station (computer).
3. Outstation Monitoring Unit (OMU).

The operator terminal and in station unit are installed in the Traffic Management Centre. The OMU's are installed at the controllers to be monitored. The communication is through dial-ups

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Geometric Design

5 Geometric Design

The geometric design requirements for setting up traffic signal are given in this chapter. For details of design of road junctions, refer to RDM 1.3.1 Geometric Design: Rural Roads and Highways and RDM 1.3 Part 3 Geometric Design: Urban Roads.

5.1 Lane Arrangements

The number and arrangement of lanes on each approach govern the capacity of the junction. To increase the capacity, lanes may need to be added.

The use of a shared straight-ahead (SA) and RT lane adjacent to an exclusive RT lane may bring advantages where RT traffic demand volumes are large. Where 2 lanes turn simultaneously, sufficient clearance distance between swept paths must be provided. Where opposing RT movements do not have sufficient physical clearance, the RT should be separated in the signal phasing.

For methodology to estimate capacity of an intersection, refer to RDM 1.3 Part 3, Geometric Design.

5.1.1 Right-Turn (RT) Lanes

Where straight-ahead (SA) and RT traffic share a lane, the following may occur:

1. A stationary RT vehicle may block straight-ahead traffic.
2. Blocking of SA lane may cause increased delays and queue lengths.

Therefore, exclusive RT bays should be provided (if there are more than 90 pcu per hour waiting to turning right) where practical, and of sufficient length (storage) to be managed by the signal cycle. If the RT Vehicle volume is low, but disruptive; then it may be desirable to ban the RT movement.

5.1.2 Left-Turn (LT) Lanes

Where LT traffic volumes are large, exclusive LT traffic lanes may be considered. In addition, slip lanes may be provided for LT movements.

LT traffic may be controlled by signals or by a 'GIVE WAY' sign. Where pedestrian cross in front of slip lanes, the design must cater to safety requirements of pedestrians.

If the slip lane has more than one-lane, a separate signal movement is required for the safety of both the pedestrians and vehicles.

5.2 Guidelines for Geometric Designs at Signal Sites

When designing a traffic signal site, the following geometric guidelines should be observed:

1. Signal equipment should be set back from the face of kerb by at least 0.5 m to avoid damage from passing vehicles.
2. For narrow junctions and opposing RT movements, wide medians may reduce capacity because of increased clearance times.
3. The tip of a median should be set back 0.6 m from the pedestrian crossing.
4. Corner kerb radii and traffic islands should be designed to facilitate movement of heavy vehicles.
5. The radius of a corner kerb should be kept to a minimum. The use of larger corner kerb radii may allow higher vehicle turning speeds which can compromise pedestrian safety.
6. Desire lines for pedestrian movements should be kept to a minimum and therefore smaller turning radii are desirable. This also allows better placement of poles and easier lantern aiming.

5.3 Design Considerations for Special Junction Characteristics

Each intersection has different characteristics that need to be considered. Fixed obstructions such as poles, trees, awnings, underground services, drains, illuminated signs imply that alternative positions and/or additional lanterns are required.

The guidelines lantern placement are:

1. If the warning or stopping function of primary lanterns is compromised by a sharp left-hand curve, an additional lantern should be provided on the right-hand side of the road.
2. If a lantern intended for one approach cause confusion to movements on any other approach, then it should be relocated or removed.
3. Where two intersections are close together, it is necessary to tilt the lantern to reduce its visibility.
4. Where a railway line is in close proximity to a signal installation, road lanterns should not be visible to the train driver. The screening and aiming arrangement must ensure that the signal does not cause confusion.
5. When a low bridge, crest of a hill, or sharp bend obstructs visibility of the lanterns, an advance warning signs should be used. The signs comprise of dual flashing amber lights.

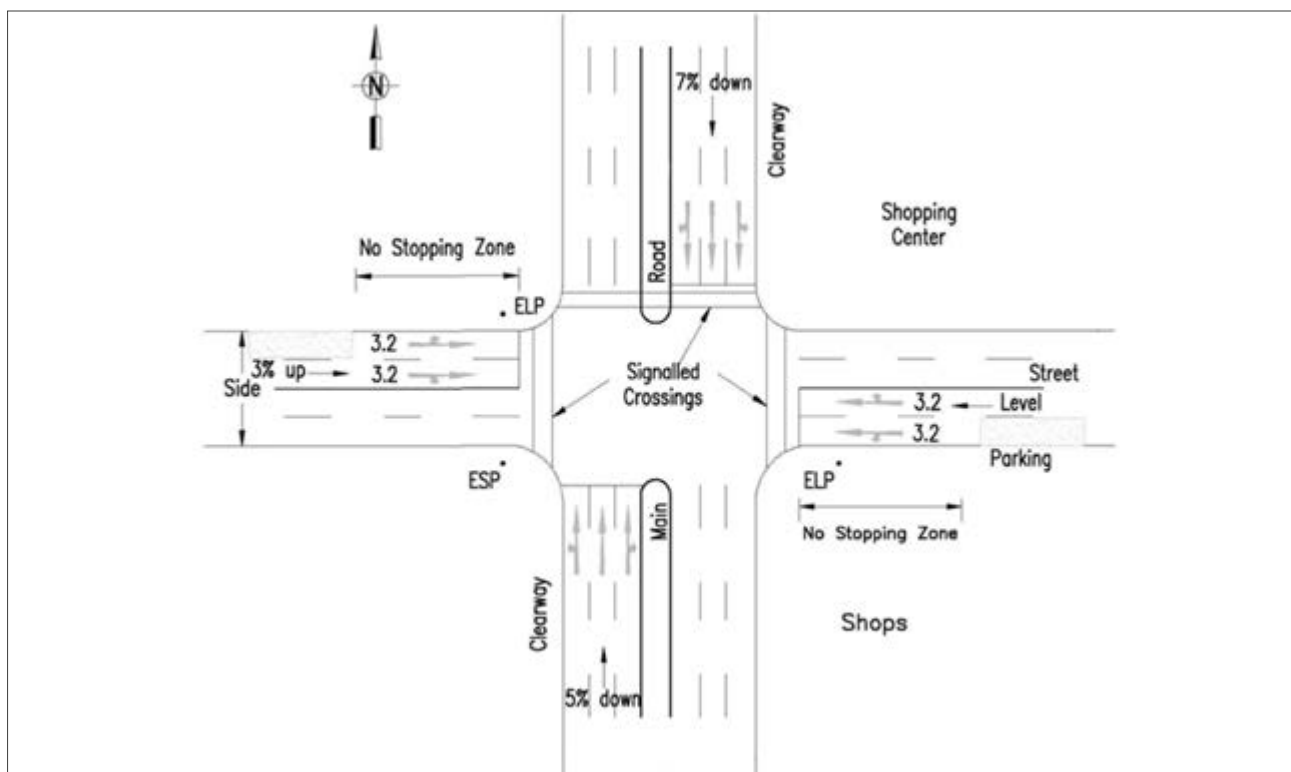
5.4 Traffic Islands and Corner Kerb Radius

Traffic Islands and corner kerb radii should be designed to accommodate the movement of HGVs. The turning paths of HGVs and other vehicles can be checked by various software packages. Vehicles of 7.5 m or longer usually require up to two marked lanes to safely make LT or RT.

Corner kerb radii should be kept to a minimum because:

1. Large radii can result in greater vehicle speeds compromising safety of pedestrians
2. Large radii mean that stop lines may have to be placed further back into the junction and this inevitable increases inter-green times

Figure 5.1 Intersection Geometric Layout



5.5 Median Islands and Median Strips

5.5.1 Divided Roads

The points to consider for design of median islands at signalled intersections:

1. The minimum distance from lantern target board to the kerb line should be 0.5 m, in order to avoid damage to the signal equipment.
2. The width of a median to accommodate pedestrians is 1.5m - 2m.
3. If an exclusive RT bay is provided, the residual width of the median should satisfy the above mentioned points.
4. Medians that are 'over-wide' should be avoided. They tend to reduce capacity where increased inter-green times are required.
5. The tips of median islands should be set back 0.6 m behind the adjacent road alignment unless a pedestrian signal crossing is provided.
6. When a pedestrian signal crossing is provided, the median should be terminated at the crossing unless the crossing is intended to run unhindered through the median. In that case there should be a minimum medium tip distance of 2 m beyond the crossing.

5.5.2 Undivided Roads with Small Median 'Strips' at the Junction

The median strips should be at least 10 m long to ensure adequate visibility. If a signal pole is provided on the median strip, the strip should be at least 1.2 m wide for single column lanterns and wider for two-column lanterns. It should be noted that this practice leaves the signal poles and lanterns particularly vulnerable to damage.

5.5.3 Painted Medians and Islands

If there is insufficient carriageway width in order to construct a kerbed median or island, or where it is desirable to permit vehicles to cross the median unhindered, then those devices may need to be painted.

Traffic signal poles should not be placed on the painted areas. Traffic signal poles should be placed on the raised walkways or medians at the recommended clearances.

5.6 Parking

The successful operation of a signal site depends on the appropriate management of parking. If parking is not properly managed, there will be:

1. A reduction in the number of effective approach and departure lanes.
2. Ineffective detector operations.
3. Obstruction of signal displays and road markings.
4. Reduced visibility of vehicles and pedestrians.
5. Decreased capacity and increased delays and queue lengths.

Parking prohibitions and restrictions may be necessary in order to ensure that the signal site functions properly. Computer programs are available for the analysis of parking prohibitions with regard to signal site performance.

5.7 Bus Zones

The effect of bus bays or bus zones placed in close proximity to signal sites may have a detrimental effect in a similar manner to that of short-term parking. Bus stops on the approach to intersections may cause detection difficulties. These factors need to be considered when designing signal sites in conjunction with bus priority schemes.

Bus stops on the departure side (or far side) of the intersection may cause vehicles to queue into the intersection when the bus is stationary. Therefore, a bus bay, or additional lane capacity or dedicated bus lane, may be required.

5.8 Signalised Roundabout

When operating within their capacity, roundabouts typically operate with lower vehicle delays than at signalised intersections. However, signalisation of roundabout may be required when flows reach saturation level.

By introducing signals at roundabouts, along with geometric improvements and road marking, it is possible to increase the capacity and improve safety. Under normal priority control at each entry, roundabouts operate in a self-regulating manner. However, signalised roundabouts have signal control on one or more of the external approaches (entry arms) and may also have signal control on one or more internal approaches (circulatory carriageway). Signalised roundabouts may also operate in a self-regulating manner to a varying degree, depending upon a combination of the extent, method and duration of the signal control conditions and the size of the roundabout. Where roundabouts require signals on most or all entries, computer programs are usually employed for modelling purposes.

The provision of traffic signals on entries to the roundabout also assists in separating the conflict between vehicular and pedestrian movements. When signal-controlled pedestrian crossing facilities are incorporated on the exit arms, designers should ensure that adequate storage length is provided to avoid traffic queuing back onto the circulatory carriageway.

For design of roundabout, both for priority control and signal control, refer to RDM 1.3. Geometric Design Manual.

5.8.1 Types of Signal Control

Fully controlled is the condition where signal control is provided on all internal and external approaches (i.e. the direct method of signal control).

Partial-control is the condition where one or more of the approaches remain under priority control. Partial-control can be used where traffic congestion does not occur on all external approaches.

5.8.2 Method of Signal Control

Direct signal control is the condition where the signals are situated on one or more of the external approaches and the internal approach immediately to the right of each signal-controlled entry. [Figure 5.2](#) shows an example of a large signalised roundabout with partial, direct signal control, where traffic flows on the minor roads (Arms B and C) are light and continue to operate in a self-regulating manner under normal priority control.

Indirect signal control is the condition where the signals are situated at such a distance away from the roundabout entry, that the entry continues to operate in a self-regulating manner under normal priority control. [Figure 5.3](#) shows an example of a signalised roundabout with indirect signal control on each entry arm.

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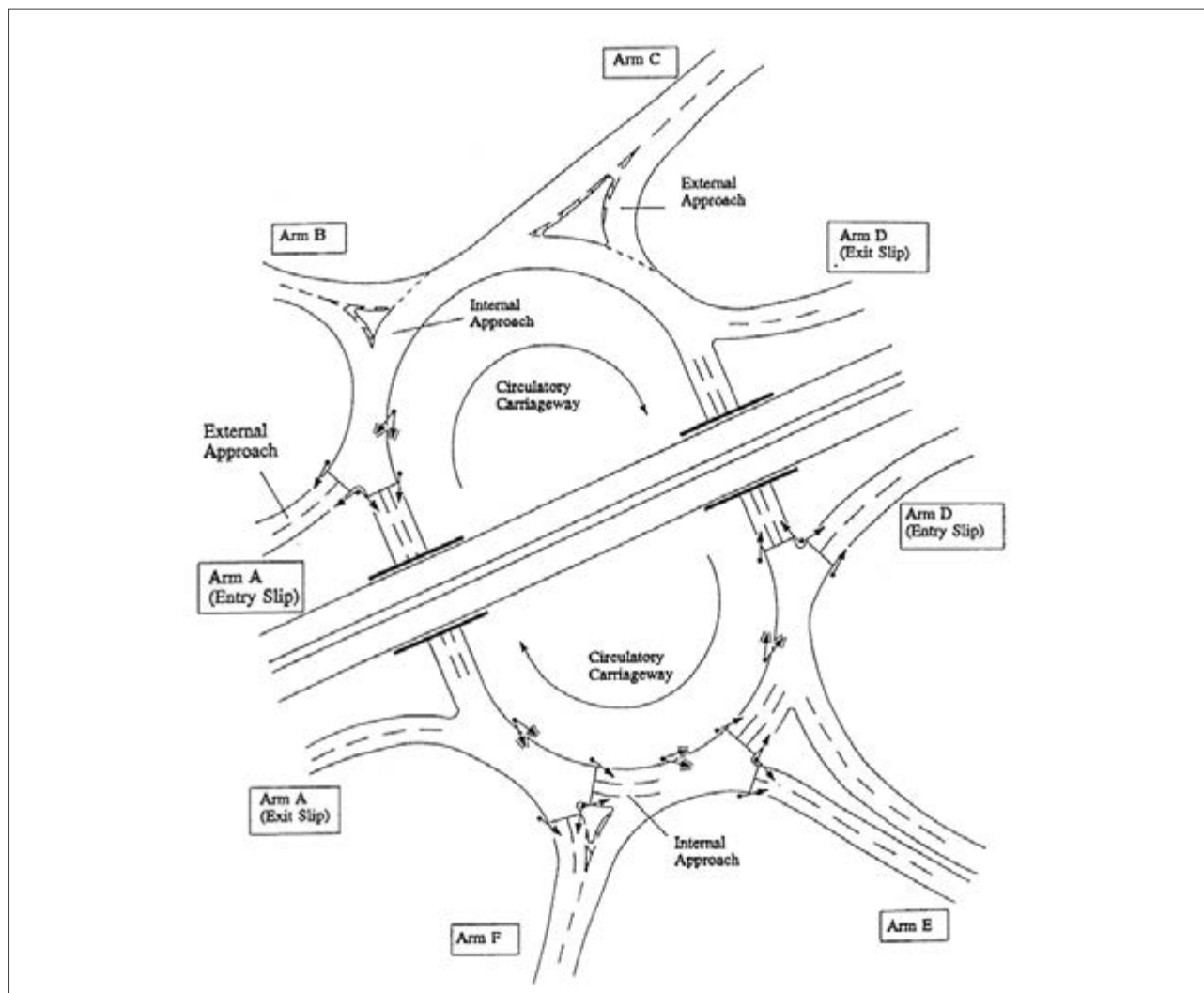
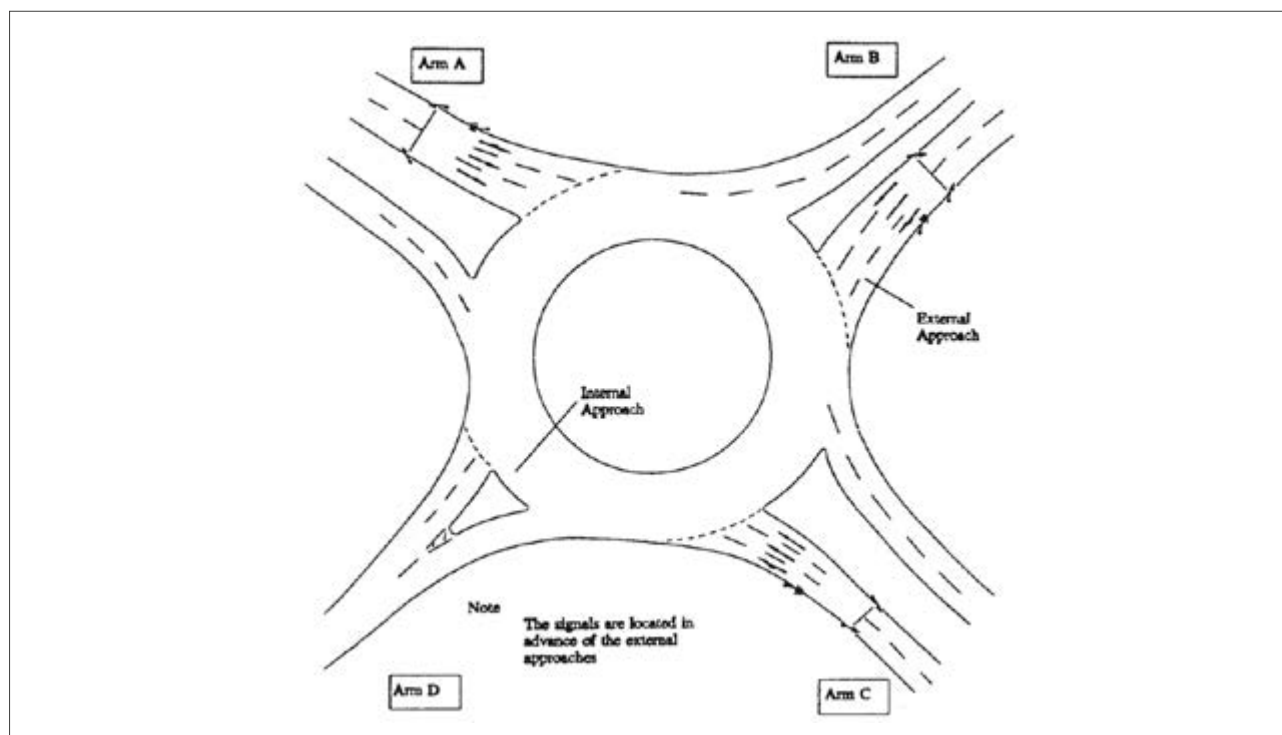
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Geometric Design

Figure 5.2 Signalised Roundabout with Partial, Direct Signal Control**Figure 5.3** Signalised Roundabout with Indirect Signal Control

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Geometric Design

Indirect signal control may be used where an entry arm with a very heavy traffic flow is preceded by (i.e. is to the left of) an entry arm with a very light traffic flow. The heavy flow may often proceed virtually uninterrupted, possibly with high circulatory speeds. This causes an unbalanced traffic flow which adversely affects the capacity of the roundabout and can lead to excessive queue lengths (and substantial delays) to other external approaches which in turn may result in congestion at preceding junctions or on slip roads.

The installation of indirect signal control can be introduced to control entry flows on one or more approaches and can provide a gap in the circulatory traffic, to favour those entry arms which were previously subjected to excessive delays and queues. In this way the capacity of the entry arms can be balanced, however increases in vehicle gap distances may be detrimental to cyclists and pedestrians crossing the arms.

5.8.3 Duration of Signal Control

Full-time control is the condition where signals are permanently operating. Under full-time signal control the give way road marking across any signalised entries to the roundabout should not be provided.

Part-time control is the condition where signals are switched on at set times (generally peak periods) or under certain traffic conditions by queue detectors. When traffic flows are light the roundabout operates in a self-regulating manner under normal priority control.

The examples illustrated in [Figure 5.2](#) and [Figure 5.3](#) could both operate under full-time or part-time signal control.

5.8.4 Geometric Modifications

Geometric modifications may be required to roundabouts in order to:

1. Increase capacity on external approaches.
2. Increase capacity on the circulatory carriageway.
3. Accommodate queues at internal stop lines on the circulatory carriageway.
4. Improve forward visibility and intervisibility in the junction intervisibility zone.
5. Provide alignment improvements and lane control measures.
6. Provide specific measures for pedestrians, cyclists and buses.

These modifications may take the form of:

1. Additional nearside or offside approach lanes.
2. Additional internal approach lanes within the circulatory carriageway.
3. Increasing the size of splitter islands in order to achieve longer internal approaches.
4. Segregated facilities or signalled crossing facilities.

6 Signal Face Layouts And Display Sequences

6.1 Introduction to Signal Displays

Traffic signals control vehicle movement by displaying green, amber or red signals. There are some variations, e.g. flashing amber arrows, or flashing pelican full amber signals. Pedestrian signals are provided to control pedestrians crossing by displaying Red 'Man' or Green 'Man' signals. Traffic signals are also used to control special vehicles such as fire tenders, ambulances, trams, buses, and trains - at rail level crossings.

Design of traffic signals should follow a consistent standard, so that road-users are not confused.

The basic design criteria are:

1. Give a logical and clear presentation.
2. Ensure uniformity of design.
3. Promote displays that are simple and easy to understand.
4. Prevent unsafe or conflicting displays.
5. Prevent unsafe sequence of displays.

The display terms used are:

1. Signal Aspect is a single optical system, usually round on a signal face, and capable of being illuminated green, amber, red, or white.
2. Signal Display consists of the aspects that are illuminated.
3. Signal Face refers to a set of signal aspects in a common lantern assembly, generally in one or two columns, together with a target board in order to improve signal visibility.
4. Signal Lantern refers to an assembly of one or more aspects, together with the means of mounting them and connecting them to signal cables.

6.2 Signal Aspects

Signal aspects are generally arranged in one or two lantern columns, making up the Signal Faces. Vehicle aspects are Green, Amber or Red, but may be White for Public Transport Vehicles.

Pedestrian aspects are Green or Red, but may also have an Amber 'Man' aspect.

Cycle aspects are usually two-aspect lanterns of Green or Red, and for three- aspect lanterns, are usually Green, Amber or Red.

Special Vehicle aspects for trams, buses or emergency Vehicles are usually White, Amber or Red.

6.3 Size of Aspects

The following are guidelines for aspect size:

1. 200 mm diameter general-purpose aspects are suitable for most urban applications except on very wide streets.
2. All aspects in the one signal face should ideally be of the same size.
3. For pedestrian aspects, 200 mm nominal diameter aspects are usually used; however larger may be required for very wide crossings.
4. 300 mm diameter aspects should be limited to Mast-arms or gantries.

Mast arms or gantries are needed when:

- a. The circular displays must provide a greater advance warning than the normal 150m from the signal stop line, or where the 85th percentile approach speed exceeds 70 km/h.
- b. An Arrow signal display must provide greater advance warning than the normal 80m from the signal stop line, or where the 85th percentile speed of turning traffic exceeds 40 km/h.

6.4 Types of Aspects

Signal Aspects are shown in Figure 6.1.

6.4.1 Full Roundel (Circular) Aspects

Full Roundel or circular aspects are the easiest to comprehend and have the greatest visual range. The signal face may be fitted with a circular Green Arrow display that permits LT and RT traffic to filter through a pedestrian movement or through oncoming traffic, unless prohibited by other controls.

6.4.2 Arrow Aspects

Arrow aspects are used in order to control particular movements at traffic signal sites including U-turn movements as seen in Figure 6.1.

Arrow aspect orientations to be used are shown in Figure 6.2.

Figure 6.1 Recommended Signal Aspects

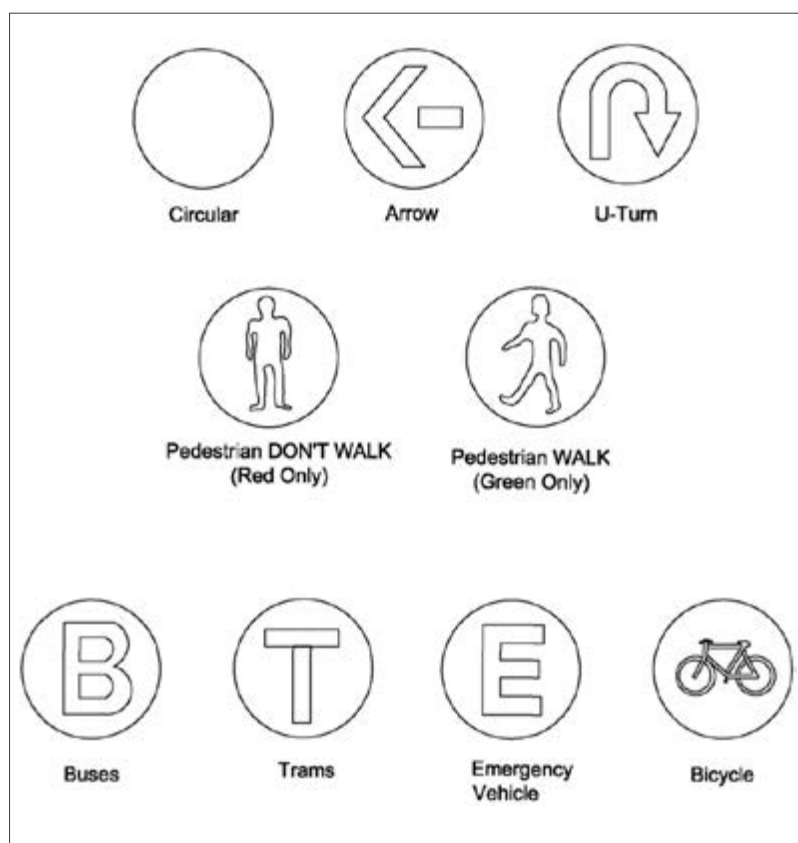
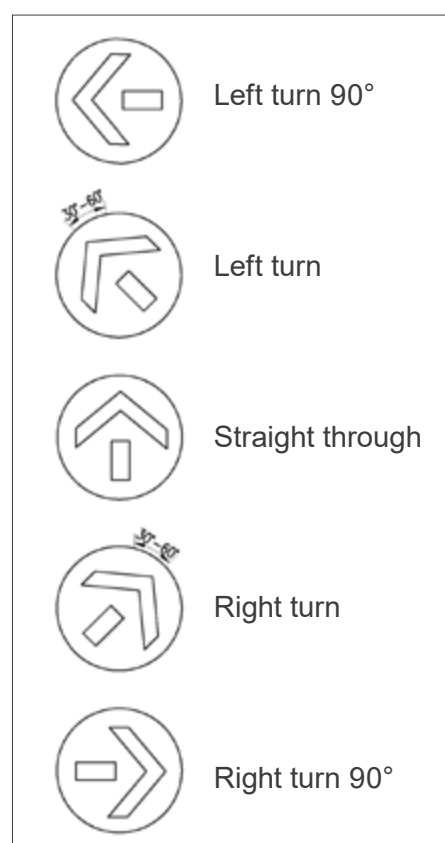


Figure 6.2 Signal Aspects Orientation



6.4.3 Pedestrian Aspects

The Don't Walk aspect is a Red 'Man' standing human figure, and the Walk aspect is a Green 'Man' walking human figure as seen in Figure 6.1.

6.4.4 Cycle Aspects

Cycle aspects are used in a similar manner to that of pedestrian aspects. The intent is to control cyclists crossing the road, or to control on-road cyclists at an intersection. The symbol for a cycle aspect is shown in Figure 6.1

Two aspects, a Red cycle and Green cycle are used for road crossings. Three aspects, Green, Amber and Red, may be used at signal sites with exclusive cycle lane approaches.

6.4.5 Special Vehicle Aspects

Special vehicle aspects are shown in Figure 6.1. Special Vehicle aspects are used to allow priority to buses, trams and emergency vehicles at traffic signals. The white T or B, and in some jurisdictions, E aspect is used to indicate that trams, buses or emergency vehicles may proceed. White arrows are also used to indicate that drivers of special vehicles may proceed in the direction of the Arrow.

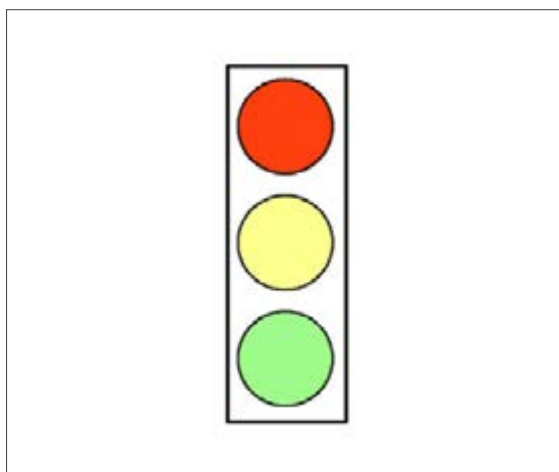
6.4.6 Combination of Aspects (Full Roundel and Arrow)

Pedestrian aspects must not be mounted or combined with vehicle aspects in the same signal face. Circular aspects may be combined. Full Roundel or Arrow aspects may also be combined with special Vehicle aspects.

6.5 Basic Signal Face Layouts

Signal Aspects are arranged in columns with the Red aspect upper-most, the Amber aspect in the centre and the Green aspect at the bottom as shown in Figure 6.3. This arrangement is universally accepted and provides unambiguous interpretation to persons with colour-blindness. In special traffic situations, two-aspect signal faces may be used. Single-aspect signals are not permissible except for overhead lane control signals.

Figure 6.1 Basic Three-Aspect Signal Face Layout



Directional 'Arrow' displays (e.g. Green 'Arrows') reinforce the provision of right of way for certain vehicle movements or reinforce the prohibition of certain movements.

Downward pointing arrow aspects should not be used on conventional signal poles or mast-arms for signal control. Their use should be confined to overhead gantries related to lane discipline.

6.5.1 Multi-Column Signal Face Layouts

Multi-column signal faces may contain four, five, or six aspects. It is usual to have an Arrow aspect column (3 Arrows) adjacent to the basic three-aspect column. A six-aspect face layout is shown in [Figure 6.4](#). This arrangement is used in order to control left-turn or right-turn movements independent of the main through movement.

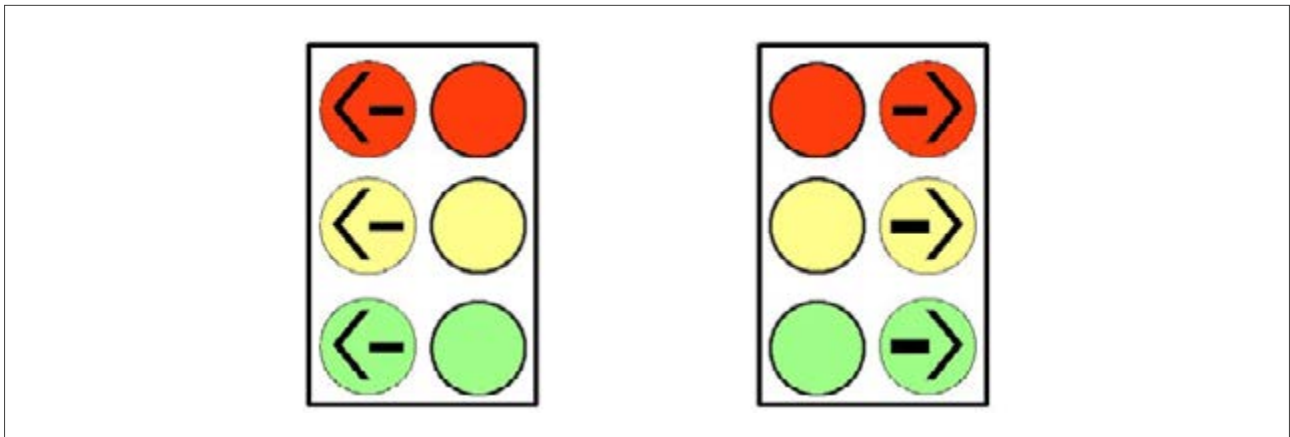
Multi-column vehicle signal faces should be arranged as follows:

1. A column shall contain only aspects of the same shape and orientation.
2. LT Arrow aspects are located to the left of the circular aspects, and RT Arrow aspects are located to the right.
3. Columns containing only an Amber aspect are not permitted.
4. Columns containing only Red and Green aspects are not permitted.
5. Aspects of the same colour must be located on the same horizontal level.
6. No column shall contain more than three aspects, except in the case of an approved 4-aspect column.
7. Each column shall have only one aspect of each colour, except for the approved 4-aspect column.

The use of three columns in a display is not allowed as this causes confusion to drivers.

The recommended arrangement is no more than 2 lantern columns, facing in any one direction, mounted on any pole or mounted at two distinct levels on a mast arm.

Figure 6.4 Six-Aspect Multi-Column Signal Face Layouts

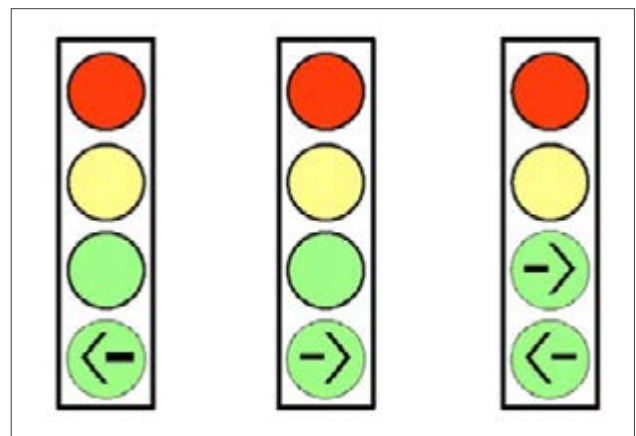


6.5.2 Four-Aspect Single-Column Signal Face Layouts

A single column of four aspects may be used consisting of Red and Amber full roundel aspects and two Green aspects (full roundel and Arrow, or with two arrows).

See [Figure 6.5](#). This is not allowed for overhead mounted signals on either Mast- arms or gantries. Four-aspect lantern columns should not be used in multicolumn displays and should be avoided altogether if possible.

Figure 6.5 Four-Aspect Single-Column Signal Face Layouts



6.6 Signal Face Layouts with RT Arrow Aspects

6.6.1 Six-Aspect Signal Face Layouts with RT Arrow Aspects

A six-aspect signal face layout with Green, Amber and Red Arrow aspects can be used in order to independently control RT movements. See Figure 6.4

A Green RT Arrow allows exclusive right of way. It should only be displayed when no conflicting traffic movements are permitted.

The Amber RT Arrow always follows the Green Arrow display in a 3 aspect Arrow arrangement. A single RT Green Arrow may be followed by a full roundel Green.

The Red RT Arrow should be displayed following the Amber Arrow display when the RT movement or a conflicting movement must be protected.

6.6.2 Four-Aspect Signal Face Layouts with LT & RT Arrow Aspects

A 3-aspect lantern column with a single RT Green Arrow aspect may be used when the RT Green Arrow display is terminated simultaneously with the Green Full Roundel display. See Figure 6.6

A 3-aspect lantern column with a single LT Green Arrow aspect may be used when the LT Green Arrow display is terminated simultaneously with the Green Full Roundel display. See Figure 6.7.

6.7 Face Layouts with Left-Turn Arrow Aspects

6.7.1 Six-Aspect Signal Face Layouts with LT Arrow Aspects

A six-aspect signal face layout with Green, Amber and Red Arrow aspects may be used in order to independently control left-turn movements. See Figure 6.4.

The Green LT Arrow should only be displayed when there are no conflicting traffic movements.

The Red LT Arrow should be displayed following the Amber Arrow display when a conflicting movement must be protected. When the Green Full Roundel is illuminated, the column of LT aspects is usually 'OFF'.

Figure 6.6 Four-Aspect Signal Face Layouts Single RT Green Arrow Aspect

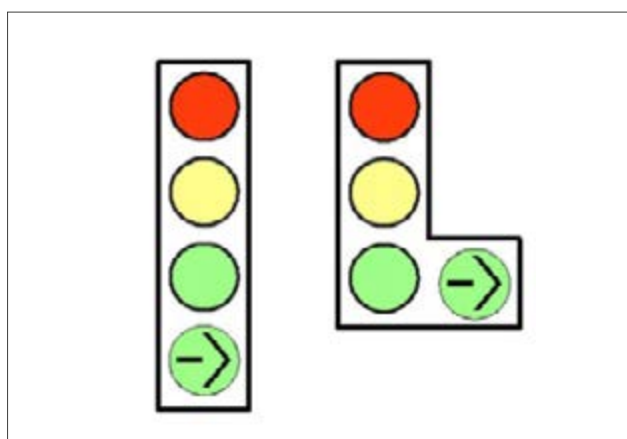
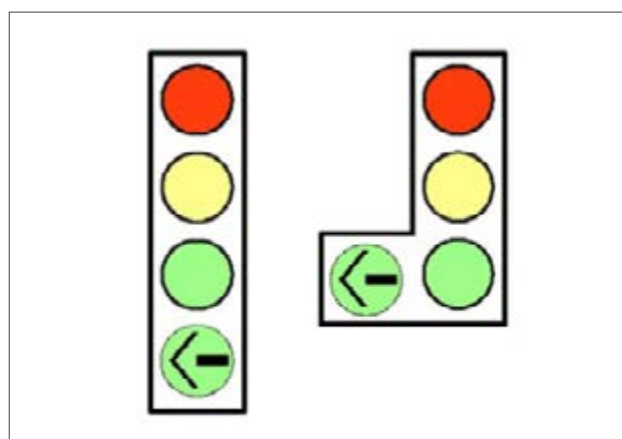


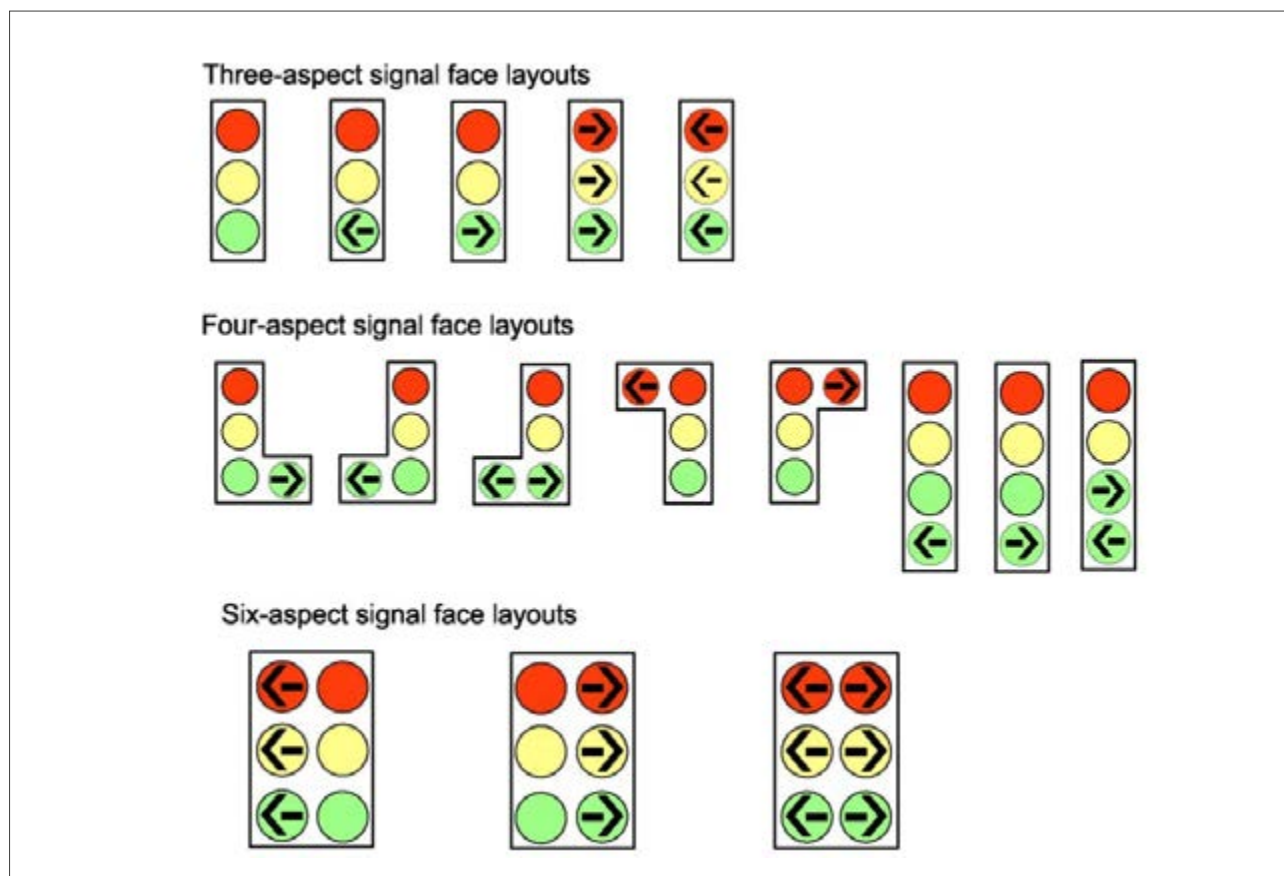
Figure 6.7 Four-Aspect Signal Face Layouts Single LT Green Arrow Aspect



6.8 Allowable Lantern Aspect Face Combinations

For '3 to 6 aspect signal face layouts, refer to [Figure 6.8](#).

Figure 6.8 Allowable Signal Face Layouts

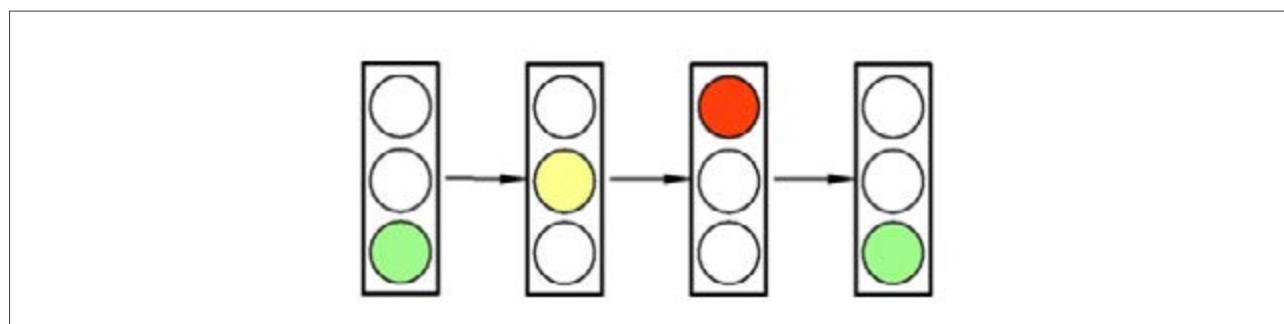


6.9 Sequence of Vehicle Signal Displays

6.9.1 Basic Sequence

The normal sequence of vehicle displays in a vehicle lantern is **Green** to **Amber** to **Red** and back to Green again. See [Figure 6.9](#).

Figure 6.9 Basic Display Sequence Three Aspect Signals



6.9.2 Sequences with Arrow Aspects

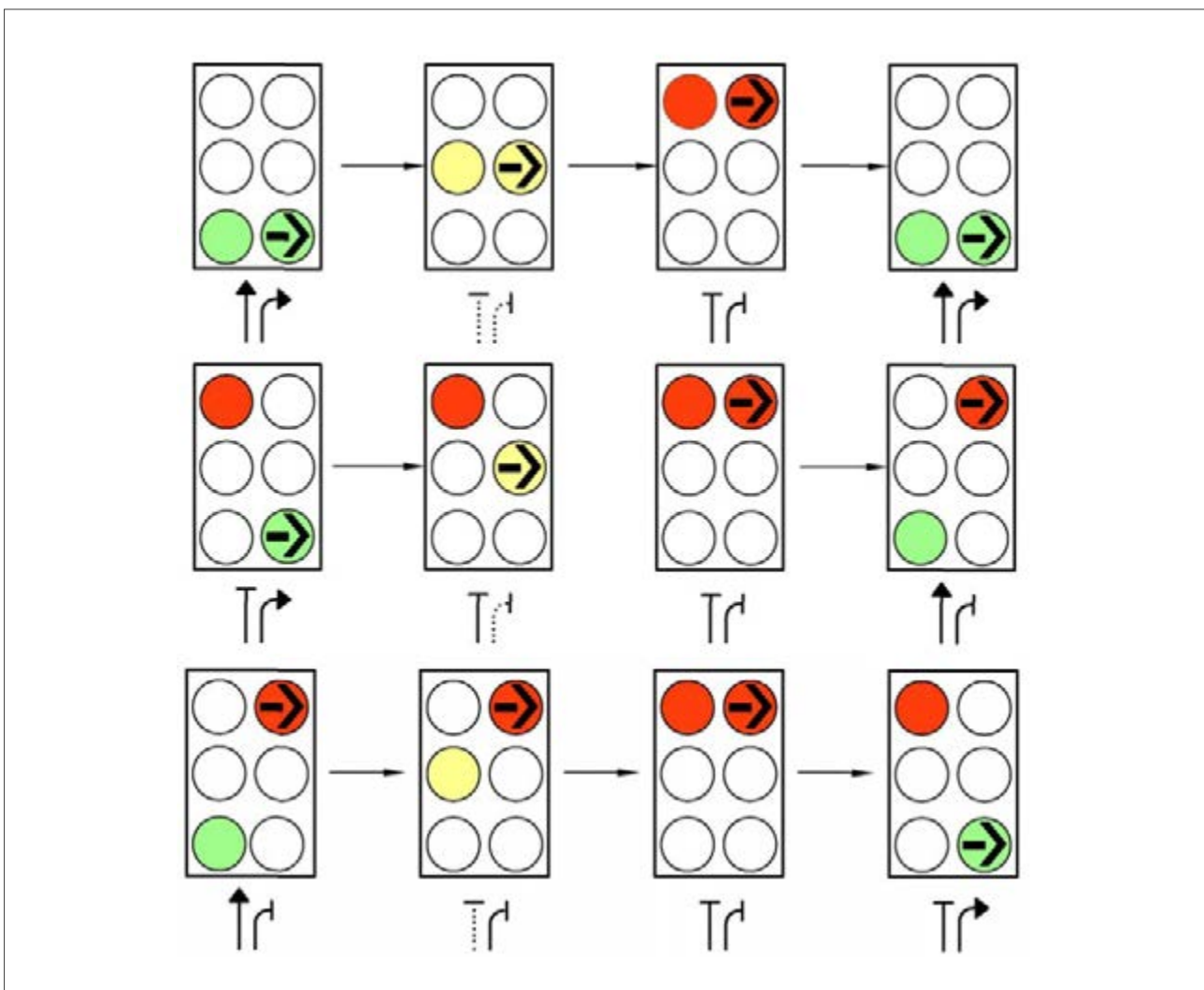
In sequences with Arrow aspects, e.g. dual column lantern displays, the requirements of the basic sequence of displays are:

1. If filter movements are allowed, then a column of Arrow aspects is **Off**.
2. Other permitted sequences for arrows on six-aspect signal faces are:
 - a. Green to Amber to Red to Off to Red to Green.
 - b. Green to Amber to Red to Flashing Amber to Amber to Red to Green.
 - c. Green to Amber to Red to Flashing Amber to Off to Red to Green.
 - d. Green to Amber to Red to Off to Green.
3. For Green displays, either Full Roundel or Arrow shall always be followed by an Amber display. LT Green Arrow may be cancelled on commencement of a Green Full Roundel.
4. A Red display must be preceded by an Amber display.

Fully-controlled RT display sequences for six-aspect arrangements are shown in Figure 6.10.

On dual column vehicle display as shown in Figure 6.9, it is not desirable for a Red or Green display on one column to change during an Amber display.

Figure 6.10 Fully-Controlled RT Display Sequences Six-Aspect Arrangements



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6.9.3 RT Sequences

Commonly used RT display sequences in order to **initiate** a RT for a six-aspect arrangement is shown in **Figure 6.11**.

Commonly used RT display sequences in order to **terminate** a RT for a six- aspect arrangement is shown in **Figure 6.12**.

Figure 6.11 Display Sequence to Initiate an Arrow - Controlled RT Movement

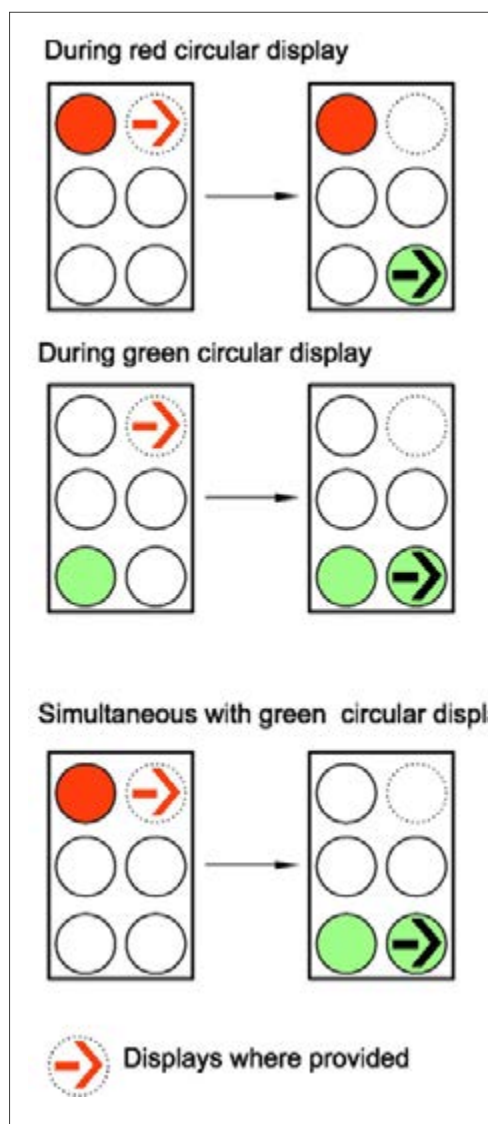
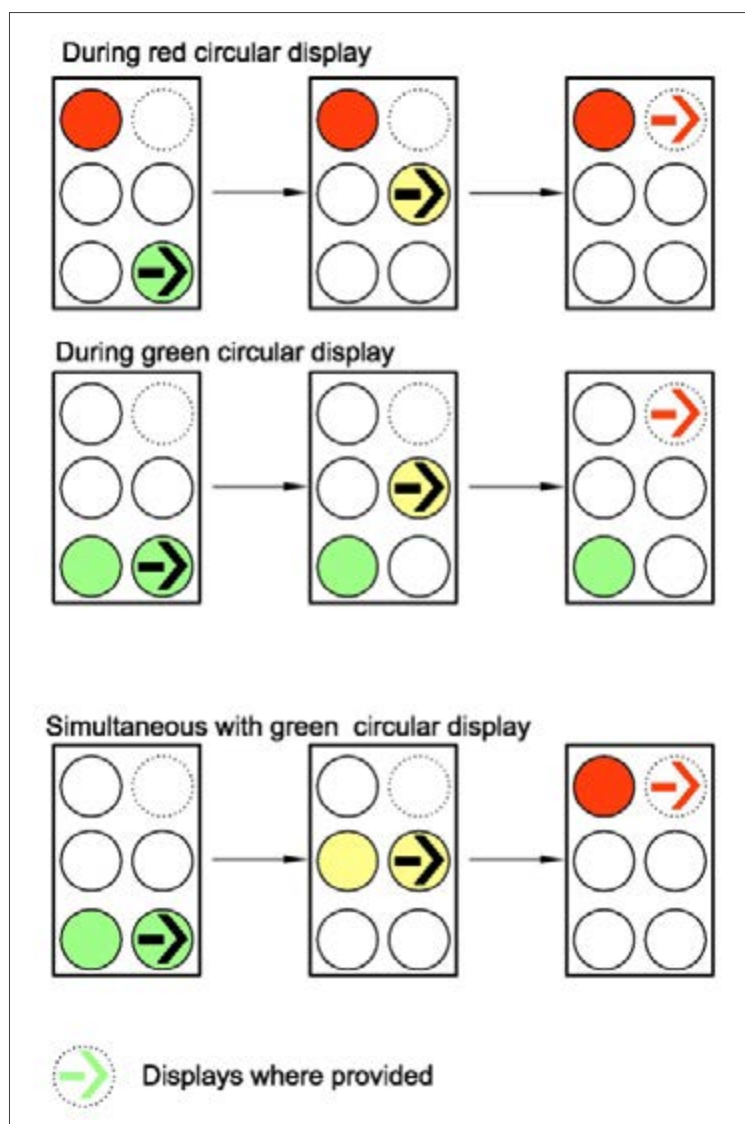


Figure 6.12 Display Sequence to Terminate an Arrow - Controlled RT Movement



6.9.4 LT Sequences

Commonly used LT display sequences in order to **initiate** a LT for a six-aspect arrangement is shown in Figure 6.13.

Commonly used LT display sequences in order to **terminate** a LT for a six-aspect arrangement is shown in Figure 6.14.

Figure 6.13 Display Sequence to Initiate an Arrow - Controlled LT Movement

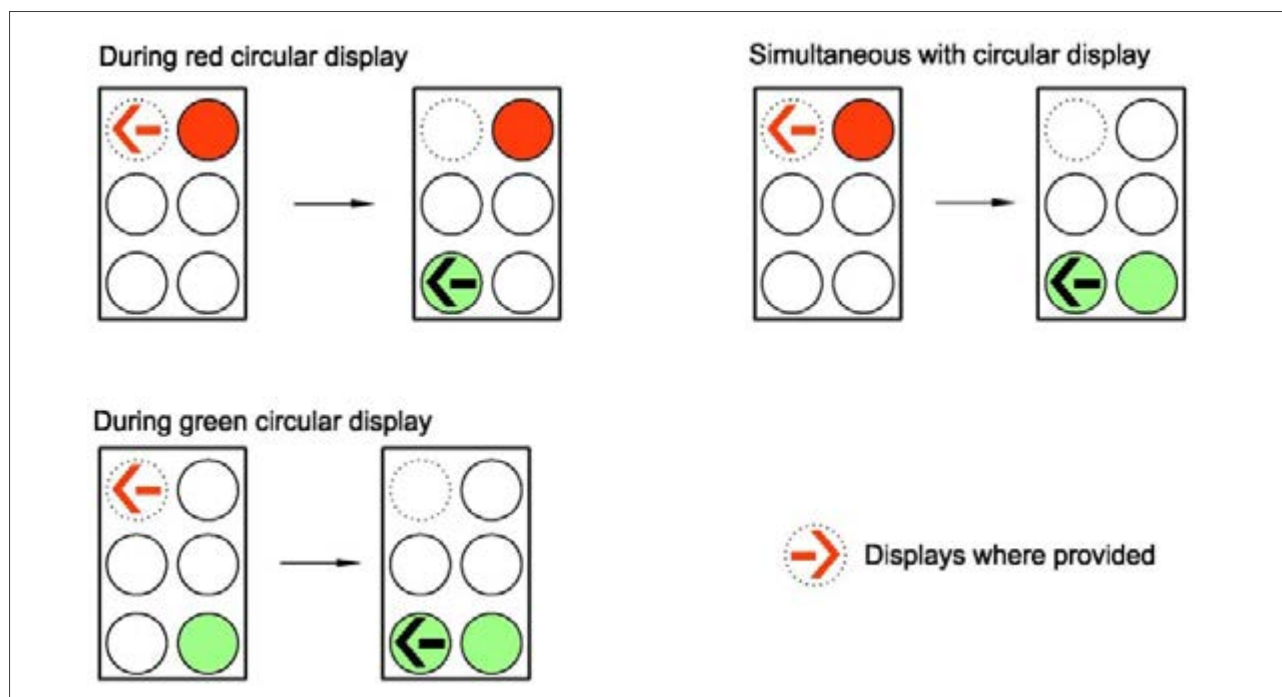
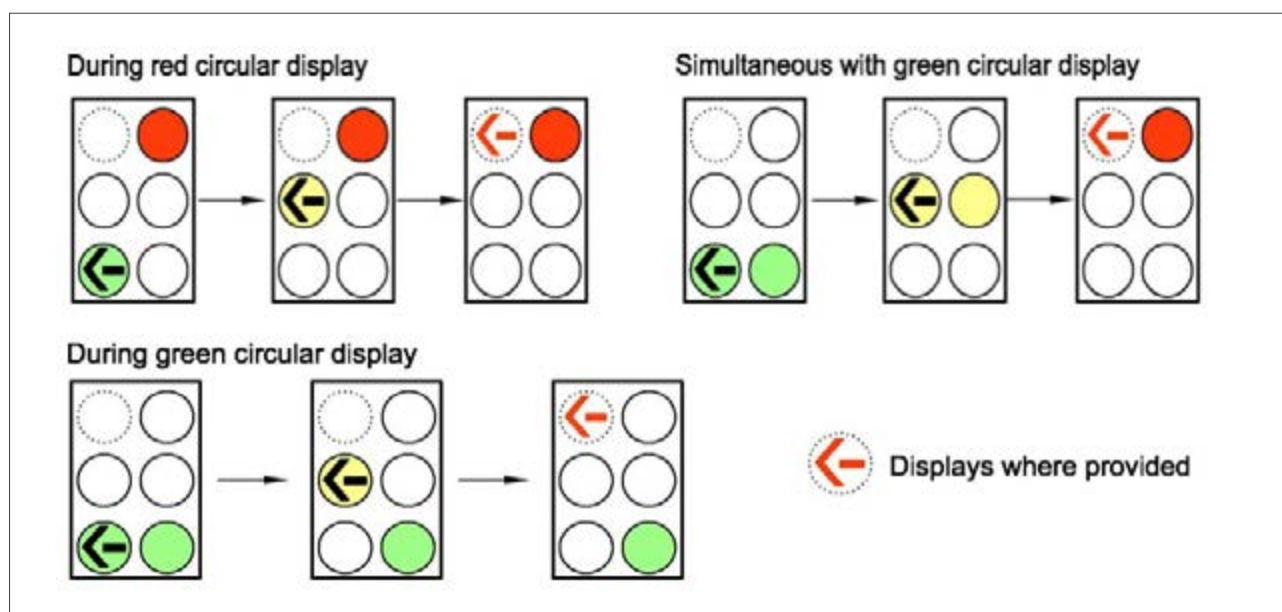


Figure 6.14 Display Sequence to Terminate an Arrow - Controlled LT Movement



6.10 Pedestrian Signals

The basic sequence for Pedestrian displays is that of **Steady Red 'Man'** to **Green 'Man'** to **Flashing Red 'Man'** to **Steady Red 'Man'**.

Pedestrian signal face layouts consist of a **Red 'Man'** Pedestrian aspect mounted above a **Green 'Man'** Pedestrian aspect. See [Figure 6.1](#).

Pedestrian aspects must always be mounted separately from the vehicle lanterns. Traffic Signal Countdown Timer (TSCT) should be installed with Steady Red to indicate the time remaining for the Green Man to come up. This helps in better control of pedestrian movement.

The Green Man, flashing and steady Red Man can be augmented by audible signals to assist elderly and physical / visually challenged persons.

6.11 Cycle Signals

The usual sequence for bicycle displays with a two-aspect arrangement is **Steady Red 'Bicycle'** to **Green 'Bicycle'** to **Flashing Red 'Bicycle'** to **Steady Red 'Bicycle'**.

The sequence for bicycle displays with a 3-aspect arrangement is **Steady Red 'Bicycle'** to **Green 'Bicycle'** to **Amber 'Bicycle'** to **Steady Red 'Bicycle'**.

A cycle signal face layout may consist of a Red 'cycle' aspect mounted above a Green 'cycle' aspect. See [Figure 6.1](#). These two-aspect cycle lanterns must never be incorporated with vehicle aspects in a common signal face. Traffic Signal Countdown Timer (TSCT) should be installed along with Steady Red to indicate the time remaining for the Green Bicycle to come up.

6.12 Special Vehicle Signals

The usual sequence for special Vehicle displays is **Red** to **White** to **Amber**, and back to **Red** again. Where white special Vehicle lanterns are displayed as single aspect units, the sequence is **Off** to **White** to **Off**.

A single White aspect does not fully control a special Vehicle movement, i.e. its absence does not compel a special Vehicle to stop.

It is usual to use White Special Vehicle aspects (B for Buses, T for Trams, or E for Emergency Vehicles) that provide control of special Vehicle movements. In addition, a Special Vehicle signal face layout consists of a single column of Red and Amber aspects may be employed. See [Figure 6.1](#).

6.13 Summary of Rules for Signal Display Sequences

The following is a summary of rules concerning the signal display sequences and are applicable to all locations regardless of the number of Phases:

1. Green Arrows should be used when there is no conflict movement.
2. Where a Full Green Roundel is the next aspect to be displayed with a Pedestrian Green 'Man' signal, the left turn must be stopped with a LT Amber Arrow, followed by the LT RA as shown in [Figure 6.14](#).
3. The LT RA is held for a predetermined period from the commencement of the Pedestrian Green 'Man' period. If this period is less than the Walk+ Clearance periods, then the Arrow lantern is blacked out. LT traffic may proceed with care on the Full Green signal. If there is sufficient Vehicle Green time following the end of the Clearance and there are no other conflicting movements, the Green Arrow may be re-introduced in association with the Full Green.

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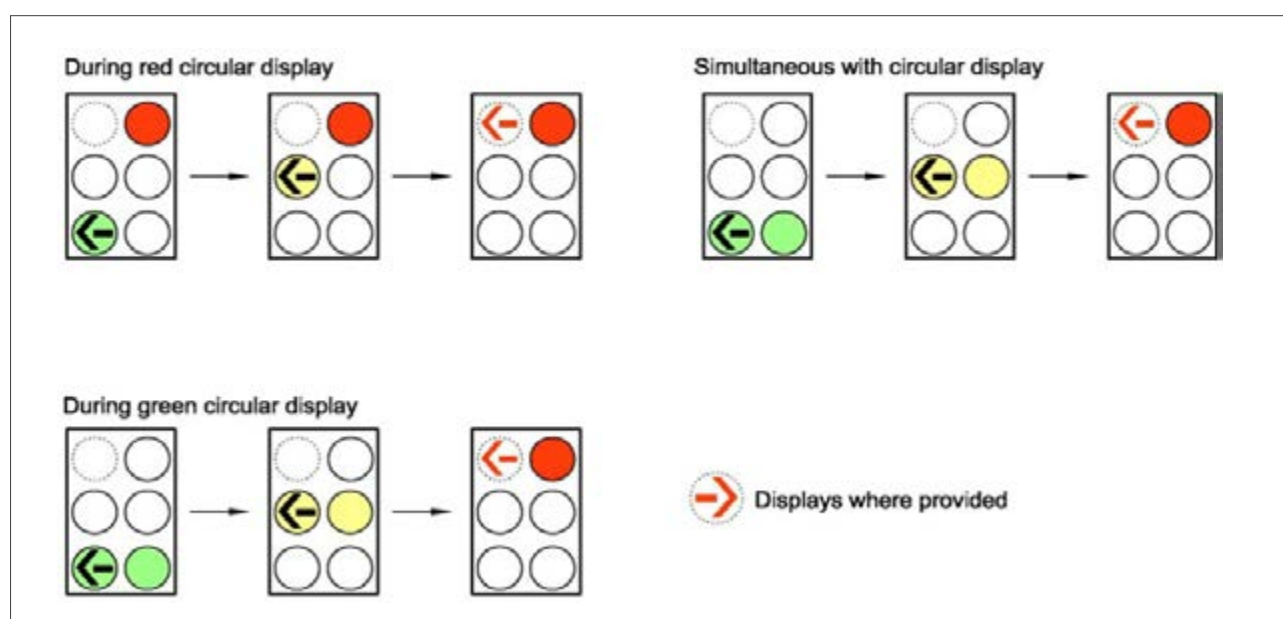
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Sign Face Layouts & Display Sequences

4. If a Full Green is the next aspect to be displayed after LT GA and the Parallel Pedestrian Green 'Man' is not demanded; the LT GA is permitted to overlap through inter-green. Where there is a conflicting RT filter movement the LT GA is blacked out as soon as the Full Green is displayed. See Figure 6.15.
5. If there is no conflicting RT movement, the GA is displayed with the Full Green. Where a Full Green is the next aspect to be displayed and there is no controlled Pedestrian crossing, the LT GA may be allowed through the inter-green. No LT RA is required under.
6. Where LT Arrows are provided, their display during the Walk and Clearance periods is as described above. RT GAs may either be used adjacent to a Full Green signal. See Figure 6.6 or with a full Red signal when the through traffic is held. See Figure 6.8.

Figure 6.15 Sequence for LT Amber



When a major fault occurs due to equipment malfunction, the recommended fallback display is Flashing Amber on all approaches at a rate of 1 Hertz with equal 'On' and 'Off' times.

6.14 Signal Displays on Start-Up

When initiating operation of a signal-controlled intersection the following sequence of displays is recommended:

1. Display **Flashing Amber** on all approaches for a minimum of ten seconds.
2. Display **Steady Red** on all approaches for three seconds minimum.
3. Display **Green** aspects to the first (usually pre-selected) movements in order to proceed.
4. Introduce at least one full cycle of all Phases and movements.
5. Proceed with normal operation.

Alternative sequence is:

1. At Power on - Display no signals to the selected Main Road Movement.
2. Display **Steady Red** on all side road approaches for 10 seconds minimum, no signals on Main Road.
3. Display **Green** aspects to the Main Road movements in order to proceed.
4. Introduce at least one full cycle of all Phases and movements.
5. Proceed with normal operation.

6.15 Traffic Signal Countdown Timer

A traffic signal countdown timer, or TSCT, is a clock that digitally displays the time remaining for the current stoplight indication – i.e., red, yellow or green. It informs the driver of the time remaining for the green and red signals. Introduction of red timer can help motorists to shut their engine if there is a long wait and be ready to utilise the green time. Green timer is not advisable because it may introduce rash driving tendency and may cause crashes at the intersection in a hurry to clear it.

Red TSCT display usage needs to be consistent across all signals on the major roads. If used intermittently or in a haphazard manner, the purpose is not served.

TSCT display is a good practice for pedestrian signals. It can be augmented by audible signals to assist elderly, physical / visually challenged persons.

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Sign Face Layouts & Display Sequences

7 Determination Of Signal Phasing (Or Staging)

7.1 Introduction and Scope

Signal Phase (or 'Stage') is the time during which one or more movements receive right of way. A Phase may have one or many movements. Phase starts with Green. The 'end' of the Phase is identified finishing of one or more movements so that a new movement can start.

Separately controlled traffic movements are arranged in a Phases that combine compatible movements in sequential and concurrent manner to form a sequence. One complete sequence of Phases is called a cycle and is typically identified by the letters A-B-C-D-A.

There are two methods of operational timing arrangements in signal controllers:

1. Phase control - where all the movements are subject to the signal timing parameters of that Phase, e.g. inter-green time, minimum Green time. Phase control minimises the number of Phases.
2. Signal Group control - where the signal timing parameters are specified for individual movements. Group control allows more flexibility to start and end regardless of the number of Phases. The movements operate with respect to each other rather than being tied to Phase transitions.

Modern Traffic Controllers are capable of employing features of both Phase and Group control methods simultaneously.

7.2 Fundamental Rules of Signal Control

The main objectives of the Signal Phase design are:

1. **Safety:** To reduce road crashes by minimising conflict among traffic and pedestrian movements.
2. **Efficiency:** To maximise capacity and to minimise travel delay, queue length, number of stops, or a combination of these performance measures.

The two objectives often conflict. Compromise are needed. Consistency of signal equipment and operation throughout a network and an understanding of driver behaviour are essential for effective design.

7.3 Determination of Phase Sequences

A single vehicle or pedestrian movement is considered as a signal group. A combination of movements that may safely run concurrently is called a Phase.

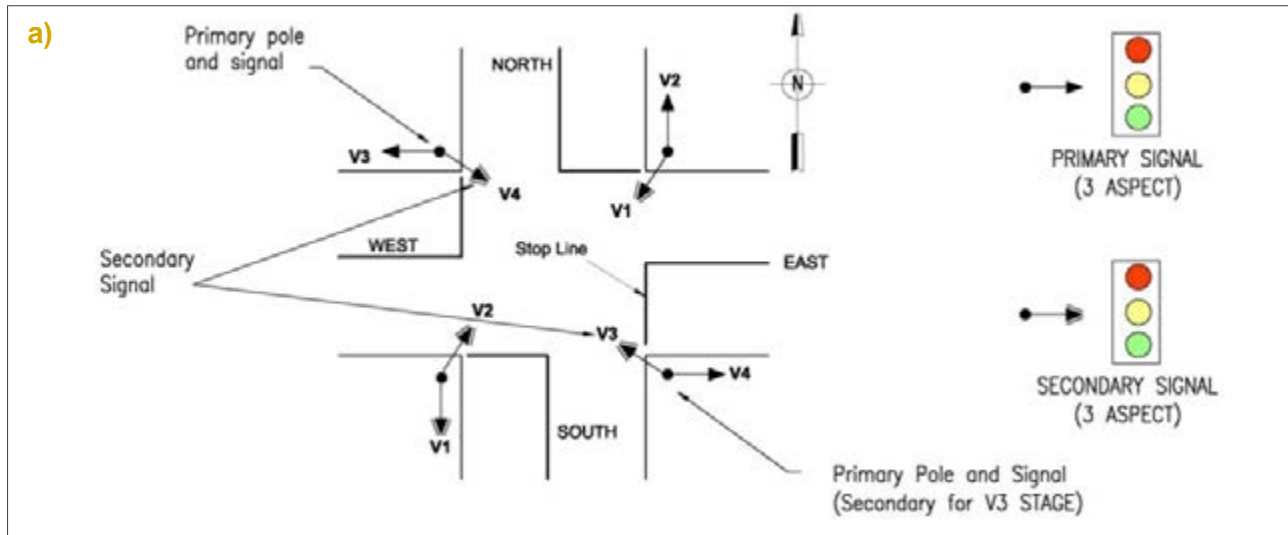
Each possible trajectory of traffic flow is called a movement. At a typical intersection, each arm or approach of the intersection can accommodate three movements:

1. Vehicles travelling straight through (SA – straight ahead).
2. Vehicles turning left (LT - turning movement).
3. Vehicles turning right (RT - turning movement).

In the simple junction layout shown in [Figure 7.1\(a\)](#) there are 12 movements in total (3 on each approach). During turning movement counts, vehicles for LT, RT, SA at each approach are counted. The counts are used to assess lane requirements.

Straight Ahead, LT and RT movements can be designated as one signal group movement 'V1'. Movements from the opposite direction can be signal group 'V2'. If there are no major conflicts between V1 and V2, they can run simultaneously as 'A' Phase.

A Phase (or Stage) is a combination of compatible signal groups.

Figure 7.1 Simple Junction Signal Layout

A pedestrian movement that is allowed to run simultaneously in Phase is also allocated a signal group output and can be called 'P1'.

In some situations, where there are several Pedestrian groups P1, P2, they may be run concurrently with signal groups V1, V2. It can be run in one separate Phase called an 'All-Round-Pedestrian' Phase. In this case, all vehicular movements are stopped.

Only one Phase can be run at a time. Phases are labelled A, B, C and are usually serviced in alphabetical order.

General guidelines for Phase design:

1. The number of Phases should be kept to a minimum.
2. As many movements as possible should be allowed to run in every Phase.
3. A Phase should consist of non-conflicting movements to the extent possible.

Figure 7.1(a) shows a basic two Phase design. Figure 7.1(b) satisfies points (1) and (2) but there are 8 conflicts. The four Phase design of Figure 7-1(c) has no conflicts but it is very inefficient.

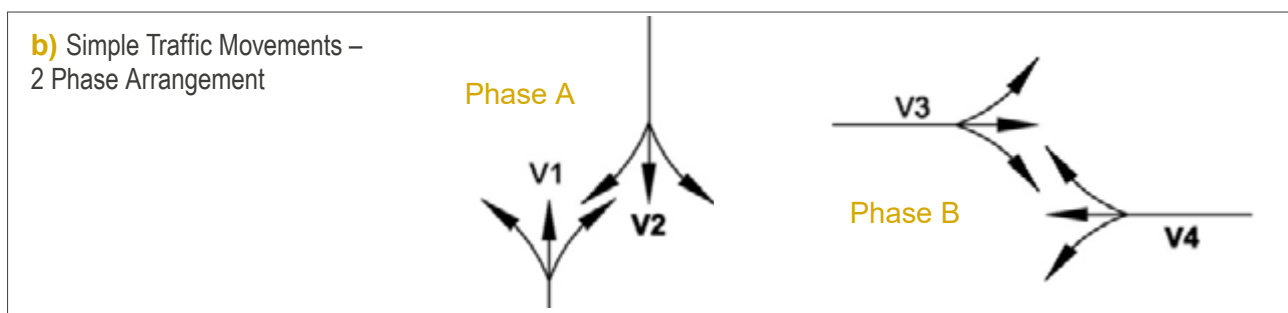
In the example of Figure 7.1(b), V1 may safely run with V2. Similarly, V3 may safely run with V4. Therefore V1 and V2 may be considered as 'A' Phase and V3 and V4 considered as 'B' Phase. This is a simple arrangement when there are no Pedestrian movements to be considered.

A Phase = V1 + V2 B Phase = V3 + V4

If V1 and V2 cannot be run simultaneously, then V1 and V2 could be allocated their own Phase. Phase sequence can be:

Phase A = V1 Phase B = V2 Phase C = V3 + V4

Please refer to Figure 7.1(b) and 7-1 (c).



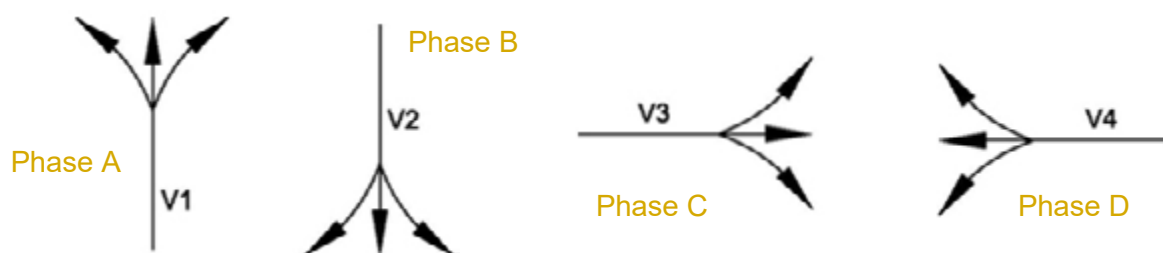
Similarly, if V3 and V4 cannot be run together, the Phase sequence can be:

Phase A = V1 Phase B = V2 Phase C = V3 Phase D = V4

Please refer to [Figure 7.1\(c\)](#)

Selection of an appropriate Phase design, including the use of arrows, overlaps and Pedestrian movements, are further discussed in the sections following.

c) Simple Traffic Movements – 4 Stage Arrangement



7.4 Signal Groups

Signal lanterns that have a particular colour sequence are called a signal group. A given signal group (e.g. V1) may control a single movement like RT, or combined movement like SA + LT. The operation of signal groups is tied to phases so that certain signal groups are green in certain phases.

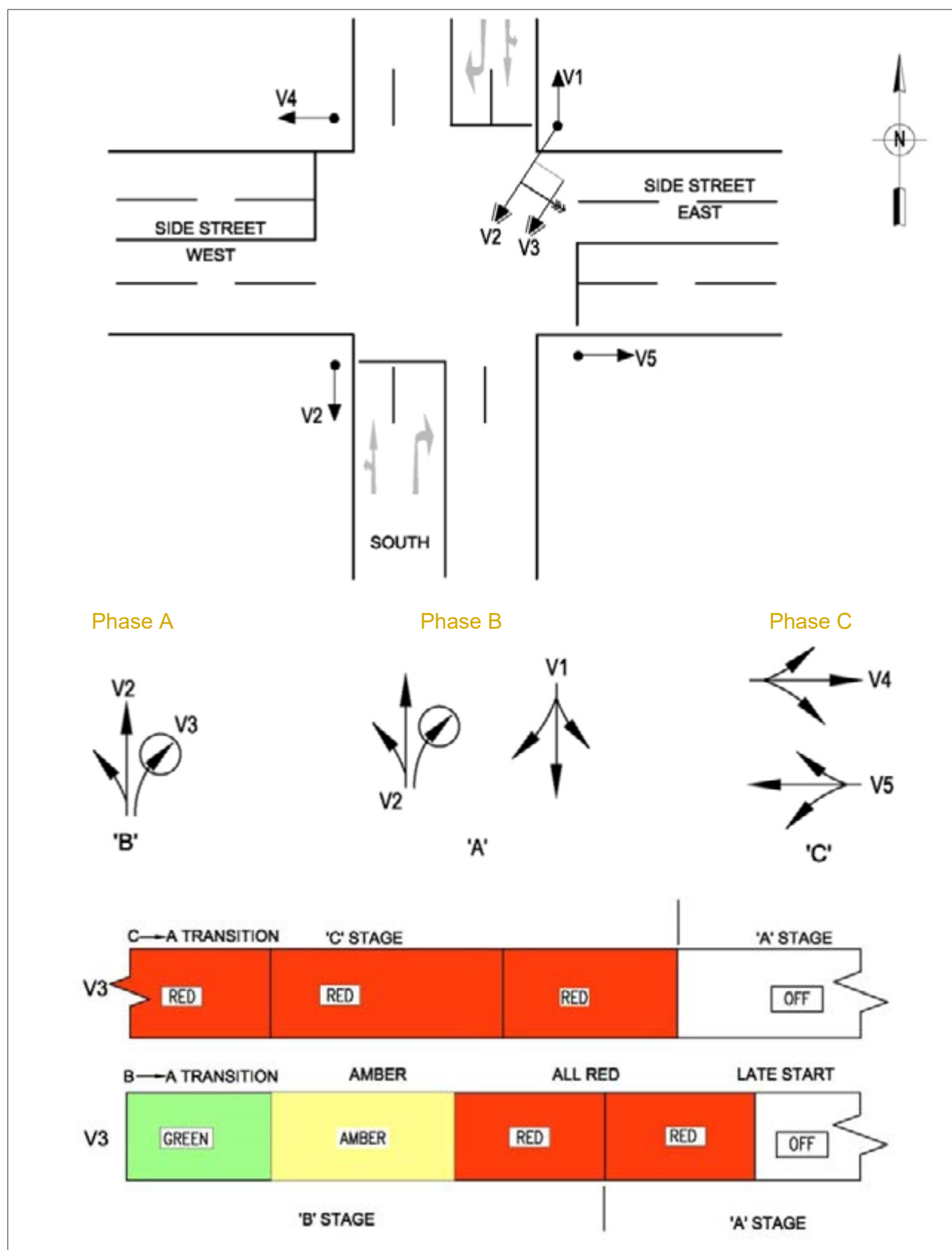
If V1 (group -GAR) runs in A Phase and B Phase, then V1 = A/B. When a signal group is green in more than one phase and also when the controller moves from one phase to another, then it is called an Overlap signal group. Those are identified by separating the phase letters by an oblique stroke, e.g., V1 = C/D.

A parallel pedestrian movement is one that runs with a vehicle movement. It should be carefully used, considering safety.

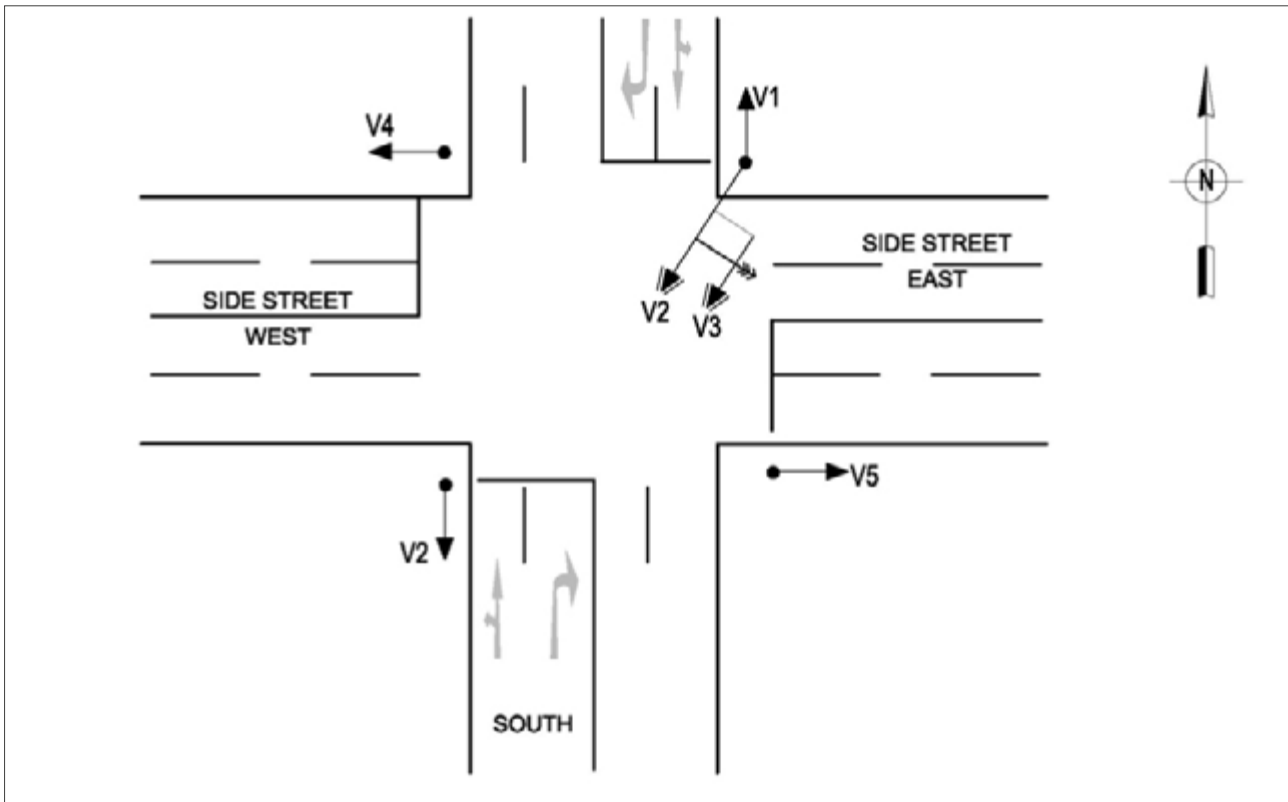
7.5 Two Phase Sequences

Inter-green period is the period of Amber + All-Red occurring at the end of a phase. It ends with the commencement of green on another phase. It is commonly set at 5 secs = 3 secs Amber + 2 secs All-Red.

The 'Late Start' period is a short period at the beginning of a phase, where one signal group green starts, but the associated signal group is delayed. It is commonly set at 2 seconds late.

Figure 7.2 Right Turn 'B' Phase Arrows Application of Late Start Control

An application of the Late Start Interval is shown in **Figure 7.3** on the next page, where V3 is started a number of seconds after V2 to allow vehicles being stopped by V1 to clear the slip lane entrance after the inter-green period.

Figure 7.3 Operation of a Late Start Interval for LT Movement

7.6 Three Phase Sequence

Where a RT movement is found to be necessary in one direction only (e.g. V3), the most efficient sequence is for the turning Phase V3 to follow the opposing through Phase V1. This is the normal sequence and is called trailing turn.

By using the reverse sequence (A-C-B) both approaches of Phase A are terminated together, and filter RT turn is permissible.

As a general rule, reverse sequence or leading turn should only be used at locations where the opposing RT cannot be banned or where it is necessary for co-ordination purposes.

In both trailing and leading turn sequences, it is necessary to provide RT Red Arrows (RTRA) as follows:

- 1. Trailing Turn Sequence** - RTRA displayed following 'B' Phase for the 'Late Start' period of 'A' Phase, if 'C' Phase is skipped.
- 2. Leading Turn Sequence** - RTRA displayed following 'B' Phase for 'Late Start' period of 'A' Phase.

When both opposing RT movements are signalled, it is called a 'Diamond Turn' or 'Single Diamond Overlap (S.D.O.)'.

In this case, where it is necessary to signal both opposing RTs, the following three Phase Diamond Turn operation may be used; 'B' Phase is introduced if either (or both) RT detectors (D1+ D2) is actuated. Refer to [Figure 7.4](#)

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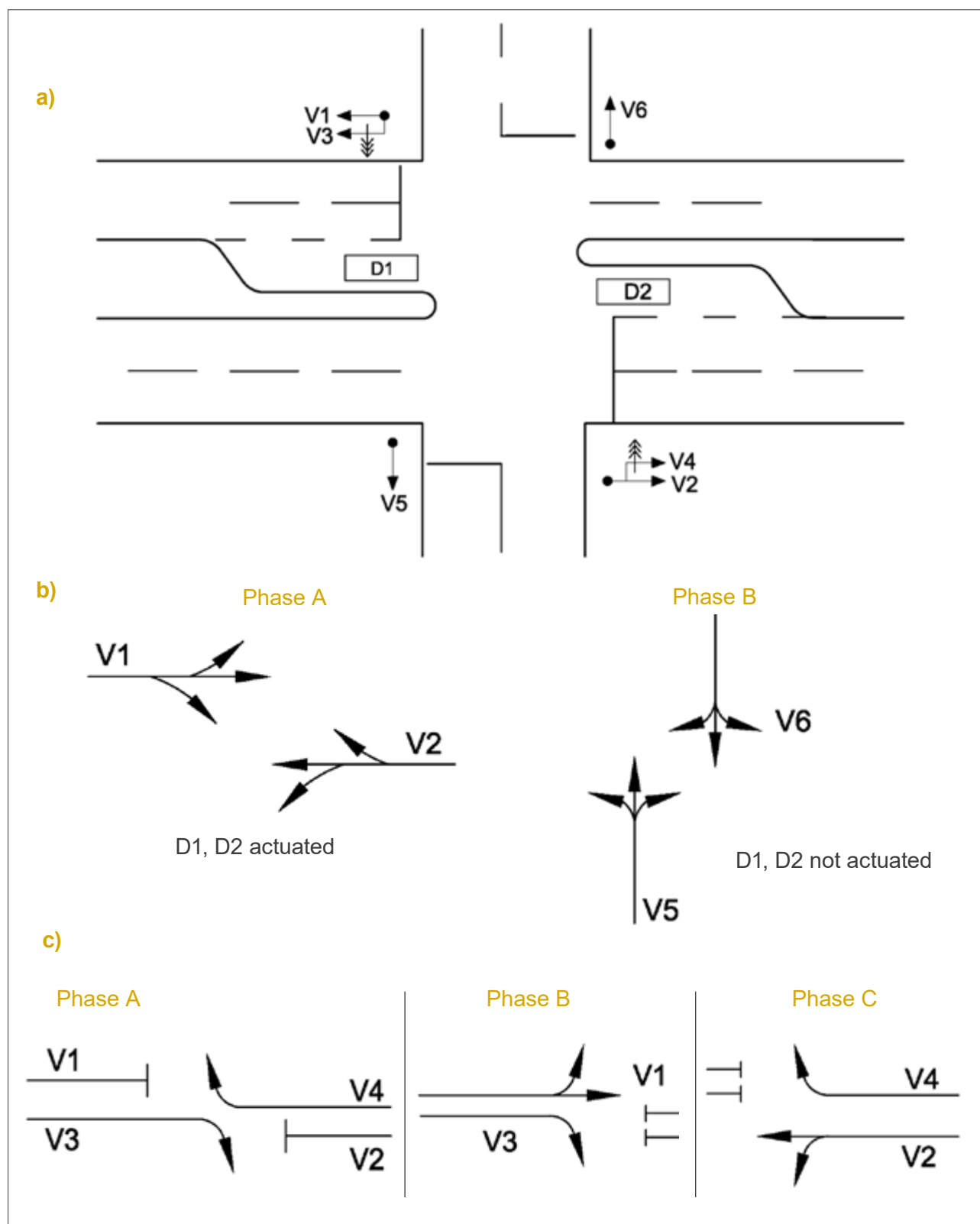
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Determination of Signal Phasing (or Staging)

Figure 7.4 Alternative Overlap Operation (Single Diamond Overlap – S.D.O)

7.7 Multi-Phase Signal Sites

The number of phases should be kept to a minimum for the most efficient site operation. However, additional phase can be introduced where separate RT bays are available, and the traffic volumes warrant more than two phases. For Diamond Turn on one street, SDO can be efficient.

In the case of Diamond Turns on both intersecting streets, the 'Double Diamond Overlap' sequence (D.D.O.) is recommended.

7.8 Evaluation of Phase Design

The relative effectiveness of alternative phase designs can be evaluated using computer software packages that offer analytical or simulation techniques for estimating operational performance measures.

7.9 Road Crash Risk Factors

Safety evaluation is a prime consideration in selecting a phase design. The factors that contribute to increases in risk of road crashes involving turning movements are:

1. Inadequate sight distance for conflicting vehicles or pedestrian movements.
2. More than one lane of turning traffic.
3. Parallel pedestrian movements are exposed to:
 - a. High volumes of turning traffic
 - b. High speed of turning traffic
4. The 85th percentile speed of the opposing traffic is greater than 70 km/h and the RT traffic has to turn across three or more lanes of oncoming traffic.
5. There is a Lagging RT with opposing Filter turn.

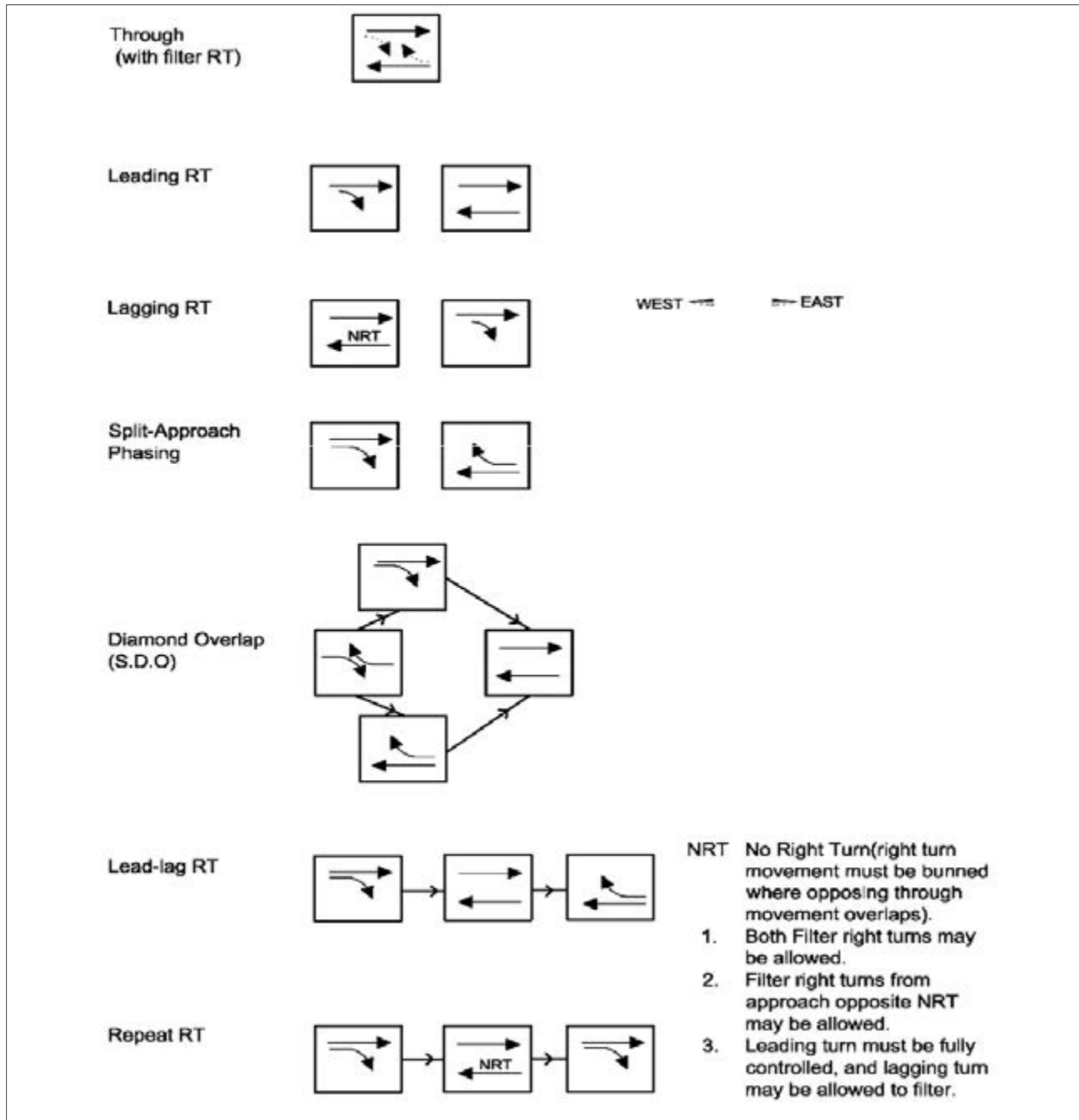
7.10 Phasing Design

Figure 7.1 and Figure 7.2 present the basic elements of phase design applicable to four-way or three-way intersections.

The choice of phase system depends on the following factors:

1. **Layout:** the number of lanes available for each movement on the approach and departure of each intersecting road.
2. **Alignment:** the horizontal and vertical alignment in regard to the intersection angle and sight distance available to allow safe filtering of RT movements.
3. **Traffic flows:** the volume of traffic including the proportion of heavy vehicles in each through or turning movement.
4. **Signal coordination:** progression considerations for an intersection within a coordinated signal system.
5. **Pedestrians:** the pedestrian movements that need to be controlled and how they will be catered for in the phase.
6. **Special Vehicles:** whether or not buses, trams, cycles need to be separately controlled, and how they will be catered for in the phase.

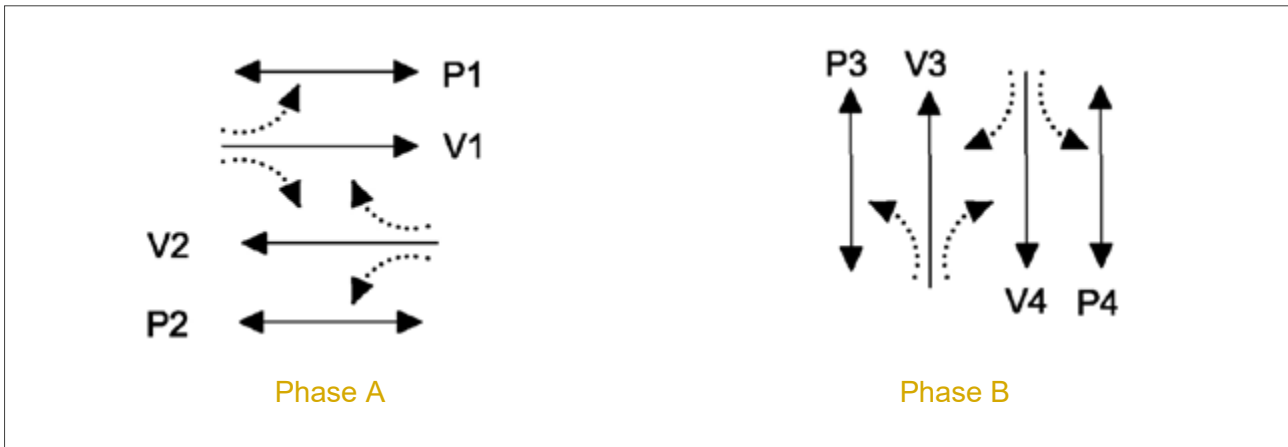
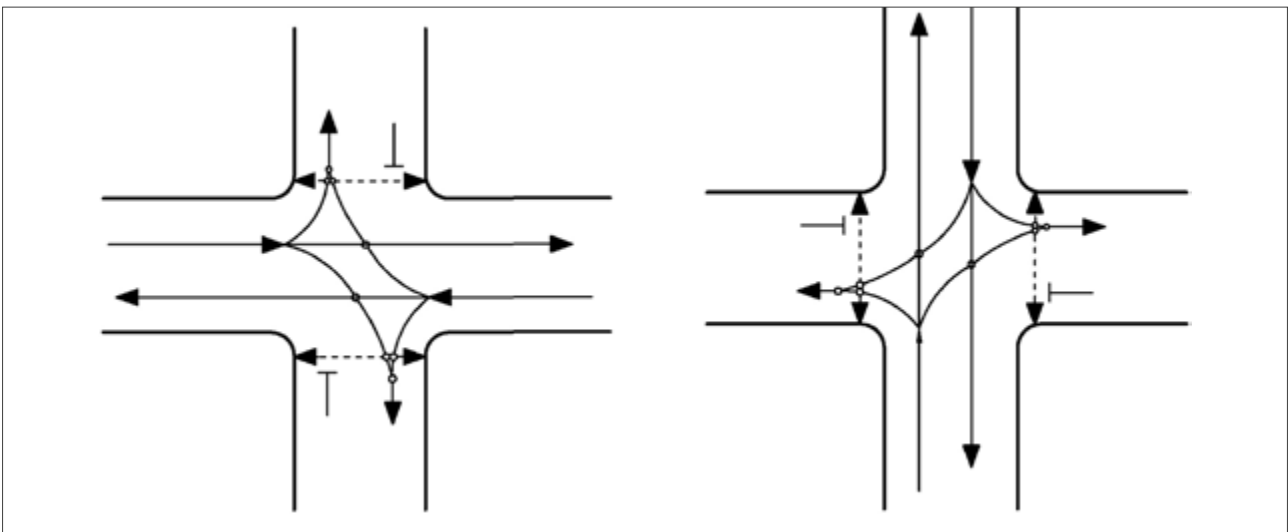
The basic elements of signal phasing are shown in Figure 7.5 on the next page.

Figure 7.5 Basic Elements of Signal Phasing

By allocating right of way to each road alternately, the two-Phase system eliminates all crossing conflicts between through traffic movements. However, it leaves four turning conflicts between RT and SA Vehicles, four merge conflicts between RT and LT Vehicles, and eight turning (RT & LT) Vehicle vs. Pedestrian conflicts.

'P' denotes Pedestrian movements.

Figure 7.5, Figure 7.8 and Figure 7.9, show two complete Phase arrangements for a crossroads intersection consisting of leading (Figure 7.8) or lagging (Figure 7.9) RT Phases on one road and a SA through Phase with filter RT on the other road. Thus, the signal cycle consists of 3 Phases in these cases.

Figure 7.6 Two Phase Control – Filter RTs and Parallel Pedestrians**Figure 7.7** Conflicts in Two Phase Control

For all phase options shown in **Figure 7.5**, except the split-approach phasing, provision of an exclusive RT lane is recommended in order to:

1. Reduce the exposure to the rear-end conflict between through and RT Vehicles.
2. Avoid lane blockage by Vehicles waiting for gaps or stopped by a Red display.
3. Isolate detection of RT Vehicles (to prevent through Vehicles unnecessarily calling the turn Phase for leading RT Phases).

A RT bay is a practical solution for providing an exclusive RT lane. An exclusive RT lane is essential for operation of Diamond Overlap Phasing, and is recommended in the case of fully-controlled RTs.

If the RT movement is Arrow controlled, the sign should preferably switch on at the same time as the Arrows change from an Amber display to a Red display or Red Arrow drop-out.

The Phase alternatives shown in **Figure 7.5** can be achieved through the following types of RT control at traffic signals:

1. **Filter Only:** No RT Arrows, three circular aspects only. Full Green alone indicates that RT traffic can 'Filter' through gaps. RT vehicles give way to through and LT Vehicles from the opposing direction.
2. **Full Control:** Three-aspect RT Arrows on a six- aspect signal face. The Green Arrow indicates that the Vehicle can RT unopposed. The Red Arrow indicates that the Vehicle is not permitted to turn. Filter RTs are not permitted at any time during the cycle.

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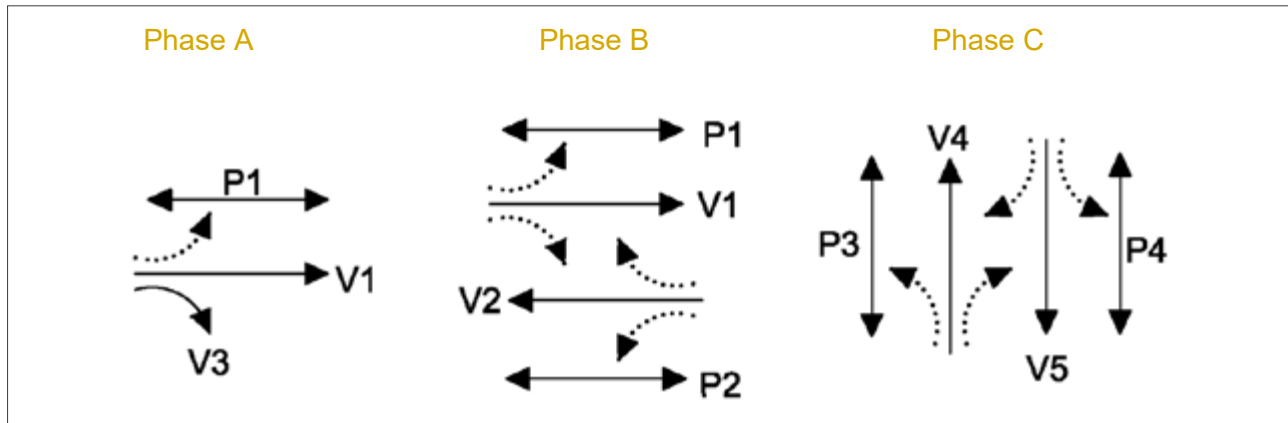
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Determination of Signal Phasing (or Staging)

Figure 7.8 Three - Phase System

3. Red Arrow Drop Out: Three-aspect RT Arrows on a six aspect signal face V1 & V3, are provided but the Red Arrow V3 extinguishes (drops out) and the adjacent Full Green display V1 permits Filter turns after a few seconds in the through Phase.

4. Flashing Amber Arrow: As an alternative to blanking the Arrow displays, a Flashing Amber Arrow is displayed after the Red Arrow. This indicates that a RT may be made but the driver must give way to other vehicles and pedestrians.

The System may be programmed to operate in Full Aspect Control during certain hours of the day, and Filter Only or Partial Control during other times of the day.

7.11 Through Phase with Filter RTs

This Phase arrangements shown in Figure 7.5 to Figure 7.9, allows the SA and LT movements and filter RTs from opposing approaches to operate in the same Phase using 3-Aspect Full Roundels signal faces.

A filter RT movement must give way to and find safe gaps in conflicting vehicle or pedestrian traffic before proceeding. A Phase alternative that uses a Green RT Arrow may be considered if:

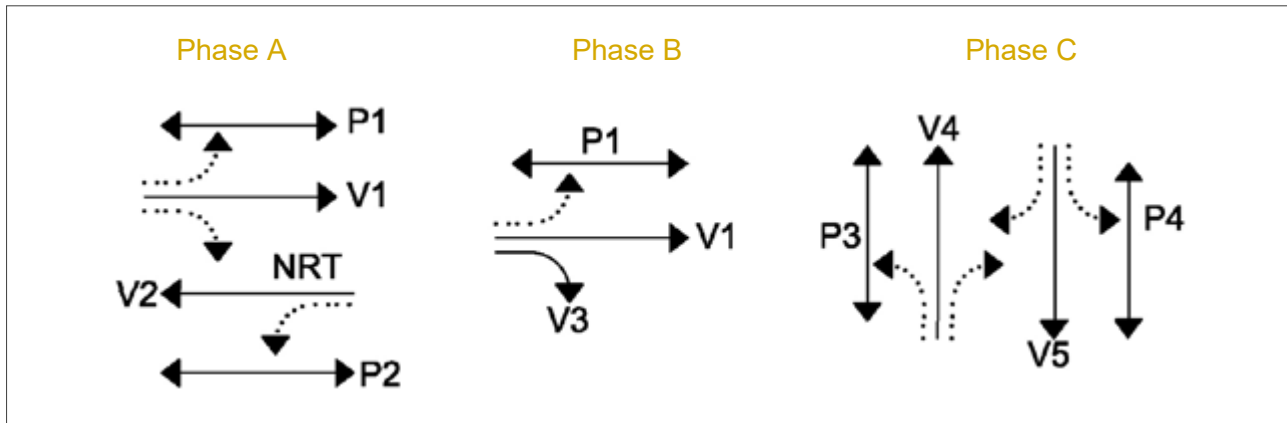
1. Intersection performance can be improved by providing a separate RT Phase.
2. The filter RT is considered unsafe due to road crash risk factors.

If a suitable phase with safe and efficient RT movement cannot be found, RT can be banned.

If only one RT is banned and the other RT cannot filter efficiently, a lagging RT sequence can be introduced.

If neither RT can be banned, a leading RT sequence can be introduced, if one of the RTs can Filter safely and efficiently and the other cannot.

If both RTs cannot Filter efficiently, then a Diamond Overlap sequence can be an option. If both RTs are required but the Diamond turn is not practical, a Split-Approach sequence may be considered.

Figure 7.9 Three - Phase System

7.12 Leading RT Phasing

A leading RT sequence is where a RT phase precedes the phase in which the opposing through movement runs (see Figure 7.5 and Figure 7.8). A leading RT Phase is controlled by 3-Aspect RT Arrows in a 6-Aspect signal face.

This Phase is a suitable option where an Arrow controlled RT has a Filter RT from the opposing direction, which cannot be banned and is able to Filter safely and efficiently.

Where it is safe to do so, the Arrow-controlled RT can be allowed to Filter through the opposing through movement.

This Phase system becomes inefficient for shared lanes with no RT bay. The provision of an exclusive RT bay is recommended for leading RT phase.

7.13 Lagging RT Phasing

A lagging or trailing RT sequence is where a RT Phase follows the Phase in which the opposing through movement runs.

A lagging RT Phase is controlled by a 3-Aspect RT Arrow in a 6-Aspect signal face. This Phase sequence can be used where a two-Phase system has safety or efficiency problems with Filter RT operation.

If the RT movement from the direction opposite to the lagging RT cannot be banned, the conflict can be avoided by:

1. Using a leading RT sequence rather than a lagging RT sequence.
2. Forcing the overlapping through movement to stop and then start up again.
3. Using Split-Approach phase or diamond overlap Phase or lead-lag RT phase.

7.14 Split-Approach Phasing

Split-approach Phasing allocates separate phases to opposing approaches at four-way intersections. The SA and turning movements from each approach operate simultaneously and RT movements are unopposed under this Phase. Split-approach Phasing is controlled by 6-Aspect signal faces.

Split-approach Phasing is appropriate where:

1. Side streets at an intersection are slightly offset so RTs cannot make a Diamond turn, or sight distance makes opposing filter RT movements unsafe.

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Determination of Signal Phasing (or Staging)

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2. Turn proportions vary significantly during the day requiring flexible shared lane arrangements.
3. A heavy RT movement has very light opposite movement. The right-turn vehicles may fail to give way to opposing through vehicles.

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7.15 Diamond Overlap Phase

Diamond Overlap Phases allow opposing RTs to operate either independently or together or with the main road traffic. If there is no demand for one of the RTs, then that RT is allowed to terminate early.

RT arrows are used to control Diamond Overlap movements. The RTs can operate as Filter movements where safety permits.

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7.16 Lead-Lag RT Phase

A Lead-lag RT Phase combines the leading and lagging RT arrangements.

In this Phase arrangement:

1. The leading RT must be fully controlled using dual 3-Aspect lanterns with RT Arrows, and the lagging RT can be fully controlled by the same arrangement.
2. A signal face with a 5-aspect signal can be used. In that case the RT Arrow aspects have only a Green and Amber Arrow mask.

Lead and Lag RT Movements are useful for signal coordination purposes. The lagging RT movement may be allowed to Filter through traffic during the preceding through Phase. If Filtering causes safety problems, then the RT must be stopped by a Red Arrow display.

The leading RT movement should not be allowed to Filter during the following through Phase.

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7.17 Repeat RT Phase

The Repeat RT Phase system allows the RT Arrow to operate twice in the same cycle. Effectively, this provides a combined leading and lagging RT Movement.

The Repeat RT Phase system may be:

1. Fully controlled using 3-Aspect RT Arrows in a dual lantern.
2. Partially controlled using dual lantern RT Arrows.
3. Partially controlled using a 5-Aspect signal face.

The RT movement in a Repeat RT system be allowed to Filter during the main road Phase (partial control), unless Filtering causes safety problems. In that case the RT must be held by a Red Arrow display.

The RT movement from the opposing direction cannot be allowed to Filter during the main road Phase in order to avoid the lagging RT conflict.

The Repeat RT Phase system may be used where RT lanes are short. It is also useful where there is insufficient storage room for vehicles.

The use of the Repeat RT system means that an additional inter-green period is needed and this may reduce the efficiency of the intersection.

7.18 LT Movements

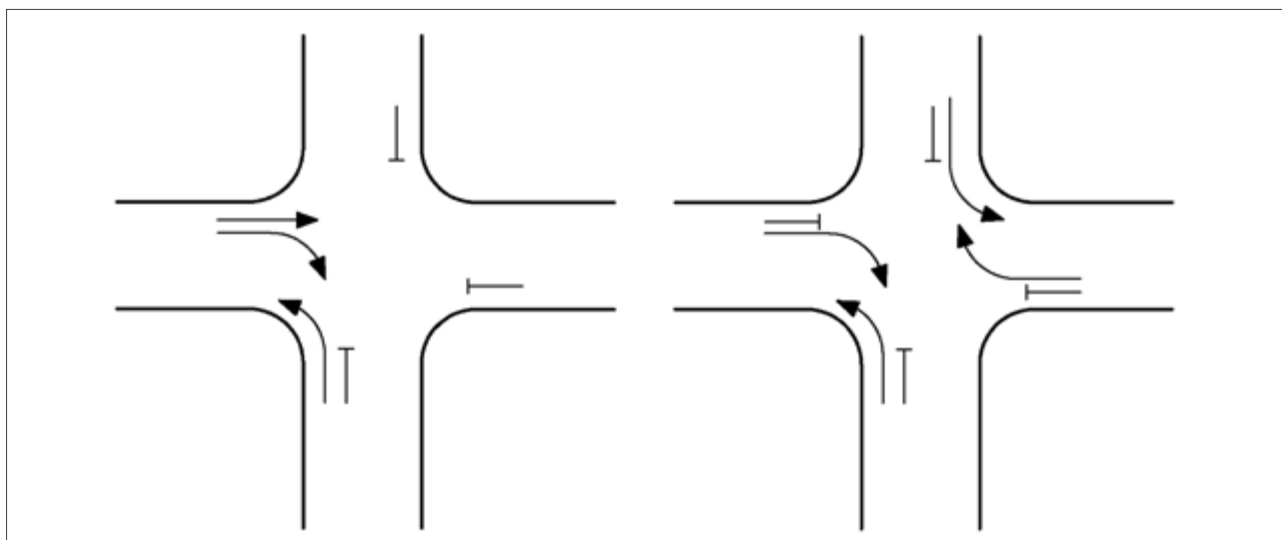
LT movements may run in one or more phases. When LT movement is controlled by 3-aspect GAR Full Roundel signal, and there is a parallel pedestrian movement, the vehicular traffic needs to filter through pedestrian gaps. If this is unsafe, then Red LT Arrows (LTRA) can be used to hold LT Vehicles.

LT Arrows are used for the following:

1. The volume of LT traffic is high.
2. The speed of LT traffic is high because of the turn geometry.
3. The visibility of pedestrians to turning vehicles is inadequate.

Green LT Arrows (LTGA) may be used when there is no conflict with other vehicle or pedestrian movements.

Figure 7.10 Complementary LT and RT Arrow - Control



A LTGA should never appear in the same Phase as a Filter RTGA of the opposing RT movement.

If un-signalled LT slip lanes are used in the intersection design, then care is needed to allow for pedestrians crossing the slip lane. If there are more than one slip lane, signals are required.

Figure 7.11 Complementary LT and RT Arrow - Control

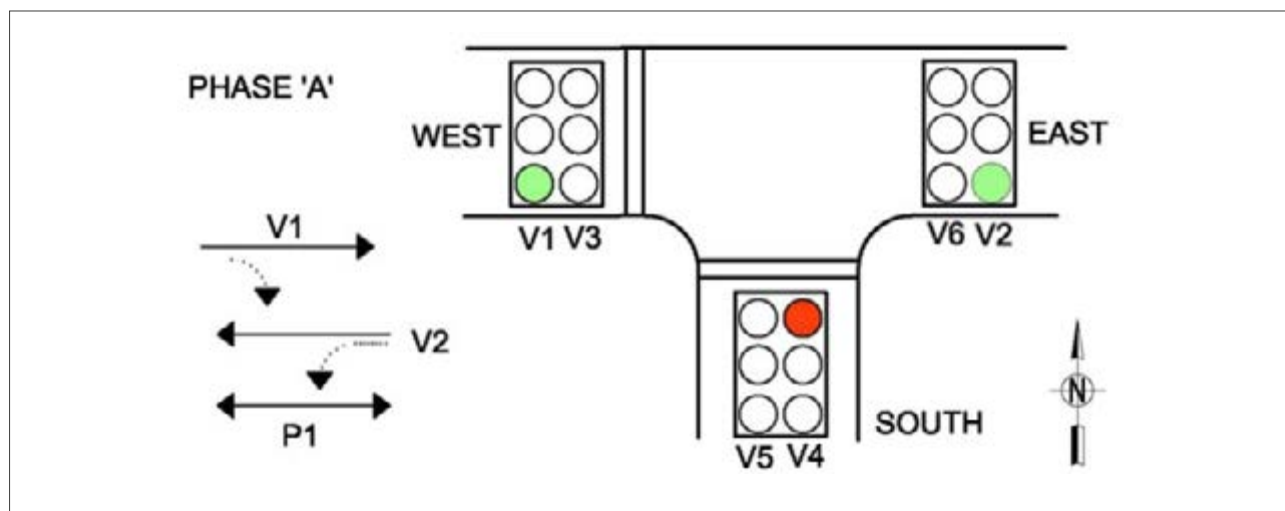
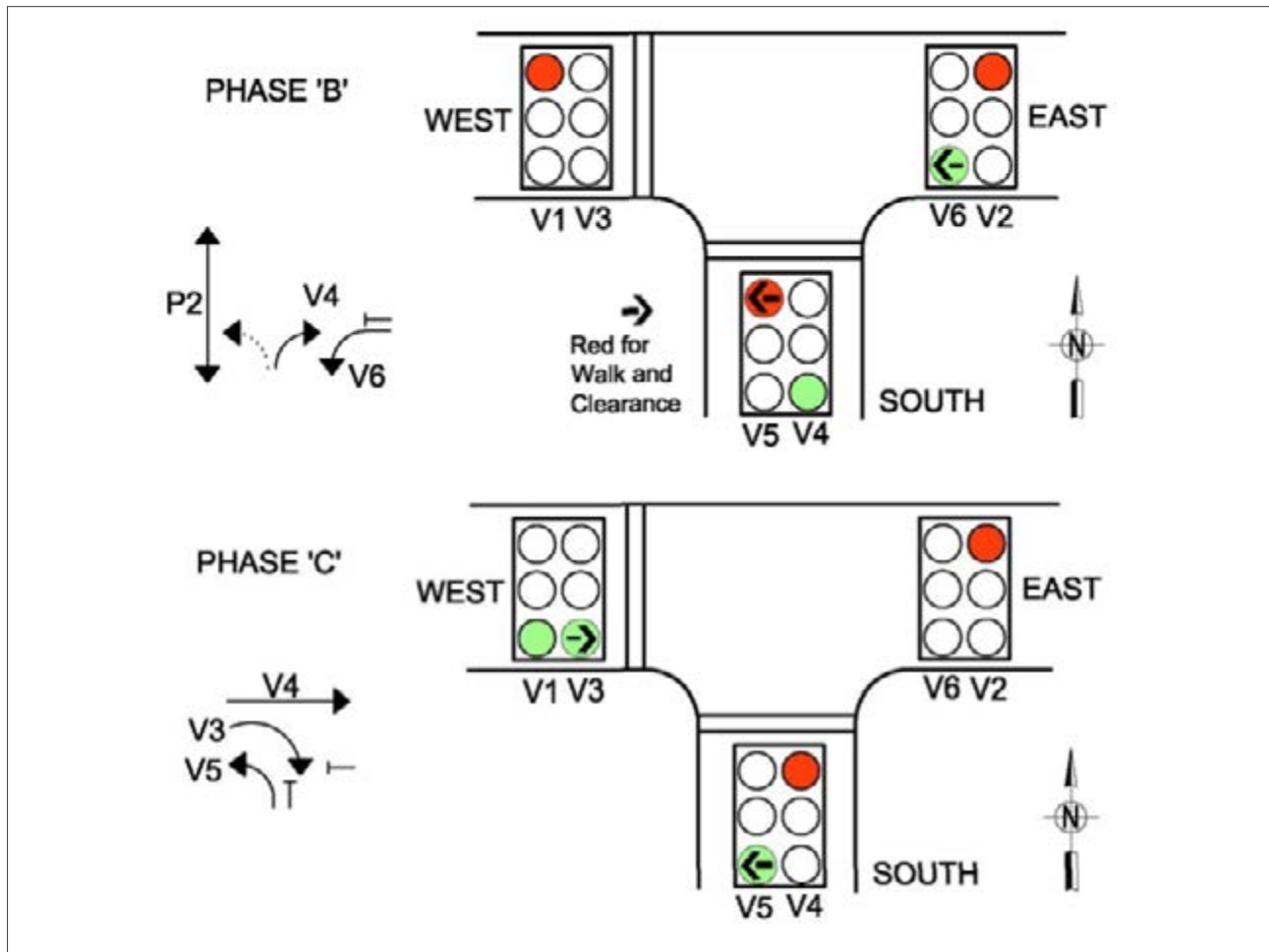


Figure 7.11 (continued) Complementary LT and RT Arrow - Control

7.19 Pedestrian Movements

7.19.1 General

Pedestrian movements are controlled by the following:

1. Green 'Man' or Walk period indicated by a steady Green 'Man' display.
2. Flashing Red 'Man' or Clearance period consisting of a Flashing Red 'Man' display that advises pedestrians already on the crossing to complete the crossing, but advises pedestrians not to start crossing.
3. Steady Red 'Man' or Don't Walk period consisting of a steady Red 'Man' display that prohibits crossing.

Note: The Flashing rate for the Clearance periods should be once per second (1 Hz) with an on-off ratio of 60:40.

7.19.2 Pedestrian Movement Operation

The duration of the Green Man signal is generally not less than 6 seconds, but depends on crossing width. The duration of the Flashing Red 'Man' should be based on the time to cross the full road width at a walking speed of 1.2m per second. The All-Red times before and after the pedestrian movements should be at least 1 second but can be increased depending on the road speed. The time allocated for crossing should consider additional time for Persons with Disabilities, pregnant women and women with children, if there is significant number of such persons crossing at that location.

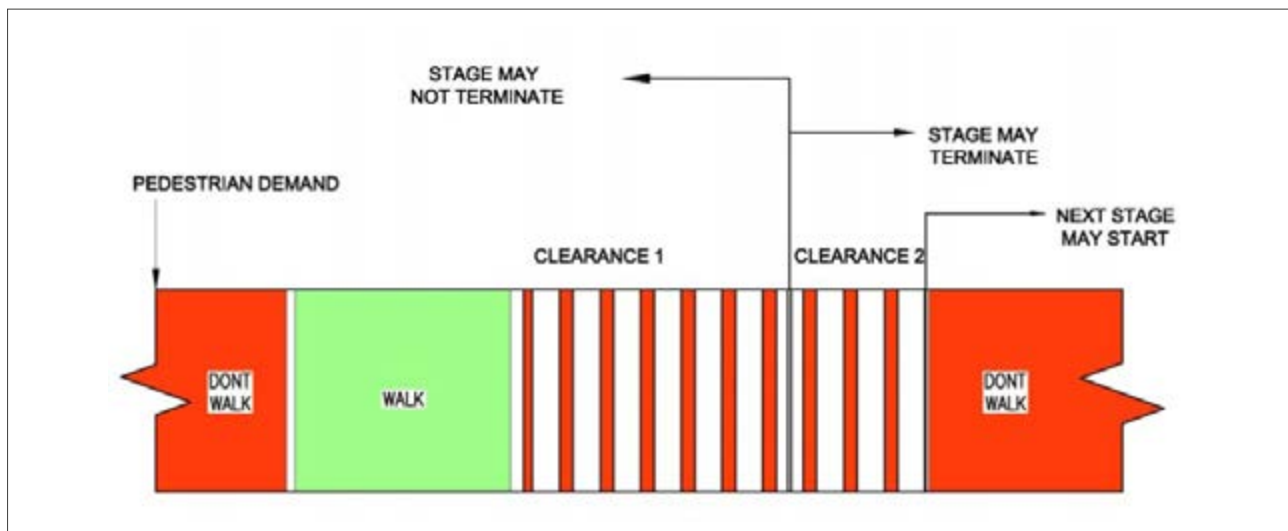
Pedestrian movements may be grouped with vehicle movements to form a Phase. Alternatively, Pedestrian movements may be grouped into one Phase without any vehicle movements. This is an exclusive or 'all-round' pedestrian phase. Where pedestrian movements are grouped with vehicle movements it is done such that pedestrian movements run parallel to vehicle movements.

The PBUs associated with a pedestrian movement demand phases in a similar manner to a 'locked-call' vehicle detector demand. They also place a demand for the operation of the pedestrian movement. Pedestrian movement may be introduced:

1. When there are no demands for other phases.
2. When there are no conflicting vehicle movements running.

Within the controller, the active interval of the pedestrian movement may be divided into three periods called WALK (Green 'Man'), CLEARANCE 1 and CLEARANCE 2, (See Figure 7.12). When not active, the pedestrian movement display is Steady Red 'Man'.

Figure 7.12 Pedestrian Movement Operation



Normally, any Phase in which the Pedestrian Movement is introduced cannot terminate until the Clearance 2 period has commenced. The Clearance 2 period represents the greatest part of the pedestrian clearance.

As pedestrians take time to cross, the pedestrian clearance can represent a significant amount of 'inefficient' time. Therefore, pedestrian time is usually introduced only by demand.

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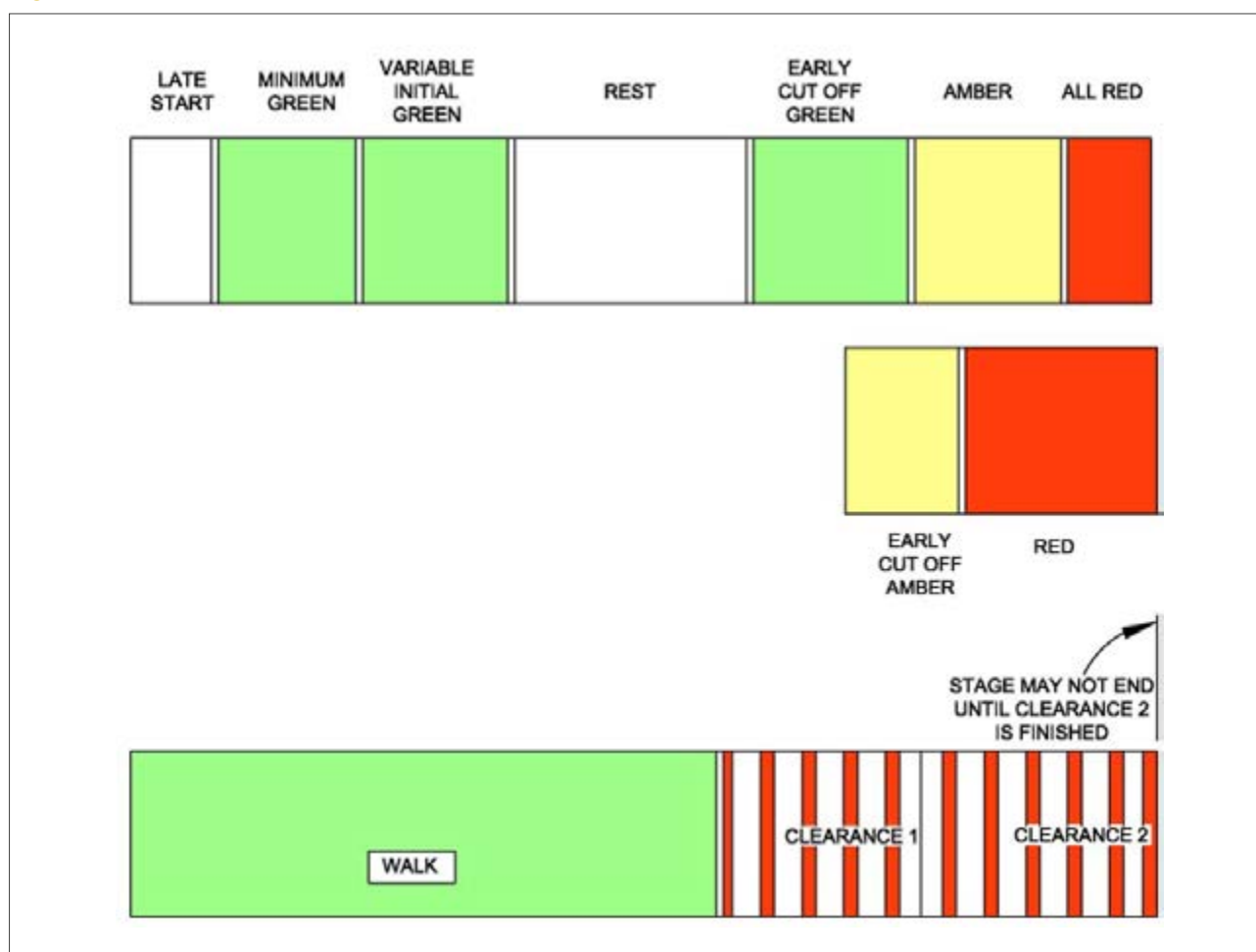
Determination of Signal Phasing (or Staging)

Pedestrian movements may be terminated by other conditions like:

1. The occurrence of the end of the rest period. If the controller is on vehicle green, then the Green Man is terminated by a call from an opposing vehicle or opposing PBU. If the vehicle green is extended, the green man extends with it. This type of termination is called 'WALK for Green'.
2. The presence of an opposing phase demand. The Pedestrian movement is terminated after the 'minimum' Green 'Man' time. This type of termination is called demand terminated.
3. The presence or release of a special signal from a linking plan.
4. On the arrival of a dedicated signal or Phase command from a master computer called command termination.

In addition, pedestrian movements may overlap with vehicle signal groups.

Figure 7.13 'WALK for Green' Operation



7.19.3 Protected Pedestrian Movements and Conditional Movements

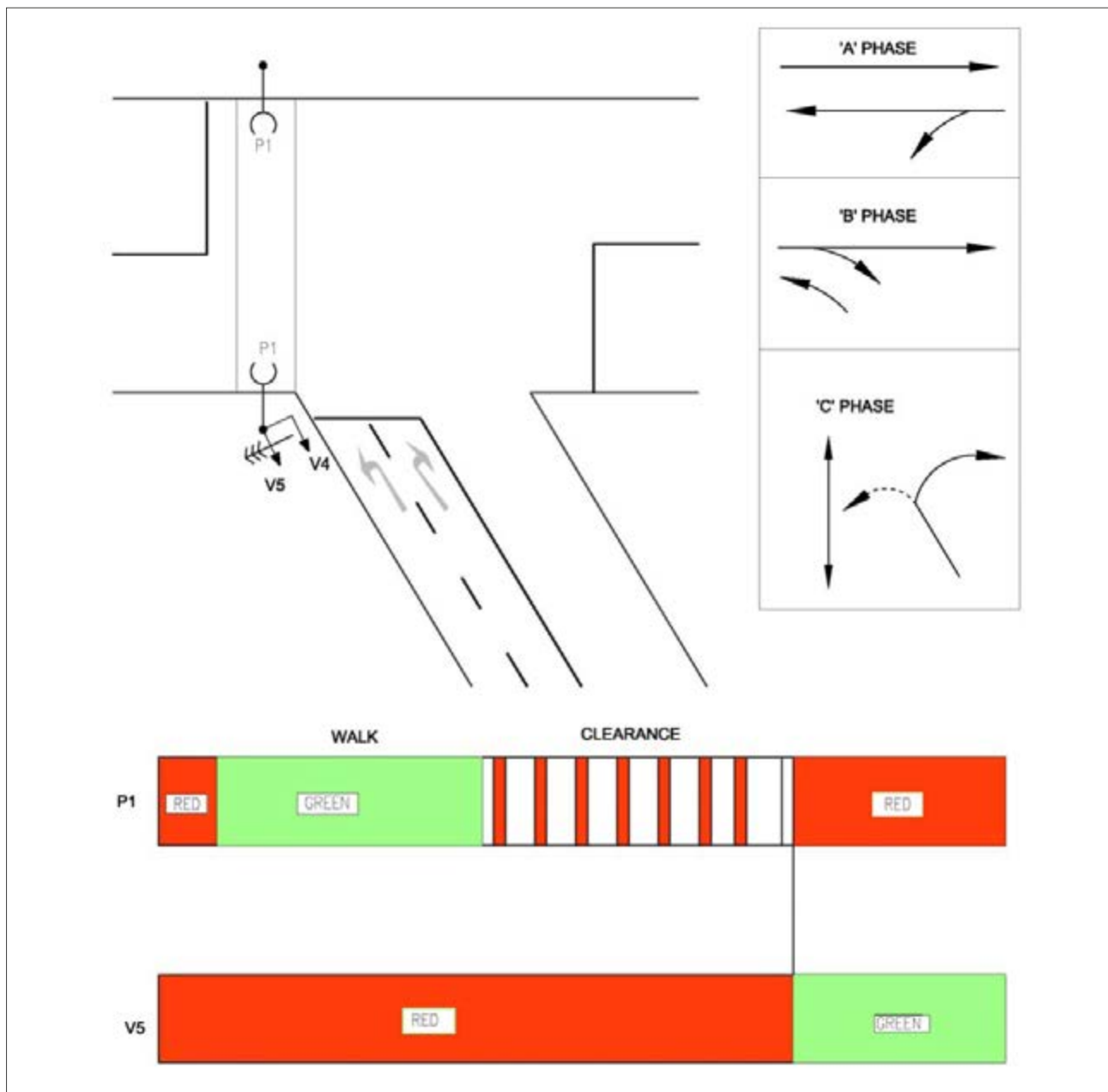
Pedestrian movements do not always run with its associated phase. It may run for part of a phase. It is more efficient to allow vehicle movements to run when pedestrian movements are not running. Depending on the severity of the conflict between a pedestrian and vehicle movement, four control schemes can be followed:

1. Allow vehicle movement to filter as necessary through Pedestrian movements.
2. Hold vehicles on a Red Arrow for part of the Green Man then allow filtering. The LT or RT Red Arrow may be timed by a special Red Arrow timer.'
3. Hold vehicles on a Red Arrow for the duration of the Green Man, allow filtering during the clearance periods.
4. Hold vehicles on a Red Arrow for the full Green Man and clearance periods.

For the four conditions, the signal group is said to be a conditional signal group.

The phasing is shown in Figure 7.14 and Figure 7.15.

Figure 7.14 Fully Protected Pedestrian Movement



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Determination of Signal Phasing (or Staging)

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7.19.4 Pedestrian Movements at Intersections

Pedestrian signals are traffic signals intended for the exclusive purpose of controlling pedestrian traffic. Their objective is to assist pedestrians to cross safely with the minimum delay to traffic.

The following points apply in the design of pedestrian movements:

1. Green Man display is based on PBU demand. When pedestrian movements are heavy, automatic introduction of the Green Man can be introduced by the controller.
2. Green Man is introduced with the parallel vehicle movements. If pedestrian - vehicle conflict is high or the speed of approaching vehicles is high, Red Arrow control is required to protect pedestrians.
3. Reintroduction of the Green Man can be done when there is sufficient time left in the phase to provide adequate Walk and Clearance times.
4. Slip Lanes: Pedestrian signals across a LT slip lane are recommended where pedestrian volumes are high or when pedestrians require additional protection, or where there are two or more lanes to cross.
5. 'Staggered' Crossing: Staggered pedestrian signal crossings can be considered where the central median is large enough to allow adequate area for pedestrians to wait. It is suitable for very wide roads, where long pedestrian Green Man and Clearance times would cause unacceptable vehicle delays.

7.19.5 Mid-block Pedestrian Signal Crossings

At a Mid-block pedestrian signal crossing, all vehicles are held on Red during the pedestrian Green Man and Clearance time.

If the crossing is a Pelican site, vehicles may proceed on the Vehicle Flashing Amber signal, provided there are no pedestrians on the crossing. This helps to reduce vehicle delays.

In the case of staggered pedestrian crossings, the median widths should provide adequate standing area, and the crossings should be offset.

7.19.6 Pelican Crossing Signals

These are independent pedestrian crossing signals, not part of a junction signal installation. These are often installed in busy suburban areas, or near schools or railway stations, or mid-way between intersections.

Pelican crossings control both vehicular and pedestrian traffic. They can operate on the same signal as a major junction (RM – GM – FRM – RM). They can also operate on a Pelican sequence, where the motorists' amber signal flashes along with a pedestrians' Flashing Green Man.

The decision on whether a Pedestrian crossing installation is to be provided is based on the warrant analysis which includes safety and other site related considerations. The decision on whether to use a Pelican rather than a pedestrian crossing is based on road width, vehicle speed and pedestrian volumes.

An audible signal device should accompany the Green 'Man period.

Pelican crossings are generally not suitable if there are more than two lanes on either approach or if there is a central median or island.

7.19.7 Crossing Width

The minimum pedestrian crossing width for both pelican and pedestrian crossings is 2.4 m. An additional 0.5 m can be provided if the volume of pedestrians per hour is above 600 (average over the 4 hours of peak pedestrian use) up to a maximum of 5 m.

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Layout of Traffic Signal Equipment

8 Layout Of Traffic Signal Equipment

8.1 Introduction to Pole Placement and Lantern Alignment.

This chapter covers the placement and positioning of signal equipment at the signal site, along with the orientation and placement of poles, lanterns, and PBUs.

8.2 Arrangement of Signal Lanterns (Faces)

The arrangement and classification of signal lanterns (faces) with respect to the type of approach are given below.

The types of approaches are:

1. Four arm junction with no central median (Undivided approaches).
2. Four arm junction with central median (Divided approaches).
3. T-junction with a central median (Divided approaches).
4. A mid-block pedestrian crossing (Undivided approaches).

Central medians allow placing of additional poles, signal lanterns and placing split pedestrian movements. Signal lantern arrangements are shown in [Figure 8.2](#)

8.2.1 Primary Signal Poles

The Primary signal lantern is mounted on the primary signal pole, usually 1 m past the stop line on that approach. The position is shown in [Figure 8.1](#).

8.2.2 Secondary Signal Poles

The Secondary signal lantern is mounted on the secondary signal pole on the far side corner, usually diagonally opposite the primary signal pole.

For traffic coming from the opposite direction, the primary pole becomes the secondary pole. The idea of primary and secondary pole is relative to the direction of travel. The positions are shown in [Figure 8.1](#).

8.2.3 Tertiary Signal Face

The tertiary signal lantern is mounted on the tertiary signal pole on the far side corner, on the same side of the road (left side where people drive on the left). The position is shown in [Figure 8.1](#).

8.2.4 Dual Primary Signal Lanterns

Dual primary lantern is placed on dual secondary pole, on the opposite side of the primary pole. The position is shown in [Figure 8.1](#). It is not recommended for two-way undivided street.

For one-way street, primary poles should be placed on both sides of the street. Secondary signals should also be used at the junctions of one-way streets.

On divided approaches, dual primary pole should be placed on the median.

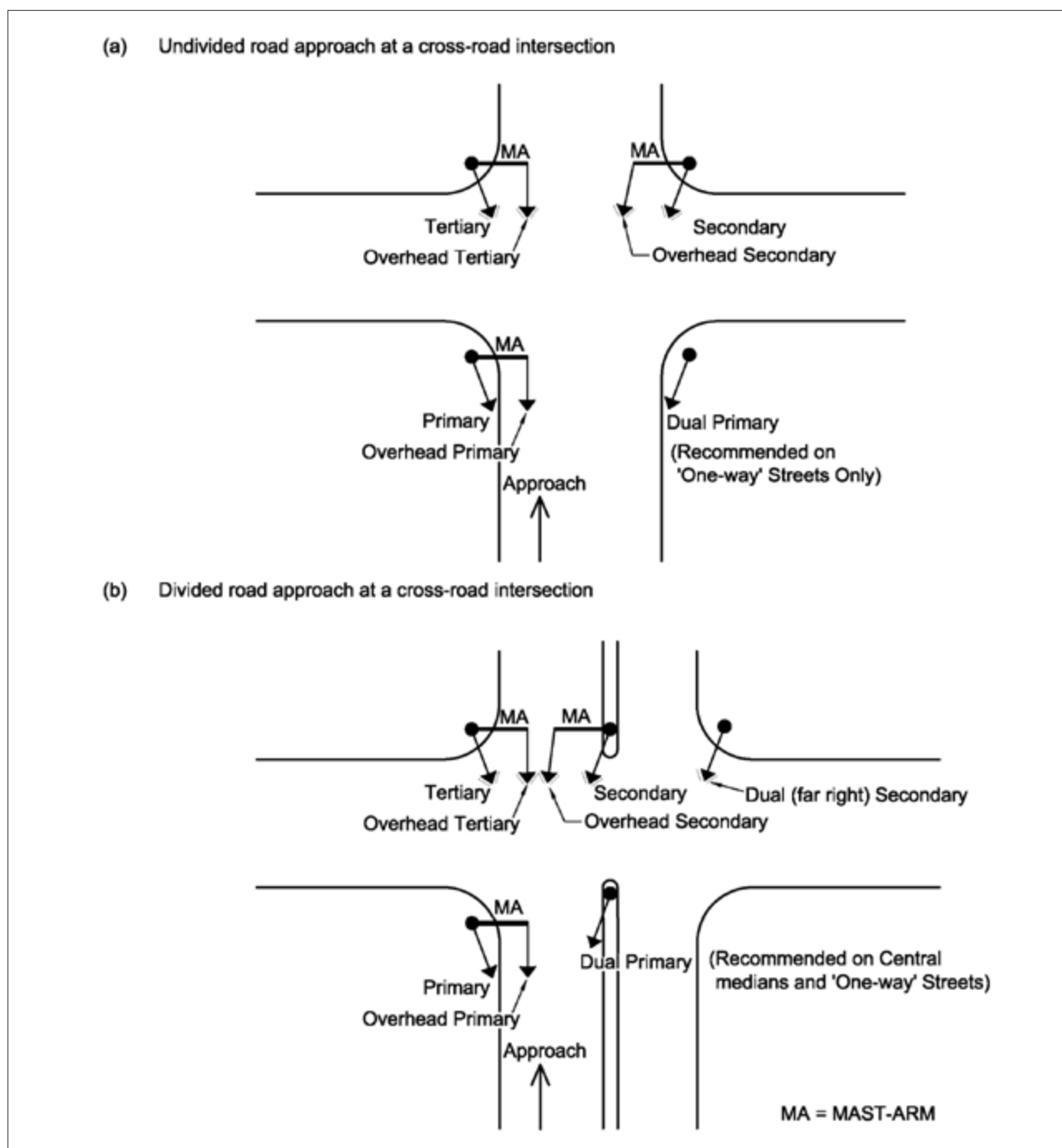
8.2.5 Dual Secondary Signal Lantern

The dual secondary signal lantern is mounted on the secondary signal pole on the far side corner, diagonally opposite the primary signal pole. If there is a median, the secondary pole should be placed on the median and the dual secondary pole on the far side diagonal corner.

8.2.6 Overhead Signal Lanterns

Signal lanterns mounted on mast-arms above the roadway are designated as Overhead Primary, Overhead Secondary and Overhead Tertiary Signals.

Figure 8.1 Signal Pole & Lantern Arrangements Four-Way Intersections



8.3 Signal Lantern Functions

A signal layout design should consider the display requirements for each approach on a case by case basis. The different types of display are:

1. **Warning Display:** This occurs where the junction signals are obscured by a sharp bend or hill crest.
2. **Stopping Display:** This is the most important signal display. It is required to inform approaching drivers that they are required to stop.
3. **Starting Display:** This is required to inform drivers at the stop-line as to when they can proceed.
4. **Manoeuvring Display:** This is required for large area junctions, where drivers may get confused about their movement.

The function of each signal display is detailed in [Table 8.1](#).

Figure 8.2 Signal Lantern Arrangements for T-Intersections & Mid-Block Signal Crossings

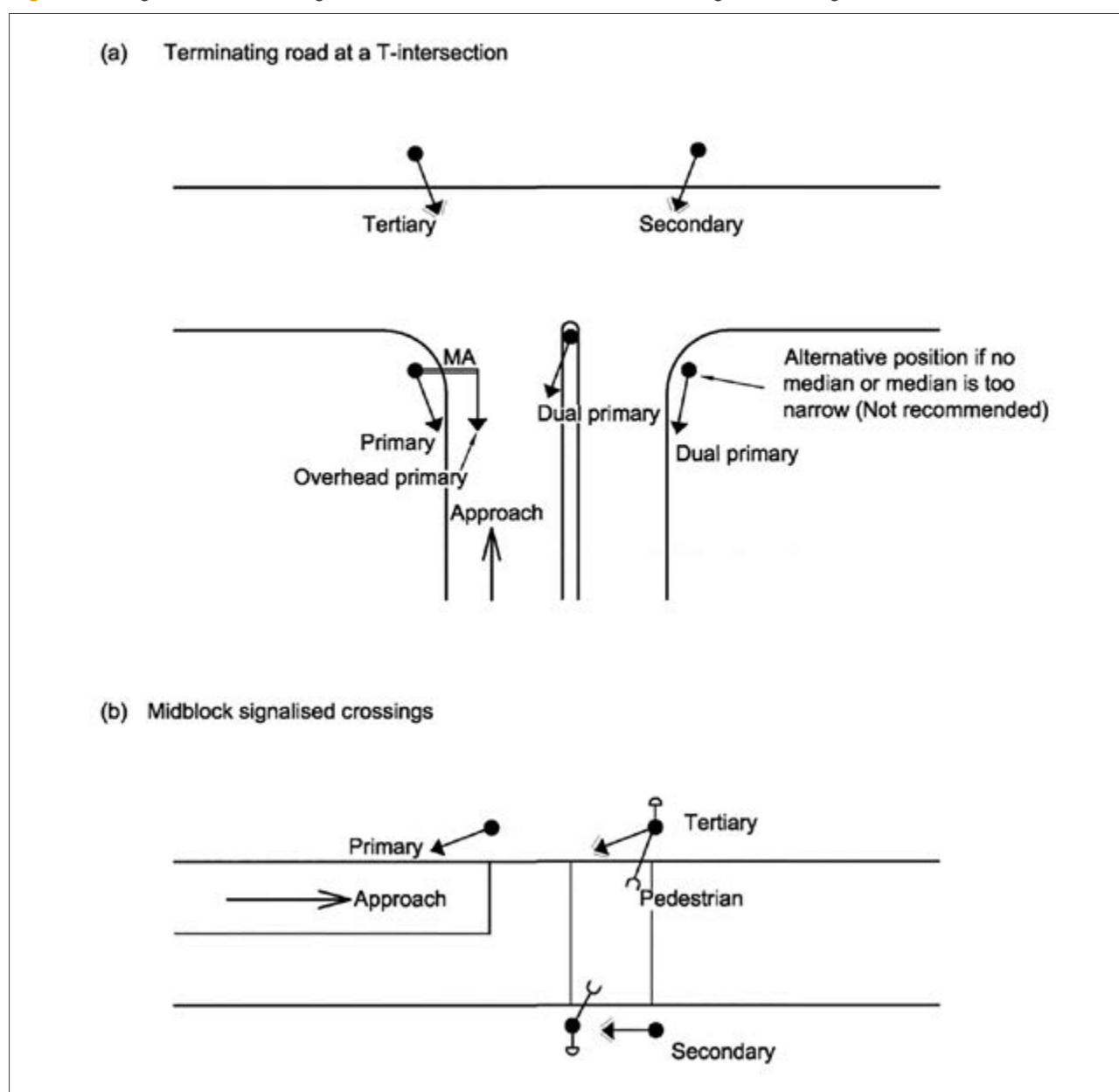
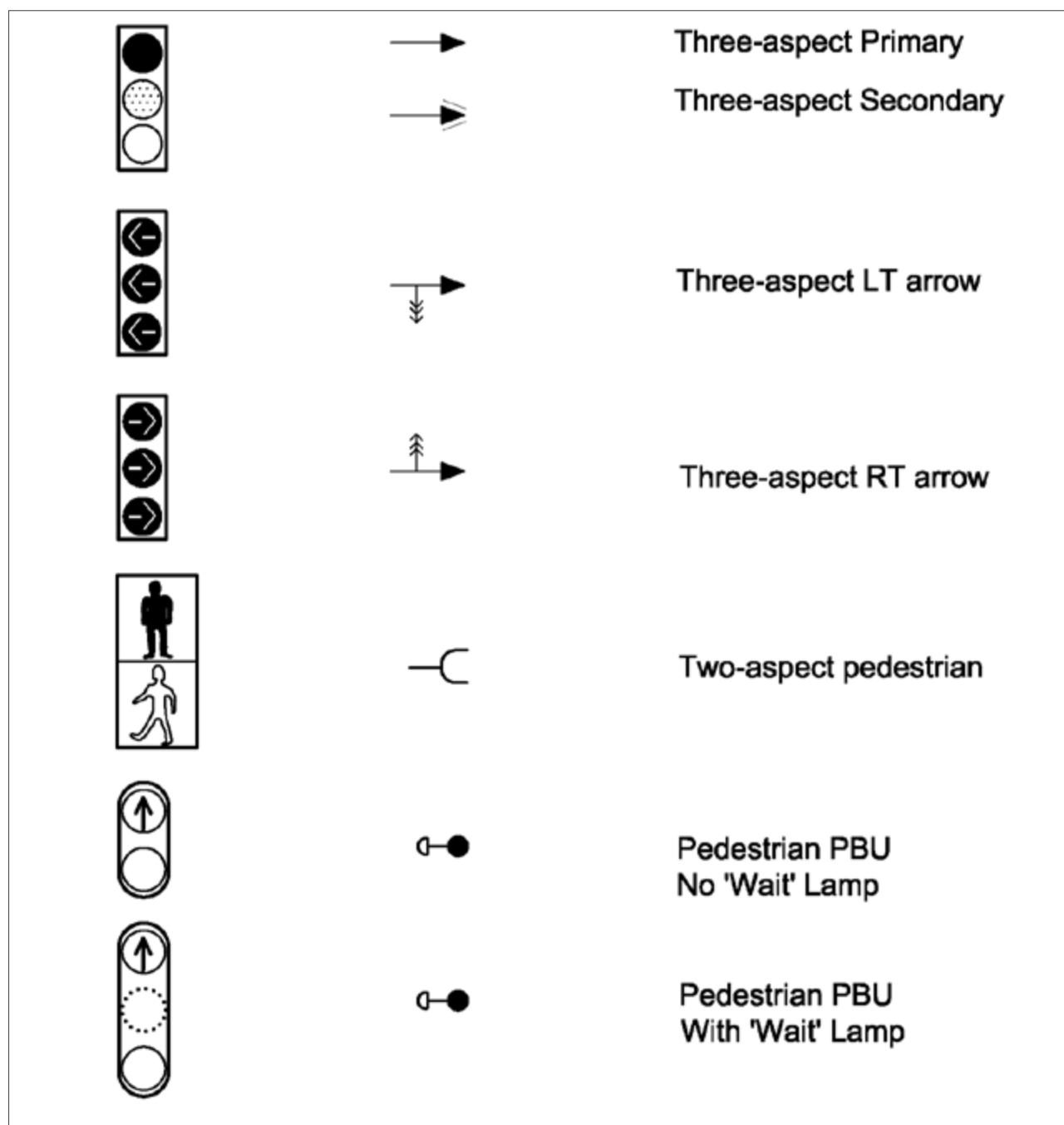


Figure 8.3 Symbols Used to Describe Signal Displays and PBUs**Table 8.1** Signal Face Functions

Location of Signal Face	Warning	Stopping	Starting	Manoeuvring
Primary	Yes	Yes	No	No
Secondary	†	†	Yes	Yes
Tertiary	†	†	Yes	Yes
Dual Primary	Yes	Yes	No	Yes

+ These functions may be provided depending on site geometry, topography and other special conditions.

8.4 Signal Displays - Site Requirements

8.4.1 Recommended Minimum Number of Signal Displays

Usually there are two signal faces at an approach - primary and secondary. A third signal face is preferable for wide intersections, with multiple lanes and high approach speeds.

The minimum number of signal faces for each LT movement is two. Where the LT vehicles turn into a slip lane or island, the tertiary signal face may be located next to the lane or island.

For RT movement, two signal faces should be provided. They should be located as follows:

1. Primary and secondary locations on a divided road with medians of sufficient width.
2. In the overhead primary and secondary locations, where medians are absent or narrow.
3. In the primary and secondary locations if the above two conditions cannot be satisfied.
4. If there are secondary lower level signals and secondary overhead signals, RT aspects should be placed on both.
5. If there are two or more lanes making a RT simultaneously, a RT signal face should be placed in the dual (far-right) secondary location.

Too many poles and signal lanterns may cause confusion and may add to the visual clutter at the intersection.

8.4.2 Dual Primary Signal Displays

Provision of a dual primary signal face should be generally avoided, except in the case of a one-way street.

8.4.3 Mast-Arm or Overhead Signal Warrants

Mast-arm signal faces are required:

1. If the stopping sight-distance to the pole-mounted signal face is inadequate, e.g. sight obstructions because of trees, verandas, or, where the junction is subject to large volume of tall and heavy vehicles.
2. If the roadway is too wide for signal faces to be seen by approaching vehicles.

Overhead signals may need to be omitted where their use would cause confusion at another intersection in close proximity.

8.4.4 Overhead Signal Displays Placement

Mast-arms should ideally be placed in the primary position near the stop-line. Mast-arms can be used on one-way streets, when a dual primary pole and signal face cannot be placed.

It is not recommended to place mast-arms in secondary or tertiary positions unless additional starting and/or manoeuvring functions are required.

Mast-arms may also be used when additional RT Arrow aspects are required and where dual primary lanterns cannot be placed.

8.4.5 Pedestrian Signal Displays

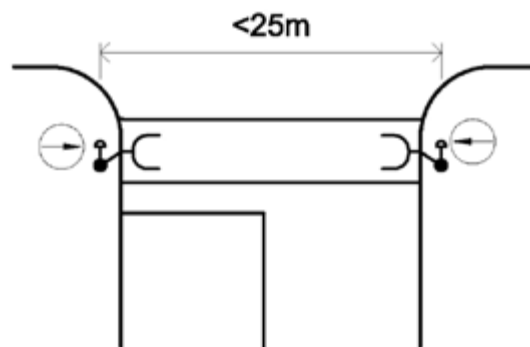
The following requirements apply for pedestrian crossings at junctions and mid-block:

1. A pedestrian signal face must be placed on each side a signal crossing. It should be aimed at the opposite side of the crossing.
2. Some modern pedestrian crossings on narrow streets do not have any pedestrian signal faces. The Green Man is visible on the nearside PBU.
3. For crossings of more than 10 m width, two pedestrian signal faces should be provided on each side of the crossing.
4. If the crossing distance exceeds 25 m in length, additional pedestrian signal faces should be installed on a median island.
5. Pedestrian signal faces on one movement should never be visible to persons using any other pedestrian movement, except in the case of an All-Round pedestrian movement.

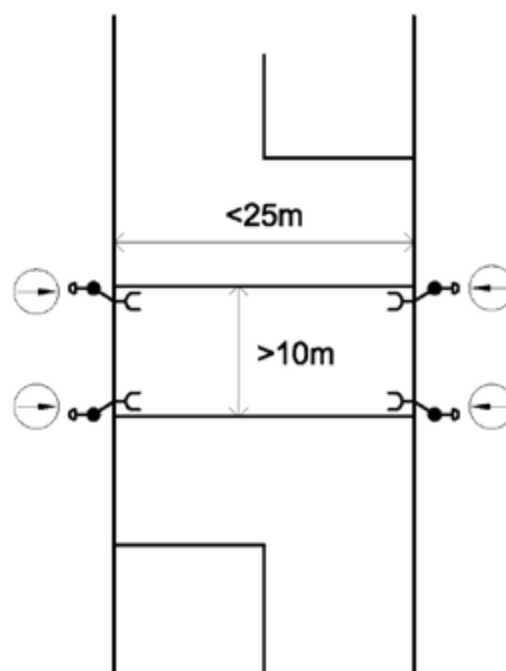
The positioning of pedestrian signal faces is shown in [Figure 8.4](#).

Figure 8.4 Pedestrian Signal Faces: Pedestrian Arrow Orientations Shown In (a) and (b)

a) Intersection signals crossing (in example, arrow legends point horizontally on the push button)



b) Wide midblock signal crossing (in example, arrow legends point upwards on the push button)



8.5 Positioning of Poles, Lanterns and Signal Lantern Displays

The ability of a driver to see signal faces is dependent on a number of factors such as signal colour, light intensity, aspect size, background luminance, as well as the position of the pole and signal in the driver's visual field.

Traffic signal poles should be placed judiciously in order to avoid knock-downs. In particular, mast-arms tend to be large rigid structures and should be placed where the probability of impact by vehicles is low.

The number of signal poles should be minimised in order to avoid visual clutter and to reduce cost. It is usual to place pedestrian and vehicle signals on a single pole.

8.6 Equipment Placement for Four-Way Intersections

The recommended positions for signal poles and mast arms are:

1. Standard signal poles and mast-arms should be placed at 1m from the kerb face (and stop-line), but not closer than 0.6 m.
2. Median poles should be placed centrally on the median. If the median is wide, additional poles should be placed at no more than 2m from the relevant kerb face.
3. Poles for primary and dual primary signal faces should also be placed approximately 1 m beyond the stop-line but not more than 3 metres. Poles should be placed not less than 1.2 m from the tip of an island or median.
4. Secondary signal lanterns and faces are placed opposite to the primary pole.
5. Poles for tertiary signal faces are placed on the 'left-far' side of the junction, or at a minimum of 2m from the corner and 1m from adjacent edges of LT slip-lane islands.

Visibility of lanterns is shown in [Figure 8.5](#).

8.7 Equipment Placement at T-Junction (Three-Way)

At T-junctions, poles (and mast-arms) for secondary signal faces should be placed to the right of the projection of the centre line (or median), but closer to the line of sight of the approaching motorist.

The preferred location of the signalled crossing is to the left of the stem as shown in [Figure 8.7](#).

If a common pole is to be used with a pedestrian signal face, then the placement of poles for secondary and tertiary signal faces is shown in [Figure 8.7](#).

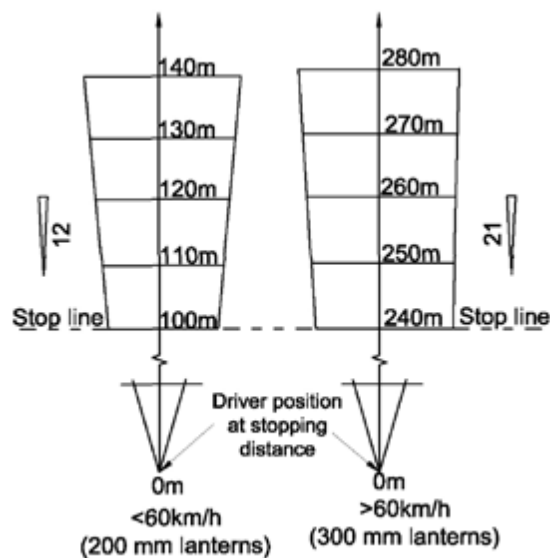
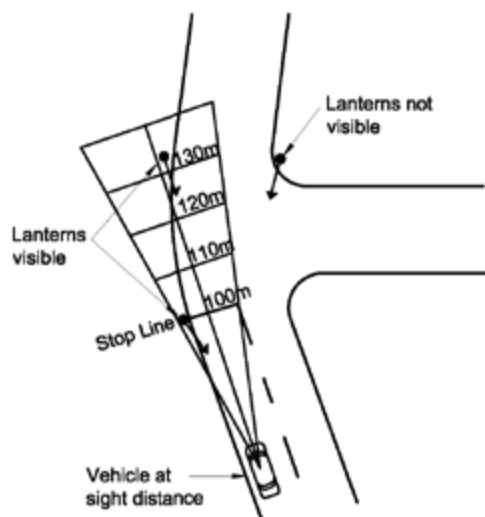
The Traffic Engineer will make placement decisions dependent on site conditions or immovable objects such as trees, drains, driveways, etc.

8.8 Equipment Placement at Mid-Block Signal Crossings

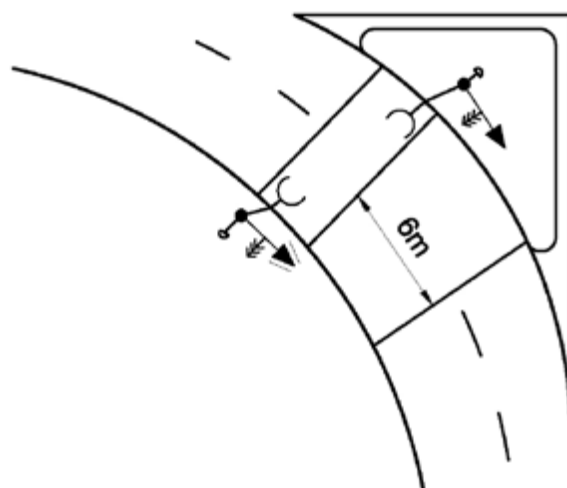
For undivided carriageways, the secondary signal faces are mounted with the opposing primary signal face on a common pole. For divided carriageways, secondary signal faces are mounted with the pedestrian signal face on a common pole located on the projection of the crosswalk line. See [Figure 8.7](#) and [Figure 8.8](#). For further details, refer to Traffic Signs Manual, Chapter 6, Department for Transport, UK and to Guide to Traffic Management Part 10: Traffic Control and Communication Devices, Austroads.

In the case of tertiary signal faces for both divided and undivided carriageways, the signal faces are mounted with the pedestrian signal face on a common pole located on the projection of the departure crosswalk line.

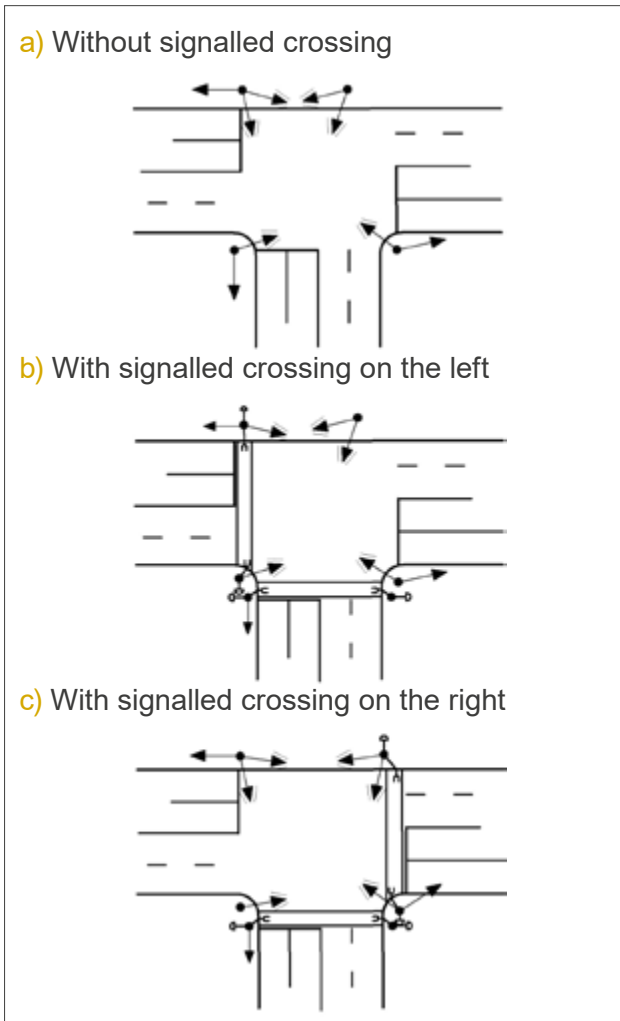
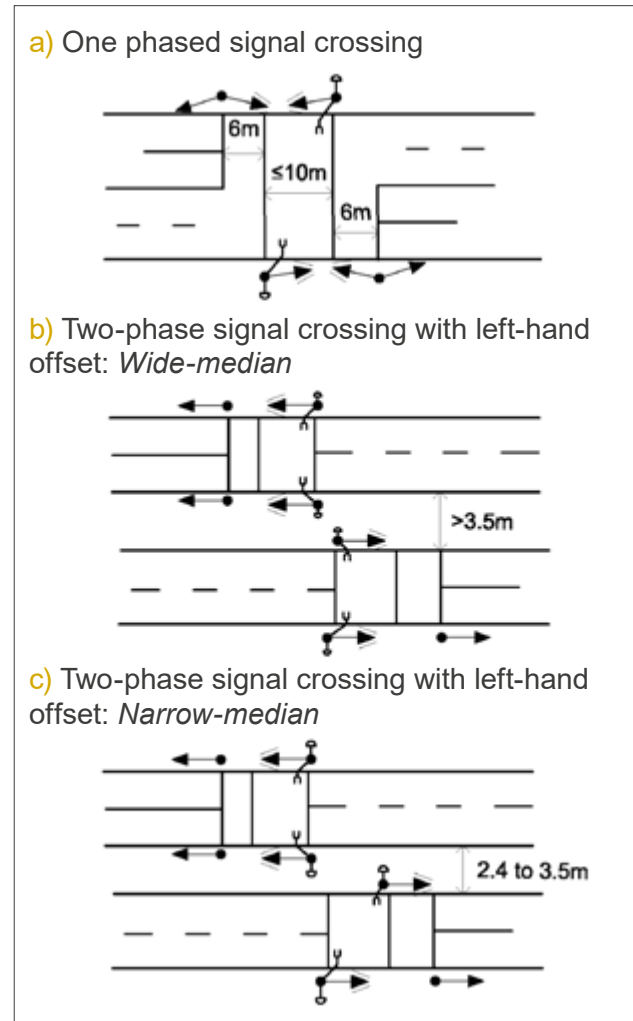
If mast-arms are used, tertiary signal faces may not be required.

Figure 8.5 Signal Lantern Visibility Templates**a) Visibility templates****b) Typical application of visibility template for stopping sight distance of 100 m**

Source: Guide to Traffic Management Part 10: Traffic Control and Communication Devices, Austroads

Figure 8.6 Signal Lantern Placement Channelled LT Control

Source: Guide to Traffic Management Part 10: Traffic Control and Communication Devices, Austroads

Figure 8.7 Intersection Signal Location**Figure 8.8** Midblock Crossing Arrangement

Source: Guide to Traffic Management Part 10: Traffic Control and Communication Devices, Austroads

8.9 Lantern Mounting Heights

The mounting height of a lantern is measured from walkway level to the top of the lantern body without target board.

The following guidelines apply to the placement of upright lantern bodies:

1. The normal height should be 4 m.
2. Primary lantern heights may be reduced to 3.2 m.
3. If there are obstructions such as awnings or trees, the mounting height should be a minimum clearance of 2 m.
4. For pedestrian lanterns the mounting height should be 3.2 m.
5. Mast-arm lanterns should be mounted with a minimum clearance of 5.5 m between the road surface and the lower end of the target board.

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Layout of Traffic Signal Equipment

8.10 Lantern Aiming

Lantern should be aimed at the traffic streams. The visibility range of each lantern is determined by its mounting position, photometric performance and its orientation. [Table 8.2](#) gives the visual coverage of lanterns. [Table 8.3](#) gives the aiming distance.

Primary signal lanterns should be aimed at a point 3 m from the stop-line at the centre of the approach. Manoeuvring signals should be aimed at the centre of the stop-line. The visibility requirement of a lantern should never be less than 8 m.

For signal lantern visibility template refer to [Figure 8.5](#).

Table 8.2 Visual Coverage of Lanterns

Aiming Distance	Visual Coverage	
Point Selected (metres from lantern)	Ground Mounted (metres from lantern)	Overhead (metres from lantern)
40	10 – 70	
60	25 – 95	
80	40 – 120	50 – 110
100	55 – 145	65 – 140
120	75 – 170	
130		90 – 170
140		105 – 195
150		125 – 225

Source: *Guide to Traffic Management, Part 9 Austroads*

Table 8.3 Recommended Aiming Distances

Approach Speed (km/h)	Stopping (metres from stop-line)	Warning (metres from stop-line)
40	40	80
50	60	100
60	80	130
70	100	150
80	120	170

Source: *Guide to Traffic Management, Part 9 Austroads*

Note: Refer to [Table 8.3](#) for lantern visibility template

8.11 Visor Types

Visors are used to cover the aspects of a lantern so that it cannot be seen by drivers from another approach. Visors restrict lateral vision. It is also used to shield the lantern from direct sunlight so that sun-phantom effect is not created.

Closed visors are used on secondary and tertiary lanterns. Long visors are often damaged by passing HGV or buses. Cutaway visors are used to provide additional visibility on the cutaway side.

The cut-off angle for various types of visors may be specified by the Engineer and can be checked with the lantern manufacturer. The cut-off angles are given in [Table 8.4](#).

Table 8.4 Cut-Off Angles for Visors

Lantern Aspect Size (mm)	Visor Type	Length (mm)	Angle for Total Cut-Off of Signal Indication
200	Open Type A	200	No restriction
200	Closed Type B	200	90°
200	Closed Type B	300	67°
200	Cutaway Type C	300	Open side see Fig 8.5 32° on closed side
300	Open Type A	300	No restriction
300	Closed Type B	300	90°
300	Closed Type B	400	74°
300	Cutaway Type C	400	Open side see Fig 8.5 37° on Closed side

Source: Guide to Traffic Management, Part 9 Austroads

8.12 Louvre Types and Placement

Louvres are used to provide visual restriction in the straight direction. Louvres should not be used on aspects with symbols. Horizontal louvres should be used in order to minimise the sun-phantom effect and to restrict signal visibility along multi – stop-line approaches.

Vertical louvres should be used to:

1. Provide more directional visibility where cut-off visors are inadequate, typically at skewed intersections.
2. Restrict the visibility to a certain lanes of an approach
3. Shield the aspects from the view of train drivers or other road-users who should not be able to see certain aspects.

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Layout of Traffic Signal Equipment

9 Traffic Detection

9.1 Introduction

An important part of traffic-responsive signal control systems is the detection of vehicle and pedestrian traffic demands. Detectors are used to register the presence and/or passage of vehicles and pedestrians.

Detectors can be grouped broadly as vehicle detectors, push-button (for pedestrian) detectors and special detectors.

Many types of detector exist including inductive loop, push button, microwave/radar, infrared, sonic, video image processing, magnetic and pressure. The most common detectors are the inductive loop detectors for vehicles and the push-button (PBU) detectors for pedestrians.

9.2 Vehicle Detection Systems

Vehicle detectors sense the presence of a vehicle and convey an electronic signal to the traffic signal controller, or centralised ATC computer. Vehicle detectors have two major modes of operation, passage and presence mode. A passage detector registers moving vehicles regardless of the speed or size of the vehicle. A presence detector sends a signal whenever there is one or more vehicle in the detection area.

Inductive loop is the most widely used detector. It consists of two parts: a loop and a sensor. A loop consists of three or four turns of wire embedded in the road surface. A sensor contains the electronic circuits. When a vehicle enters a loop, a message is sent to the sensor unit, which registers a vehicle.

Loop detectors are installed by cutting the loop pattern into the road surface, placing an insulated copper cable, then backfilling and sealing it with an epoxy resin. The looped cable ends are connected to the local traffic controller unit. One loop is placed in each approach lane. If the traffic volume is high, more than one loop may be required in each lane. In such cases, the spacing is guided by the approach speed and gradient.

Another way of detecting vehicles is by installing cameras on mast-arm. The camera can detect speed, volume and occupancy of vehicles.

9.3 Characteristics of Inductive Loop Detectors

The important characteristics of a loop detector are:

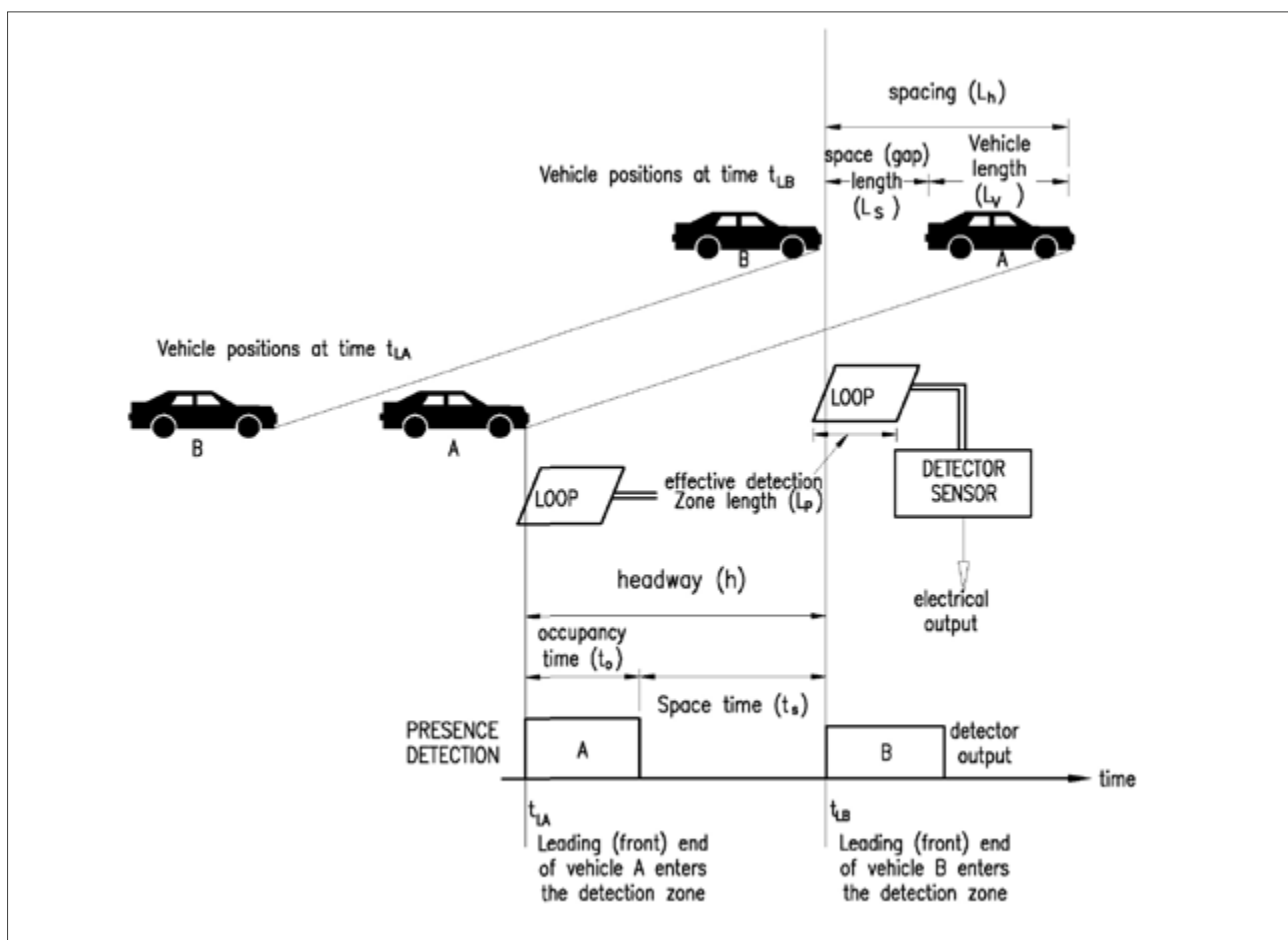
1. It gives one actuation per vehicle.
2. Detection may be passage or presence.
3. Detection area is the loop area.
4. Detection occurs over normal traffic speed range.
5. Installation is relatively simple, cheap and fast.
6. Sensor unit can be automatically self-adjusting.

The limitations are:

1. Loops are affected by environmental factors (temperature, rain, electromagnetic noise, etc).
2. Its sensitivity is limited. The more sensitive it is to vehicles, the more sensitive it is also to environmental factors.
3. As sensitivity is increased, unwanted detection of vehicles not in the loop area is a problem.
4. Presence time is limited (to about 20 minutes).
5. Loop wire is vulnerable to damage (loop wire needs to be close to surface for optimum sensitivity).
6. Distance between loop and sensor unit (controller) is limited to approximately 300 metres.
7. Some types of sensor unit channels 'interact' or 'crosstalk' with each other.

Figure 9.1 shows Inductive Loop Detection and Outputs.

Figure 9.1 Presence Vehicle Loop Detector System Relevant Measurements to Traffic Parameters



Source: Guide to Traffic Management, Part 9, Austroads

9.4 Traffic Detection During the Signal Cycle

During a signal cycle, traffic can be detected under two considerations:

1. Initial detection: Arriving traffic (vehicle or pedestrian) at a red signal registers an initial demand.
2. Subsequent detection: Vehicle traffic approaching a green signal registers that it requires the green signal to continue.

The type and location of detection systems used determines the parameters of the initial and subsequent detections.

9.5 Vehicle Actuation – Detector Demands

In a Vehicle Actuated (VA) controller, vehicle detectors are used to introduce a phase and/or determine the duration of phases.

When a traffic signal controller receives an actuation from a vehicle, a demand is placed for a phase which contains the movement. There are three types of demand:

1. Locking.
2. Non-locking.
3. Presence timed.

These are illustrated in [Figure 9.2](#). A locking demand is one where a demand is dependent on a vehicle which registers an actuation. This type of demand is mandatory for passage detectors.

A non-locking demand requires the detector actuation to be maintained until the phase is serviced or the demand is lost. This is used where a vehicle may legally escape either by left turn on Red or by filtering.

A presence timed demand is a non-locking demand with the additional condition that the vehicle must be present for a time period (typically 2-3 seconds) before a demand is placed. This type of demand is used where a lane is shared by vehicles. The purpose of the time delay is to ensure that moving vehicles do not register a demand unless their speed is below a desired value.

9.6 Vehicle Detection Modes

The sensor units for vehicle loop detectors have a switch to allow the detector to be operated in one of two modes.

1. Presence mode: In this mode, the sensor unit produces an output whenever a vehicle is detected. The duration of the output depends on the length and speed of the vehicles. Both moving and stationary vehicles can be detected. Presence detectors can measure flow rate, occupancy time, space time, saturation. It is the preferred type of detection.
2. Passage mode: In this mode, the sensor unit produces a brief pulse when a vehicle enters the detection zone. It detects only moving vehicles irrespective of length or speed. Passage detector can measure headway and flow rate.

[Figure 9.1](#) shows the basic traffic parameters that are measured by a loop detector system. These are:

1. Spacing (distance).
2. Headway (time).
3. Occupancy time.
4. Space (or gap) time and distance.
5. Vehicle length.

Other parameters such as speed, flow rate and density, can be calculated by the controller and ATC traffic system.

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9.7 Inductive Loop Vehicle Detection

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9.7.1 Detector Types

At intersections, detectors are meant to detect traffic on the lane where it is installed. Detection zones must be clearly defined so that interference from other lanes do not occur. Inductive loops are the most widely used type of detector. Loop detectors vary in terms of placement and size of loops.

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9.7.2 Other Detection Devices and Methods

Other Devices and methods of detection are:

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1. **Microwave Vehicle Detection (MVD):** Detection occurs when a beam of microwave is disturbed by a moving vehicle. The beam is usually emitted from a device on the primary traffic signal pole towards an approaching vehicle. The microwave device is generally fixed on top of the signal head or below the green aspect. The disadvantage of MVD is that stopped and slow moving vehicle are not detected.
2. **InfraRed Vehicle Detection (IRVD):** Infra Red light unit is placed above the signal head. Vehicles travelling through the detection zone are detected. Use of IRVD is limited. It has the same disadvantage as MVD.
3. **Video Detection Imaging (VDI):** is carried out by a camera device mounted above the carriageway, usually on a mast-arm. Stationary or moving vehicle images are captured and processed by the roadside controller or remotely at the traffic control centre. VDI technology is now popular.

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9.7.3 Demands for a Phase

When a detector is activated by a vehicle movement the traffic controller registers a 'call' for a Phase. This is usually made for Vehicles arriving at a Red signal and that are waiting on a green signal. The demand is stored until the green signal commences.

The location of detectors for stored demands (or 'locking calls') is chosen on the basis that only Vehicles serviced by the demanded Phase of a particular 'loop' are detected.

Presence or passage detection mode can be used for a particular 'loop'.

The demand can be a non-locking call, in which case the call is not stored, but cancelled when the vehicle leaves the loop area.

Non-locking calls are used for approaches or lanes where vehicles may escape before the called phase occurs.

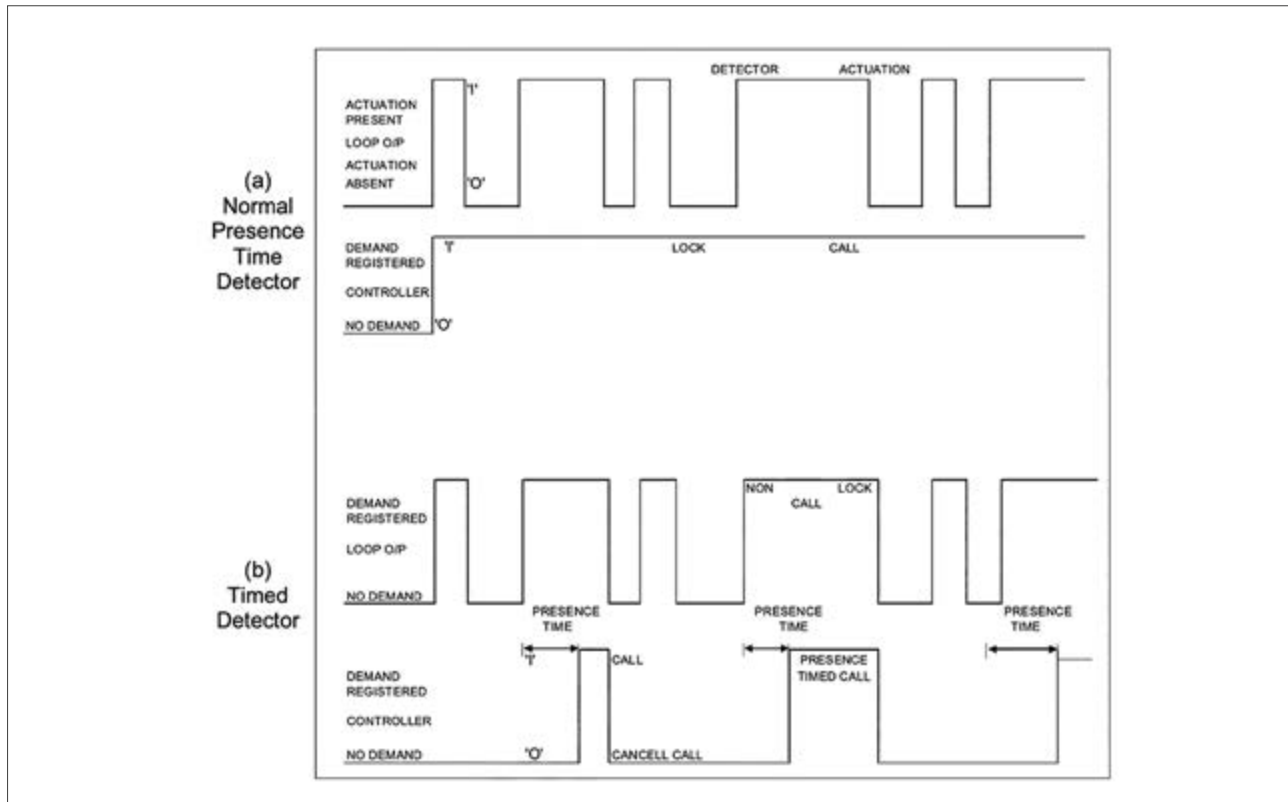
Vehicles may leave a detection area under the following conditions:

1. A RT movement is no longer needed when RT Vehicles Filter away before the RTGA commences.
2. At a 3-Phase T-junction where LT Vehicles Filter Left in 'B' Phase and there are no Vehicles waiting to RT in 'C' Phase.

Non-locking calls can only be made by presence mode detection.

Figure 9.2 shows the Inductive Loop output and Controller Interpretation.

Figure 9.2 Detector and Demand Operation



9.7.4 Green Extension

When green signal is in operation, the detector demand for green signal is not stored. The detector demands are used to extend the green time. The time extension is by 3 secs per vehicle detected. If there is a continuous vehicle stream, the green time is extended indefinitely, unless vehicles on other approaches lock a demand.

9.7.5 Detector Functions in an ATC System

In a wide area or centralised traffic control system controlled by a master computer, detectors can be assigned to operate in two modes:

1. **Tactical:** Detectors call and extend green signals for the local controller only.
2. **Strategic:** Detectors count vehicles and measure the gaps between the vehicles in order to enable the central computer to calculate the optimum duration of cycle length, phase splits and signal offsets for system coordination control.

9.7.6 Inductive Loop Shape and Size

Detection is defined by two factors which are linked to loop shape and size:

1. **Longitudinal Response:** The length of a loop is critical. If too short, the magnetic field may not be effective. If too long, the individual vehicles and the distance between them cannot be detected.. The length depends upon traffic composition. It varies between 1 m (for smaller vehicles) and 12 m (for HGVs and buses). The most effective loop length is 4.5m.
2. **Transverse Response:** The width of a loop is determined by the width of the. The width may also depend upon the dimension of the target vehicles.

Various loop shapes are shown in **Figure 9.3** are:

1. Rectangular.
2. Symmetripole.
3. Quadrupole.
4. Slanted.
5. Double Diamond.

The various applications of the rectangular loop are shown in **Figure 9.3**, including its use in a single lane and across several lanes. The slightly rotated loop (angle of 20 degrees) is more effective in detecting cyclists.

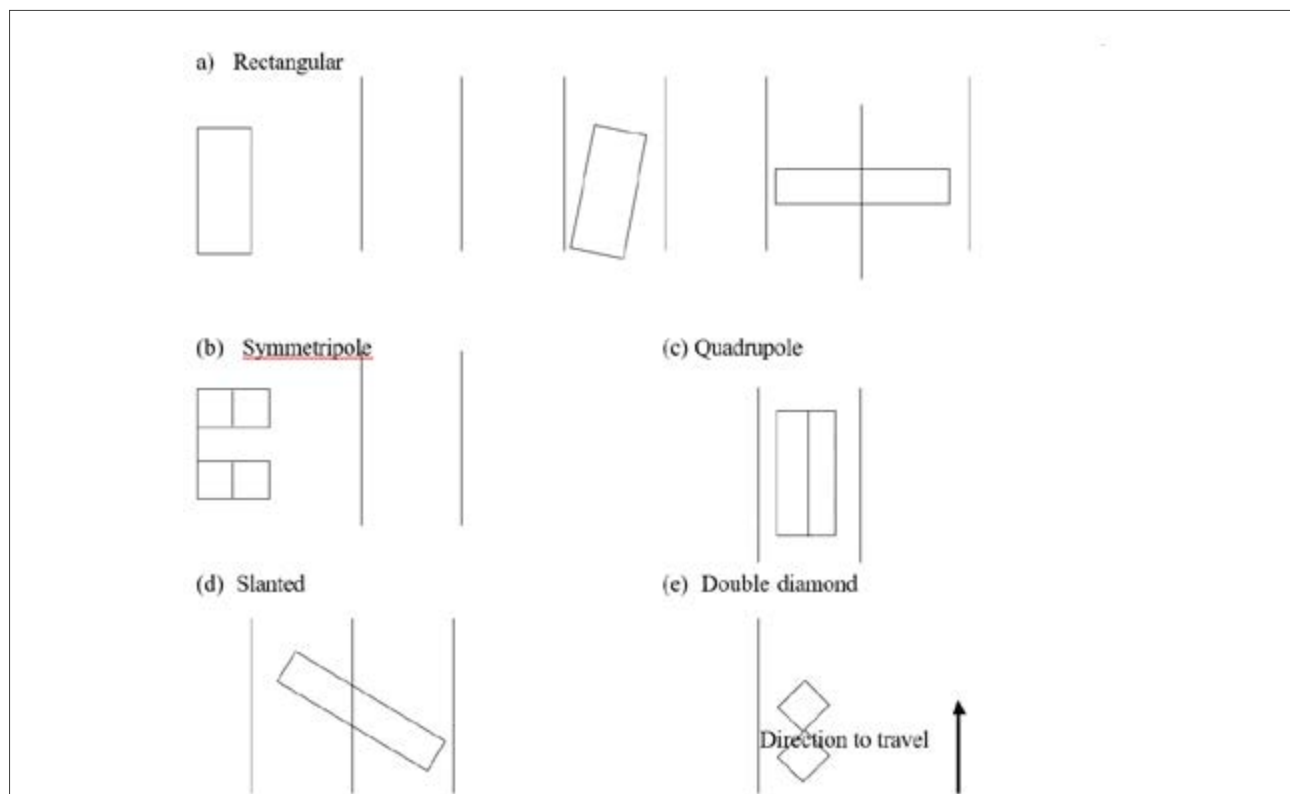
Commonly used loop shape is the Symmetripole. The Quadruple has similar characteristics as the Symmetripole, but the detection zone is a little bit less concentrated than the Symmetripole. The Quadruple loop is easier and cheaper to install and is less likely to damage the road surface.

The slanted multi-lane loop shown in **Figure 9.2** is used extensively for Isolated junction control.

The loop size and vehicle dimensions are the determining factors for a particular system location. As vehicles vary in size, a compromise is required in loop size. In order to adequately detect small vehicles, a loop size of ten square metres is considered a practical maximum. Also, a loop in one lane should be free from interference by loops in another adjacent lane. The sensitivity of detection within and between lanes is determined by the placement, shape and size of the loop.

The efficiency of the detection system is usually maximised by placing one loop per lane, regardless of the type of detection system employed.

Figure 9.3 Typical Loop Shapes



Source: Guide to Traffic Management, Part 9: Traffic Operations, Austroads

9.7.7 Placement of Detectors

For any particular detection system, the location or placement of the detector affects its function. The main types of vehicle loop detector are:

1. Stop-line detectors.
2. Advance detectors.
3. Queue detectors.
4. Emergency Vehicle detectors.

9.7.8 Stop-Line Loop Detection

Stop line detection is used to detect presence of vehicles near the stop line. Greater sensitivity of loops is required for detecting slow or stopped vehicles.

The requirements of stop line detectors are:

1. The setback distance is usually 1.5m but varies from 0.3m to 2.5m.
2. The loop length is generally 4m to 4.5m.

The placement and layout of 4.5m loops, and the recommended loop configuration is shown in Figure 9.4.

The recommended loop 'widths', which are dependent on placement from kerbs and medians, is given in Table 9.1.

Table 9.1 Lateral Dimensions of Symmetric Pole

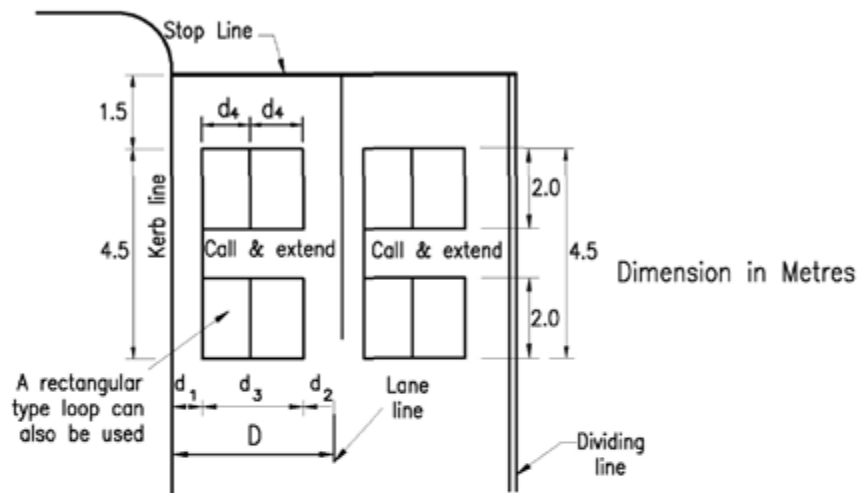
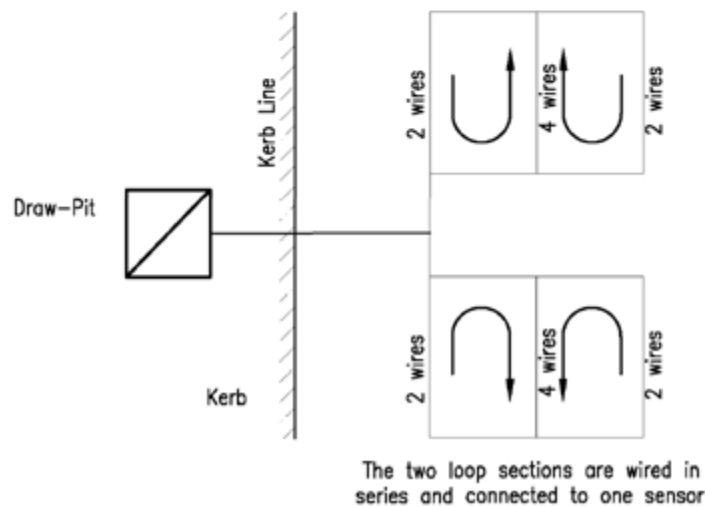
Loops Spacing from Each Outer Wire of Loop to Lane Edge*			
Lane Width (m)	General (m)	When Adjoining	
		Median Strip (m)	Kerb Side (m)
2.5 - 3.0	0.74	0.55	0.83
3.0 - 3.5	0.73	0.54	0.82
3.5 - 4.0	0.72	0.53	0.81
4.0 - 4.5	0.71	0.52	0.80
4.5 - 5.0	0.70	0.51	0.79
5.0 - 5.5	0.69	0.50	0.78

* d1 and d2 in Figure 9-4.

Source: Guide to Traffic Management, Part 9: Traffic Operations, Austroads

Figure 9.4 Stop-Line Loop Configuration and Layout**(a) Lay out of 4.5m Loops**

D = lane width
 d_1, d_2 = distances from loop wires to lane edge
 $d_3 = D - d_2 - d_1$
 $d_4 = 0.5d$

**(b) Configuration of loop**

Source: Guide to Traffic Management, Part 9: Traffic Operations, Austroads

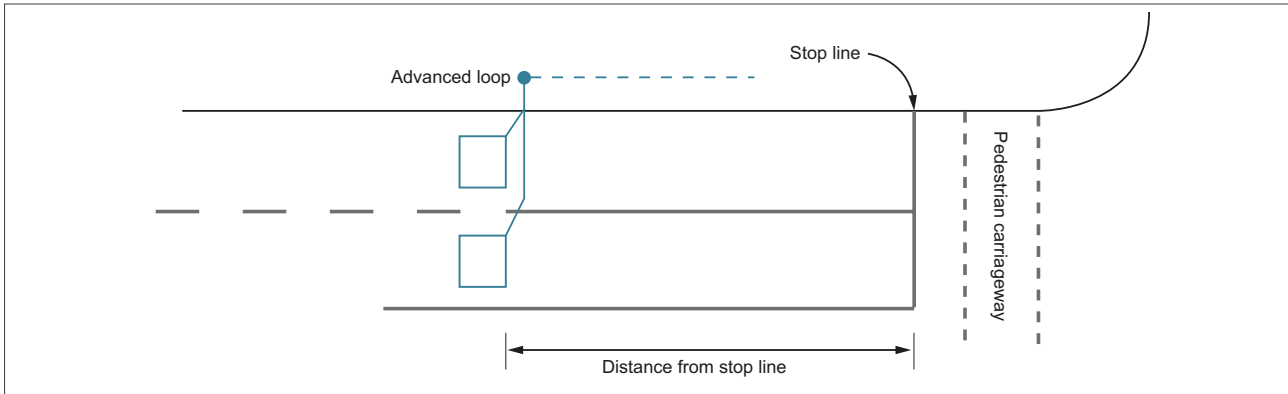
9.7.9 Advance or 'Passage' Detection

Advance detector loops are used to supplement stop-line detectors and can be set in presence or passage mode. Advance detector loops may also be used without stop line detectors as shown in Figure 9.5.

The following considerations apply to advance detector loops:

1. If advanced detectors are placed too far back from the junction, they cannot accurately detect the vehicles.
2. If vehicles enter or leave the junction in the area between the advance detector and the stop-line, they cannot be counted.

Figure 9.5 Advance Loop



9.7.10 Queue Detection

'Timed' presence detectors are used to detect queues. If the timer is set for 5 seconds, when the loop is occupied for 5 secs or longer, a demand is registered. If the vehicle passes the loop within the 5 secs, no demand is registered.

Queue can be detected:

1. If a RT queue develops in a RT bay and is too long to adequately Filter (see).
2. If a RT queue develops at the centre of the junction owing to lack of gaps in the oncoming traffic. A timed presence detector at the stop-line can call for a RT Green Arrow.
3. Queues across a railway level crossing.

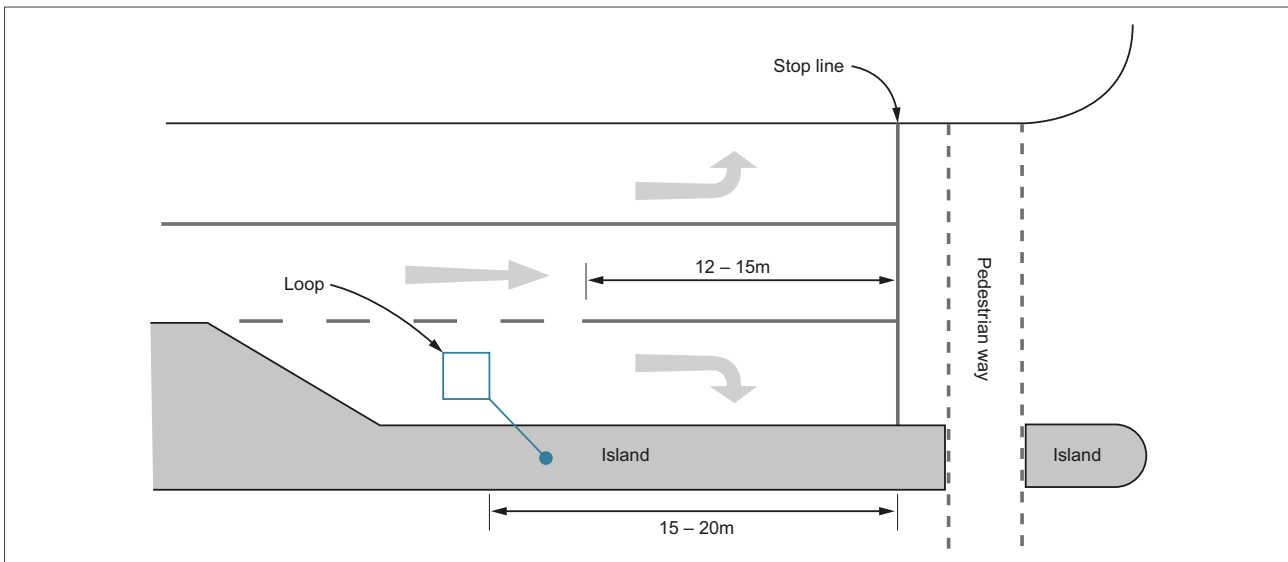
9.7.11 Operational Characteristics of Loop Detectors

The recommended loop layout and configuration for stop-line detectors is shown in Figure 9.3. The spacing between the outer loop conductors and the lane boundary or kerbs should be chosen in order to minimise:

1. Unwanted detection of vehicles in the adjacent lane, called overcounting.
2. Missed vehicles such as small vehicles or two-wheelers which do not travel through the detection zone, called undercounting.

The optimum loop width is achieved when over-counting errors are equal to undercounting errors.

Figure 9.6 Queue Loop for Detectors Demanding a RT Phase



9.8 Pedestrian Traffic Demands

9.8.1 Pedestrian Detection

The most common detection is by PBU demand. The pedestrian must press the PBU, which stores a demand in the Controller Logic. The Controller activates a lighted panel indicating that the signal system is aware of the need to cross. The illuminated panel or 'wait' lamp switches off as soon as the Green Man commences.

If a substantial number of visually-impaired or elderly pedestrians use a particular crossing, then the signal PBU should be fitted with an Audio-Tactile sensor.

The presence of pedestrians on the crossing should be detected by additional detector units in the walkway or overhead (usually infrared or microwave) in which case the crossing time or clearances should be instantly modified, e.g. Puffin Crossings.

9.8.2 Automatic Introduction of Pedestrian Movements

In addition to PBU detection, controllers should be programmed so that pedestrian movements run in each cycle. This method is usually employed in urban areas where pedestrian volumes are high and where the cycle length is long enough in order to accommodate all phases with pedestrian movements.

Automatic introduction should be set in the local controller programme or by centralised control.

9.8.3 Pedestrian Push-Button Placement

Pedestrian and cyclist PBUs are mounted on traffic signal poles, mast-arms, or special short poles. Each pedestrian crossing normally has a PBU on each side of the road. If the road or crossing is divided (or staggered), then there will be separate PBUs on the median. A special short PBU pole should be installed if no traffic signal pole can be located in a suitable position. The PBU should be located not more than 1m outside the projection of the signalised crossing nor more than 2m back from the kerb line at the signalised crossing.

Refer to [Figure 9 7](#).

PBUs for a particular crossing should be placed so that there is no confusion as to which crossing is demanded by a particular PBU.

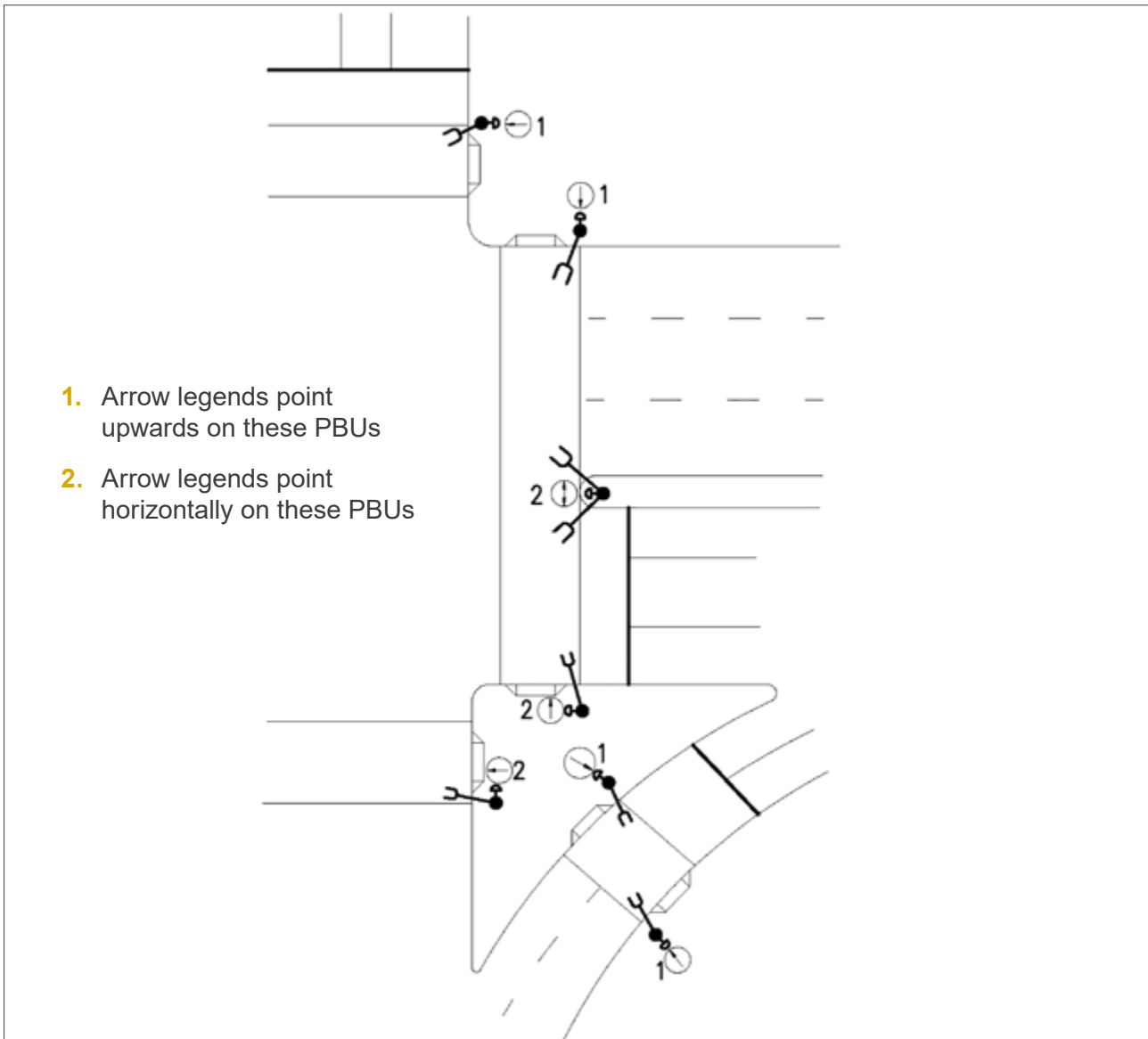
Height Above Walkway: The push button should be mounted at a height of approximately 1.1m $\pm$ 0.1m from the ground.

Orientation: PBUs should be mounted in a consistent manner and should be orientated as follows:

1. Perpendicular to the signalised crossing and facing away from the crossing, or parallel to the crossing and facing towards pedestrians.
2. In narrow medians, one push button may suffice and should be mounted parallel to the crossing.

Directional Arrow Legends: An arrow legend may be provided in a PBU to give guidance to people with visual disabilities and to benefit pedestrian and cyclists in general. The placement of arrow legend are:

1. Where the face of the PBU panel is at right angles to the crossing lines, the arrow legend should point upwards.
2. Where the face of the PBUs button is parallel to the crossing lines, the arrow legend should be horizontal.
3. Where one PBU is mounted in a narrow median, the arrow legend should have a horizontal double-headed arrow and the face of the PBU should be parallel to the crossing lines.

Figure 9.7 PBUs Location and Orientation

Source: Guide to Traffic Management, Part 9, Austroads

Audio-Tactile Units: These should be placed more than 2m from one another as the audible tone can confuse impaired pedestrians. Two PBUs with audio-tactile facilities must never be mounted on one pole.

9.8.4 Cycle Detection

Cyclists often share lanes with other vehicles. In that case it is not possible to ensure detection of cycles, owing to their narrow width. PBUs can be placed at selected points to assist bicyclists to lodge a demand.

Where separate cycle lanes are provided, special cycle sensitive loops can be used. The loops need to span the whole width of the cycle lane.

Cycles can be controlled by 2-aspect red and green signals with cycle masks, or by 3-aspect red, amber and green cycle masks.

At major junctions, a stop-line for bicycles should be placed beyond the normal vehicle stop line so that LT motor Vehicle and HGVs are aware of the presence of cyclists waiting for a green signal.

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Traffic Detection

9.9 Special Traffic Detection

Emergency Vehicles and other Modes of Transport may have to be controlled at a crossing or junction. Emergency Vehicles such as Fire Tenders or Ambulances need 'immediate' priority, whereas Transport Modes such as Trams and Buses might only get priority in certain circumstances. Trams and Buses will only get priority at certain times in the signal cycle. Railway traffic is treated differently insofar as trains will get absolute priority; however all other traffic must be stopped and thereafter released in a safe manner. Special detection methods are used in each case.

9.9.1 Emergency Vehicle Detection

Emergency vehicles such as fire tenders usually have a priority PBU within the fire-station. When activated, the nearest traffic signal junction holds all other traffic for a preset time until the emergency vehicle has passed. In some systems the activation of the fire PBU alerts the central control operators, which gives emergency vehicle priority along a certain route.

Emergency Vehicle can be detected by fitting a special transmitter called a transponder to the vehicle.

9.9.2 Buses

Transponders are often used on buses so that a bus is identified by a loop detector. If the bus shares lanes with other traffic, then transponder helps in identification bus. The transponder can identify the particular bus and relay information regarding schedule, in addition to other bus information. In some applications, a bus may only be identified if the transponder is activated by the driver. This is particularly useful for BRT operation.

Other techniques that can be used for detecting buses are:

1. Classification detectors to identify buses by special loop detectors.
2. Combination detectors to identify buses by its length and height.
3. Video based detection or number plate detection.

9.9.3 Trams / LRT

The techniques for tram / LRT control are:

1. **Loops:** Trams / LRT may be detected by inductive loops placed between the rails. There are usually many approach loops at junctions controlled by signals which are subject to passage, presence, and timed actuations. Some interface methods can be quite complex, depending on factors such as junction complexity, priority for different directions, proximity of stations, etc.
2. **Transponder/Transmitter Systems:** If trams / LRT need to be uniquely identified owing to shared road space with other vehicles, then inductive loops may not be able to differentiate trams sufficiently. In this case, it is usual to fit a transponder or transmitter system as described for buses and emergency vehicles and buses in [sections 9.9.1 and 9.9.2](#).

9.9.4 Tram / LRT Operation at Signal Sites

Operations may be categorised as passive and active priority measures. Passive priority measures imply that the signals are not aware of the tram presence. The signals are programmed to include reduced cycle time, and phase design has bias in favour of bus and tram movements.

Active public transport priority techniques at traffic signals include:

1. Phase extension - Holding the green display until the tram passes.
2. Hurry Call – start the green display when a tram arrives on red.

3. Provision of special tram phases at complex intersections.
4. Phase Skipping - a non-tram Phase is omitted from the sequence.

Tram / LRT detection by single frequency transponder allows loops to selectively detect trams in a mixed traffic stream.

9.9.5 *Railway Traffic*

Railway level crossings are frequently close to traffic signal junctions and therefore must be coordinated.

In the event of an approaching train:

1. Any traffic queued across the railway tracks must be cleared immediately.
2. Vehicles about to enter the level-crossing must be stopped.
3. If the intersection signals have not attained a 'safe' condition, and the level-crossing has not cleared, the traffic signals are switched to flashing amber.

9.9.6 *Railway Linking Requirements*

If linking of traffic signals with the railway level crossing is required, the railway authority should provide the necessary trackside equipment interfaces. Special queue-clearing sequence can be initiated before the flashing red train signals commence operation.

After the queue-clearing phase terminates, no phase is to be introduced for vehicular traffic until the train has passed. When the level crossing opens, priority should be given to the held traffic, before the traffic signal control reverts to normal operation.

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Traffic Detection

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Traffic Signal Controllers

10 Traffic Signal Controllers

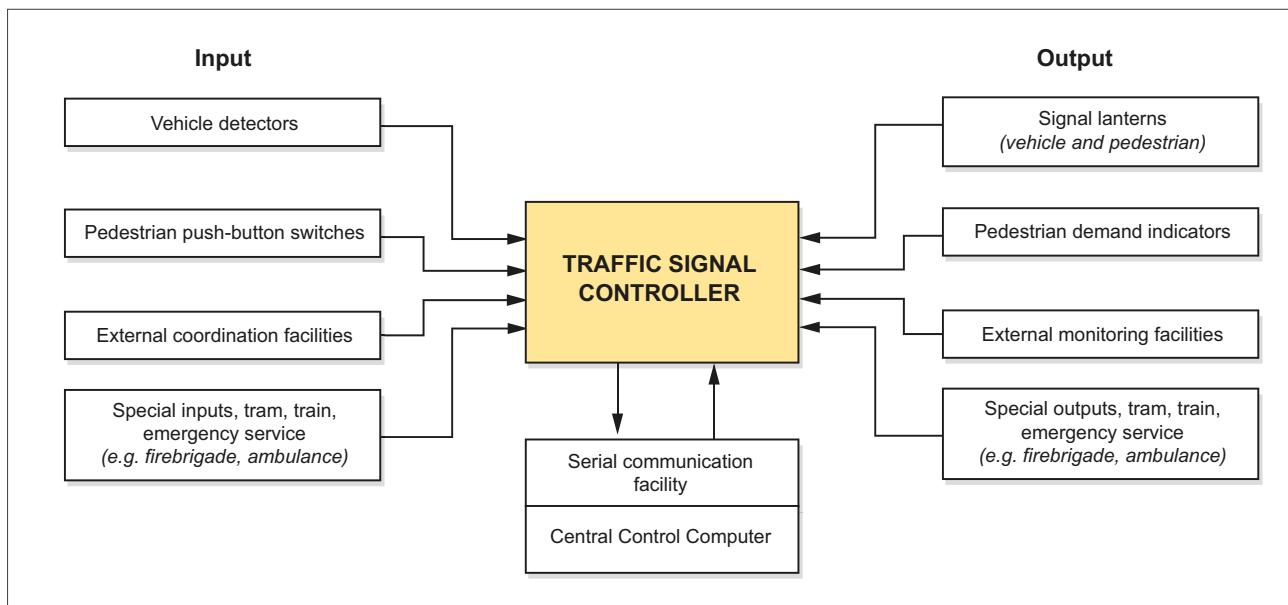
10.1 Introduction to Traffic Signal Control

Traffic Signal Controller operates traffic signals. Input and output of Controller are show in Figure 10.1. The various control facilities available in the signal controller determine the nature, duration, and sequence of signal displays. The controller hardware required is determined by the number of separate traffic movements, detectors, and the number of PBUs, which are also related to the intersection geometry, the phase design, and as to whether or not the controller may be connected to centralised control.

Controller is housed in a metal casing with an access door. This equipment is usually placed on a concrete pedestal, but may also be mounted on a pole.

The casing contains a logic rack that switches power to the signal lanterns and controls the duration and sequence of signal displays. It controls detection equipment and may also communicate with a central master computer.

Figure 10.1 Typical Interfacing Requirements for Traffic Signal Controllers



10.2 Local Controller Operational Modes

A Traffic Signal Controller operates in three basic modes: isolated, linked, or under Central Computer control. It may have many other modes available, such as control of emergency vehicles or manual intervention. There are other in-between modes such as Master - Isolated, where a controller operates in isolation, but is monitored by the Central Master Computer. In 'Link-Isolated' mode the controller responds to vehicle detections, but performs tasks according to schedules in the controller memory.

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Traffic Signal Controllers

Available modes are:

1. Fixed Time Mode (preset timings and not influenced by detection).
2. Isolated Mode (or Vehicle Actuated - VA).
3. Link – Isolated Mode ('in-between' Mode).
4. Full Link Mode (Cable-Less Link or Flexilink).
5. Hurry Call Mode (Emergency Vehicle or Railway).
6. Computer Control Mode (Master Computer).
7. Manual Control Mode (Police – Phase, All-Red, Flashing or OFF selection).
8. Maintenance Mode.
9. Fallback Mode (Flashing Amber in the event of a fault or as programmed).

These modes can be invoked remotely from the central computer, manually by means of a key, or automatically as programmed by the regional computer or by the 'time-of-day' clock within the controller.

Computer Control Mode is the highest level of control and may even override police manual control. Emergency mode usually overrides all other modes. The mode priority for a particular region is specified by the Authority.

Some controllers have a backup battery UPS system to deal with power failure.

10.2.1 Fixed-Time Control

Fixed-time control is considered to be the lowest level of control where the timings of each phase are preset in a fixed sequence and duration.

This method of control is useful for basic coordination between intersections. If each cycle is duration is known, it becomes easy to arrange the beginning of a phase at one site with respect to an adjacent site.

10.2.2 Vehicle Actuated (VA) Control (Isolated Mode)

In isolated mode, the phase sequence and the maximum duration of each phase is specified in the local controller time-settings. The local controller may terminate any phase under the control of the local vehicle actuation timers. It may skip a phase if there is no demand for it.

In Vehicle-Actuated (VA) control, the vehicle signal timings are dependent on the volumes of turning vehicle movements. The pedestrian timings are dependent on the presence of pedestrians. This causes variable sequences and variable durations of signal timing. This type of control is called 'fully actuated'.

10.2.3 Vehicle Semi-Actuated Control

In 'Semi-Actuated Control' detectors are placed on the minor roads only. If the main road is determined to be busy for most of the day, the main road phase do not need any detectors. The green period on the main road extends indefinitely until a vehicle is detected on the minor road.

Mid-block pedestrian signals operate in this manner. The main road green continues until a PBU activation is done by a pedestrian.

Each time the main road commences green, the green time runs for a preset period depending on the time-of-day. Side road vehicle or PBU demands are serviced after the preset period expires in each cycle.

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Traffic Signal Controllers

10.2.4 Local Link Control

Local Link Mode is often referred to as 'Cable-Less Linking' or 'Flexilink'. The phase sequence, duration and Green Man time are determined by the current linking plan, which is controlled by an internal clock.

In Local Linking all controllers should have their internal clocks synchronised. All cycle times (CT) in the network need to be equal.

The local controller may terminate any phase under the control of the local vehicle actuation timers or skip a phase unless there is demand for it.

10.2.5 Central Computer Control

In Central Computer Control Mode the phase sequence, the maximum duration of each phase and the duration of Green Man are determined by the regional computer. The local controller may terminate any Phase under the control of the local vehicle actuation timers or skip a phase if there is no demand unless it is prohibited by the regional computer. The regional computer controls the phase transition points in the local controller.

10.2.6 Fallback Mode

In 'Fallback Mode' the signal displays flashing amber to all approaches. Other flashing displays can be: flashing red to minor roads and flashing amber to the main road. Optionally, the signal displays can be completely off. This mode is triggered when there is a fault in the system, or a pole is knocked down by a traffic.

10.2.7 Maintenance Mode

In 'Maintenance Mode' the signal displays continue operating as normal. A technician can change the controller to any other mode to examine faulty operation, or to carry out maintenance. On completion of task, the technician can disable maintenance mode and the controller returns to its previous mode.

10.3 Controller Programming

The task of configuring a controller to the specific requirements of a particular site is known as adaptive traffic engineering. Each controller has a unique programme called a 'personality' which configures the controller to the specific operational design of the site.

The personality specifies which movements (signal groups) can run in each Phase, the Phase sequence, the Phase timings, and detector functions.

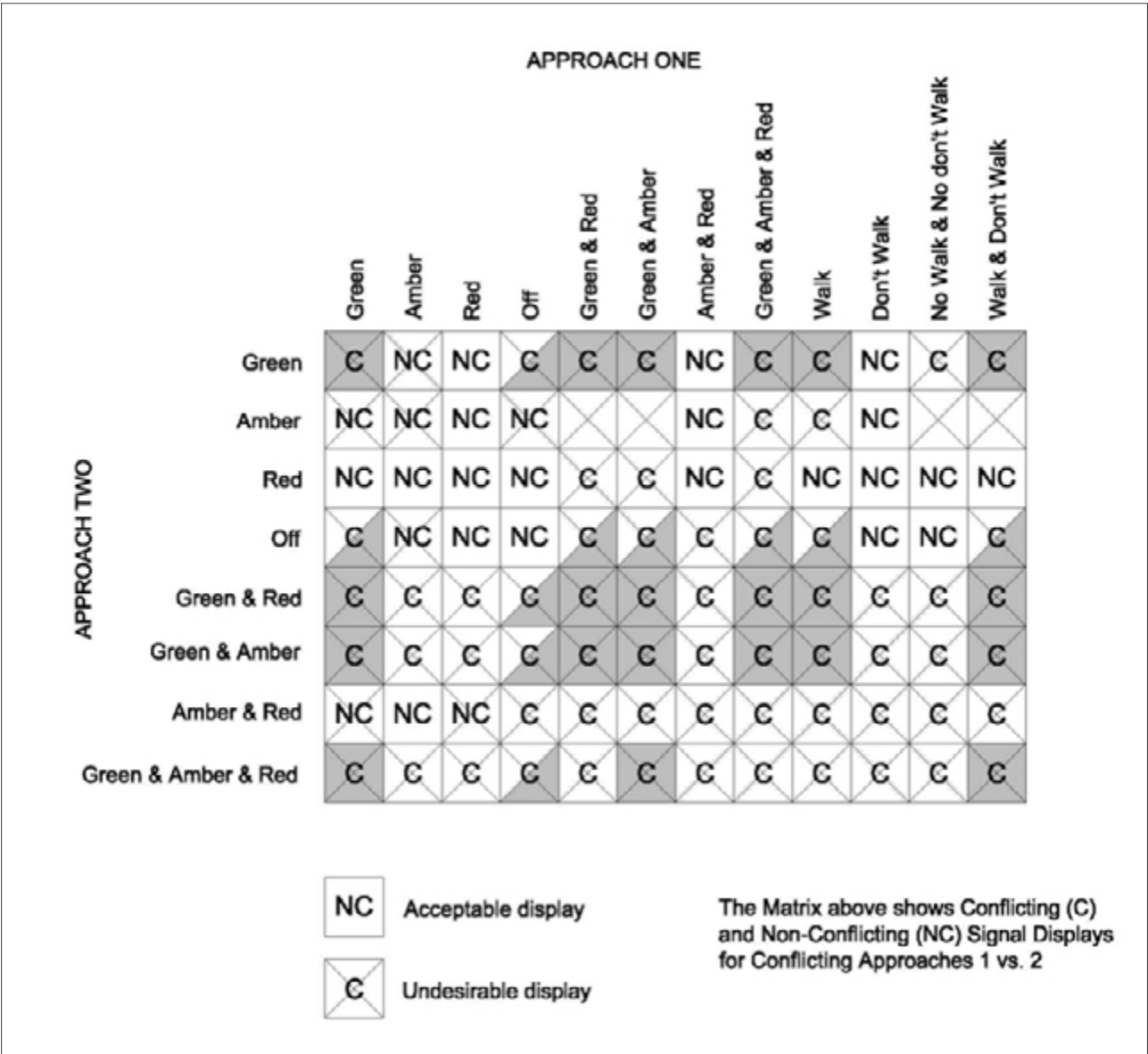
10.4 Preventing Conflicting (Hazardous) Displays

There are a number of steps in controller programming that reduces the risk of wrong programming. Once the controller is installed, there are mechanisms to switch off the signals or commence flashing amber in the event of an unsafe or conflicting display.

Conflicting displays arise from failures to switch power to the signal lanterns or by damage to the signal lanterns and cabling. Conflict monitoring is mandatory in all modern traffic controllers.

Figure 10.2 shows the acceptable (safe) and unacceptable (unsafe) lantern displays.

Figure 10.2 Safety of Signal Displays



Note: 1. Any signal display with more than one lamp lighting is considered unsafe.
2. Red/Amber is considered unsafe if any other display is showing more than one lamp lighting or if any other display is 'OFF'

10.5 Controller Timing Intervals and Movements Strategies

The main timing intervals are:

- 1. Late Start.
- 2. Minimum Green.
- 3. Maximum or Extension Green.
- 4. Rest.
- 5. Amber.
- 6. All-Red.

Other Timing Intervals such as Variable Initial Green (VIG) and Early Cut-Off Green and Early Cut-Off Amber are discussed in Chapter 14.

For the Pedestrian movements the main intervals are:

1. Green 'Man'.
2. Clearance 1.
3. Clearance 2.

Other Timing Intervals such as Pedestrian Delay are discussed in [Chapter 14](#).

10.5.1 Vehicle Phase Intervals

Each Phase is divided into the several main consecutive timed intervals as shown in [Table 10.1](#) and [Table 10.2](#).

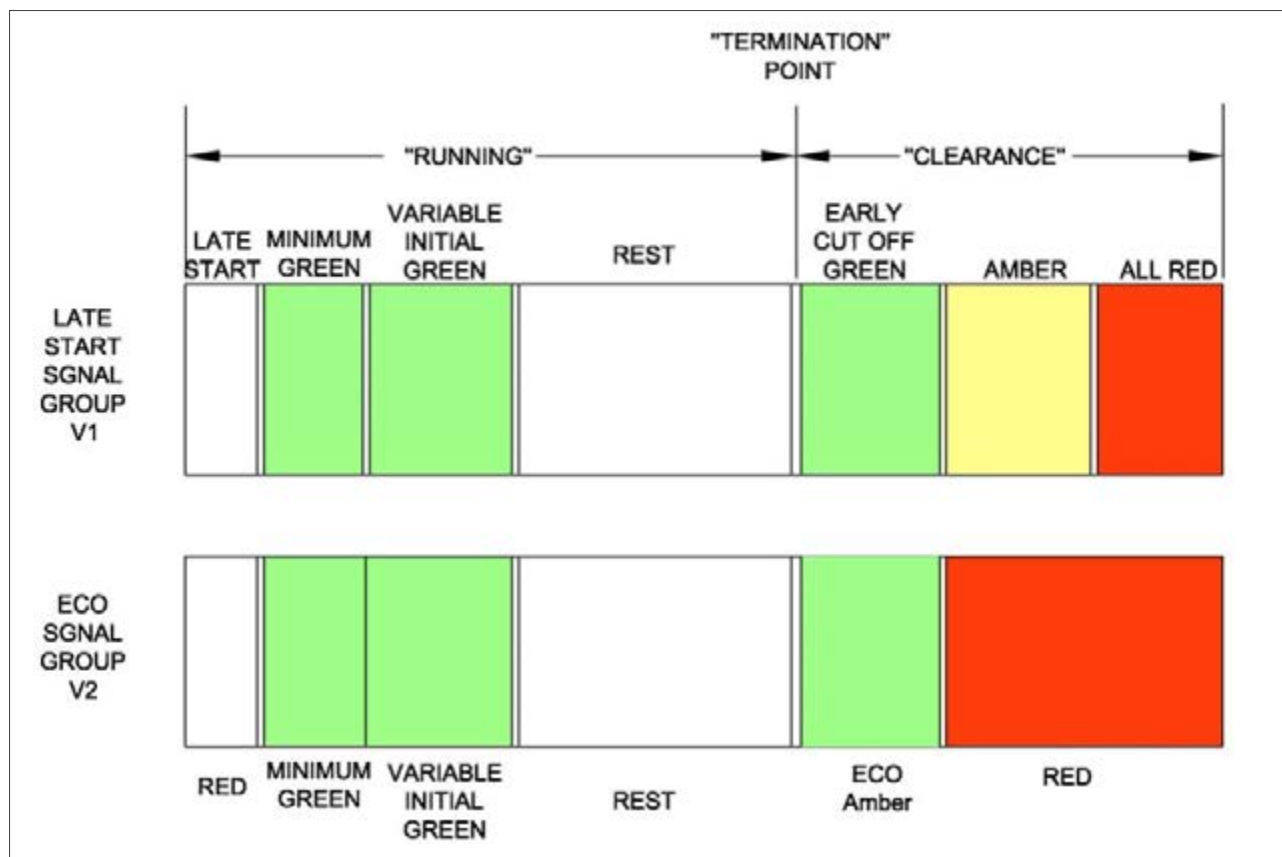
Table 10.1 Interval Description

Interval	Description
Late Start Green	The time from the start of the Phase for which the introduction of Green on any defined Phase may be delayed
Minimum Green	The time during which the Phase cannot be terminated
Rest	The time during which the Phase awaits termination
Maximum Green	The limited time during which Green may be extended by detector actuations
Amber	The time during which an Amber signal is displayed to any terminating Phase
All Red	The time during which Red is shown to any terminating Phase before Green is introduced on the next Phase. The Amber and All-Red intervals are collectively known as the inter-green period

The durations of the timed intervals for each Phase are stored in the local controller and are adjustable within the following ranges. See [Figure 10.3](#) for intervals

Table 10.2 Interval Range

Interval	Range
Late Start Green	0 to 20 seconds in 1 sec steps
Minimum Green	0 to 20 seconds in 1 sec steps
Maximum Green	0 to 250 seconds in 1 sec steps
Amber	0 to 6 seconds in 0.1 sec steps
All Red	0 to 10 seconds in 0.1 sec steps
All Red	The time during which Red is shown to any terminating Phase before Green is introduced on the next Phase. The Amber and All-Red intervals are collectively known as the inter-green period

Figure 10.3 Vehicle Timing Intervals

10.5.2 Pedestrian Phase Intervals

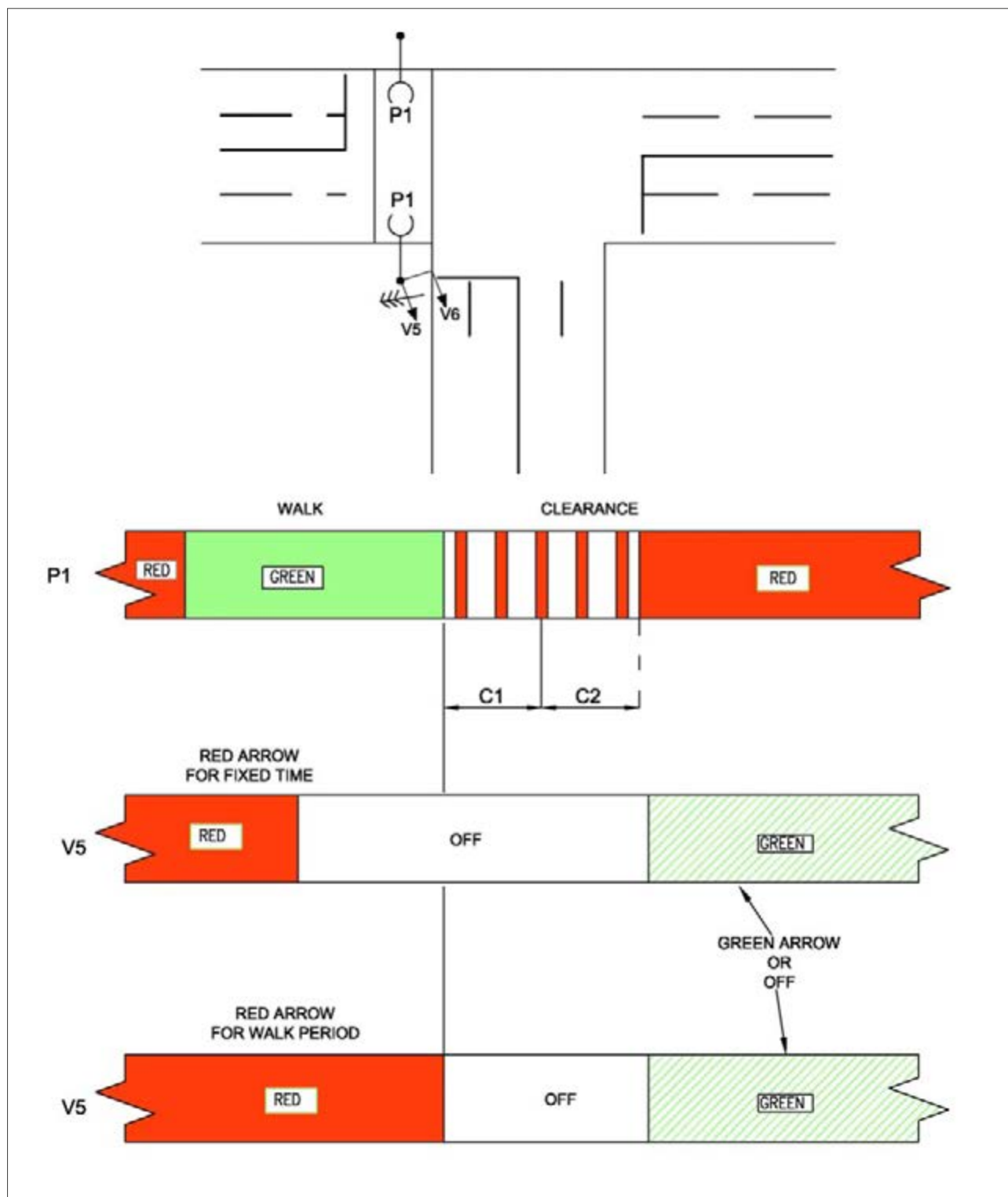
One or more independent pedestrian movements can be displayed if a PBU demand is registered. Pedestrian movement display is controlled by the following consecutive timed intervals.

Table 10.3 Pedestrian Movement Display Timed Intervals

Interval Name	Description	Interval Duration
Pedestrian Green 'Man' (Walk) Interval	The time during which a Green Man signal is displayed under isolated mode or the minimum Green 'Man' time under 'linking' or Computer Control mode.	0 to 40 seconds in 1 second steps
Pedestrian Clearance 1 Interval	The portion of the Flashing Pedestrian Clearance interval which must be timed before any concurrently terminating Vehicle Phase can step to Amber.	0 to 40 seconds in 1 second steps
Pedestrian Clearance 2 Interval	The portion of the Flashing Clearance interval which is timed concurrently with the Vehicle Amber and All-Red intervals of the Phase.	0 to 40 seconds in 1 second steps

Schematic diagram of pedestrian walkway and red arrow protection are shown in **Figure 10.4**.

Figure 10.4 Pedestrian WALK and Red Arrow Protection



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Traffic Signal Controllers

10.5.3 Vehicle Actuation Timers

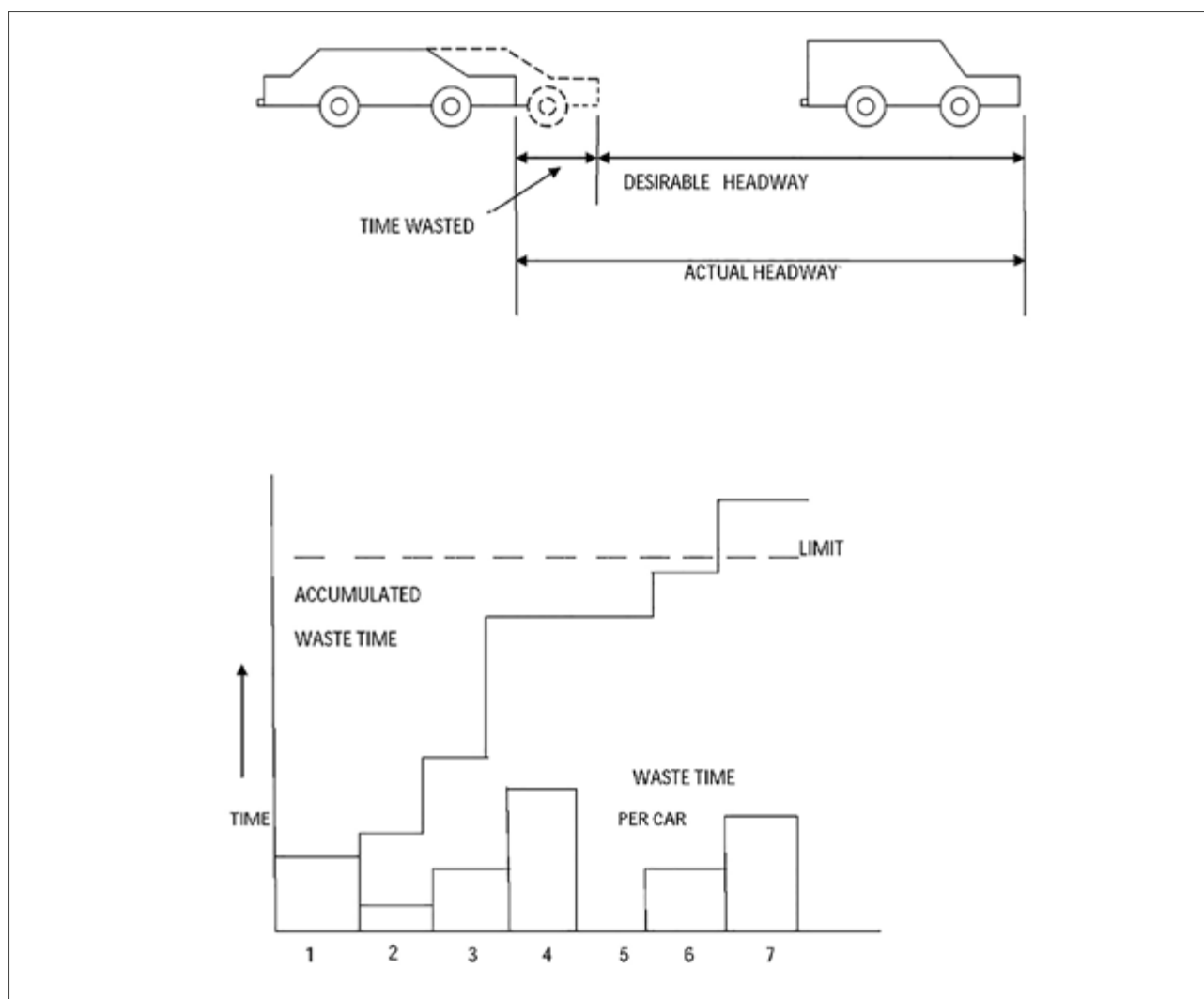
The Vehicle actuation timers operate in the following manner:

Table 10.4 Vehicle Actuation Timed Intervals

Name	Description	Interval Duration
Gap change	If the time between successive actuations of detectors on an approach with green time exceeds a preset interval, the green time expires.	0 to 10 seconds in 0.2 second steps
Waste time change	A waste time counter accumulates the time between detector actuations, after first subtracting a preset headway time. When the waste time counter reaches a preset value, the green time expires.	0 to 50 seconds in 1 second steps
Maximum change	In the case of a controller operating in 'Isolated' mode each Phase has a Maximum Green time set. If the Phase has not been terminated by Gap and/or Waste time, the phase terminates on reaching the Maximum Green time.	0 to 5 seconds in 0.1 second steps

Figure 10.5 shows the operation of headway and waste time.

Figure 10.5 Headway and Waste Time Operation



10.5.4 Control of Movements

Within the framework of phases and intervals there is flexibility for the control of vehicle and pedestrian movements in signal groups. Provision can be made for green time to overlap. Vehicle and pedestrian group greens can be introduced and terminated at any point in any Phase.

Approach timers can be associated with group movements rather than phases to enable early termination of a movement by Gap or Waste. The structure of the controller personality is such that the introduction and termination of Signal Group Greens can be based on any possible combination.

10.5.5 Detector Logic

All detectors have a tactical function even though this function may be inoperative under ATC Control or Linking plan. A detector can place a demand for a phase. When the phase is introduced, the detector operates the Gap, Headway and Waste timers. It is possible, to define an extensive range of conditional functions referred to as 'detector logic'.

There are software configuration packages for traffic controllers that can automatically take care of detector operations, amongst other features.

10.5.6 Conditional Demands

Demands for a Phase may be 'locking' or 'non-locking'. A locking demand is one where the demand, once placed, remains even if the detector does not sustain an output (i.e. the Vehicle leaves the detection zone).

A non-locking demand is cancelled if the detector output indicates the Vehicle has left the detection zone.

In some situations, a conditional demand is required to detect stopped or slow-moving traffic. A demand is not registered until the detector is occupied for a minimum set time ('presence time'). The demand may be locking or non-locking. Presence detection mode is required for conditional demands. The length and location of the sensor loops are critical for this purpose.

10.5.7 Conditional Extension

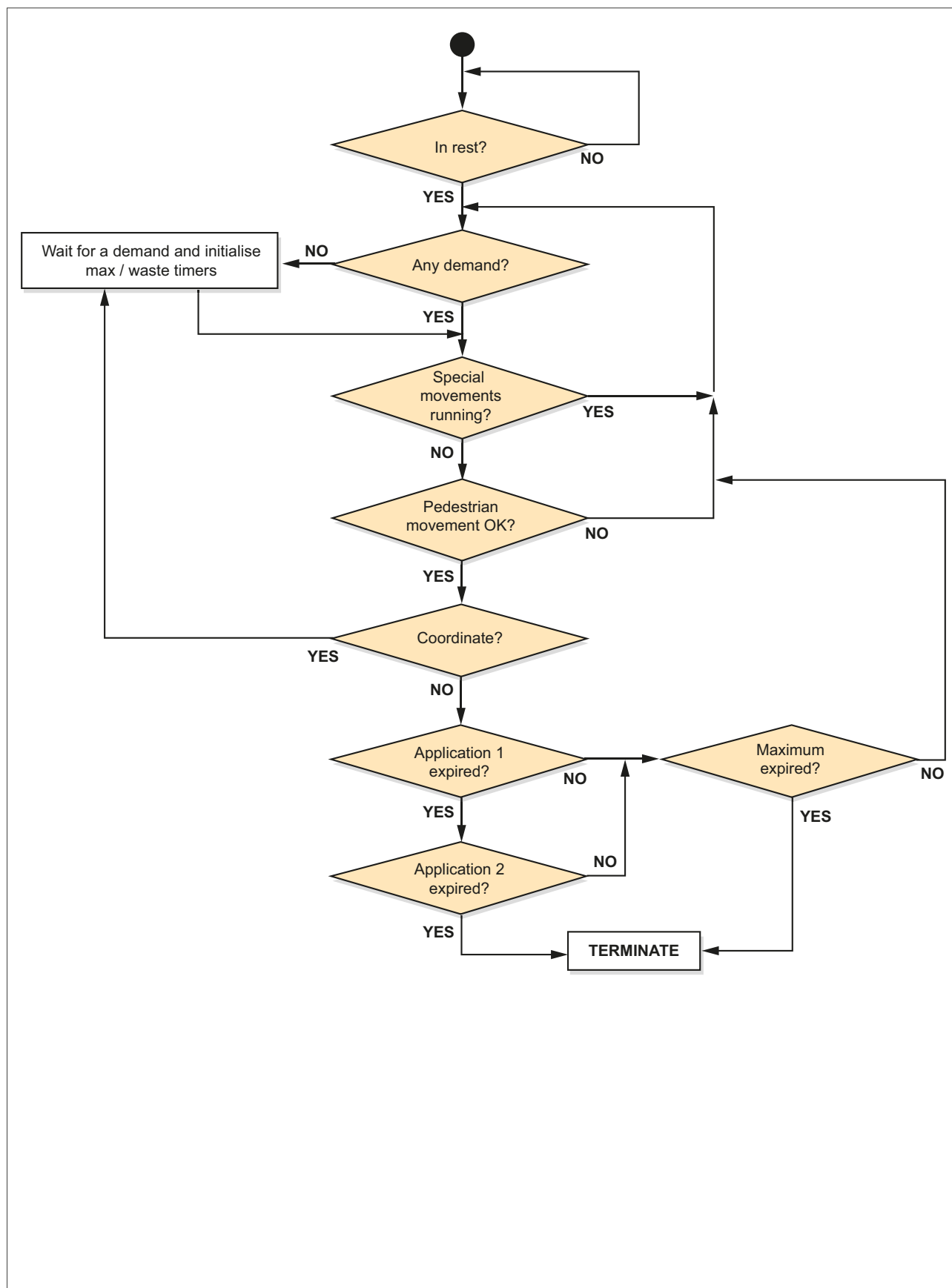
The detectors on an approach having a green signal normally extend the Phase. If, the approach continues to receive a Green signal in the next Phase, then the detector on that approach may be assigned to cease extending the Phase.

This logic is used on an approach which receives Green in two consecutive Phases. The approach does not extend the Phase if the next Phase in the sequence also presents a Green signal to that movement.

10.5.8 Conditions for Phase Termination

A Phase may terminate under a number of conditions such as Gap Expiry, reaching Maximum Green, or command from a central computer. Conditions for Phase Termination are outlined in [Figure 10.6](#).

Figure 10.6 Conditions for Phase Termination



11 Electrical Design

11.1 Introduction to Electrical Design

Each signal site shall have an electrical design so that connection is provided in a proper manner to all the signal system components. The components include lamp circuits, detection units, sensors, CCTV, controllers, lanterns, PBUs and other devices. Cabling diagrams should provide information on the following:

1. Controller connection to the main power supply and UPS.
2. Circuits for cabling to each pole and lamp.
3. Extra Low Voltage (ELV) 50 Hz circuits for pedestrian PBUs and demand indicators.
4. Circuits for external vehicle detector outputs.
5. Data links associated with coordinated signal systems.
6. Leased telephone cables or fibre optic links from the telecommunication authorities for the connection of controllers to a central computer.
7. The loop detector connection to the auto-tune circuit in controllers.

Details of duct routes, duct sizes, draw-pits, pole footings, detector duct and draw-pits, etc, are to be shown on the civil works drawing for the site.

Cable connection charts are required for core of each cable, showing its function, connection details and cable routing. These drawings are to be prepared by the signal contractor during the functional design process and updated to 'as-built' after installation and commissioning.

11.2 Cable Connection Design

The cable connection design determines the function and routing of cables. The circuits connecting lanterns, PBUs to the controller are to be optimised to reduce long cable runs and to provide an economic solution.

The number of cores in the multi-core cable is usually 20 or 30 per run. There should be sufficient circuits for each colour of signal group and PBU.

The wait lamp should be an LED type display. The cable connection chart should document the cable connection design showing the precise cores used for the terminations at each pole and mast-arm.

11.3 Cable Colours and Numbers

Cables to be used should be as per KS: 194 Specification for PVC insulated cables for electricity supply. For reasons of aesthetics and operational benefits, cables should be installed underground. Cables need to be rugged and waterproof.

Multi-core traffic signal cables should have core insulation colours based on the four-colour coding system as follows:

1. **Earth core:** Green/Yellow striped.
2. **Neutral core:** Black or Blue.
3. **ELV return:** Grey.
4. **Live feed for Signal Displays only - numbered cores:** White.
5. **Live feed for external equipment:** Brown.

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Installation techniques will depend on local practice and policies. Common ring bus or radial routing system can be used. Cable length between controller and signals should be kept to a minimum.

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The number of circuits required shall determine the number of cores required in each cable. Provision must be made for:

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1. One active cable core for each colour of each signal group, vehicle and pedestrian.
2. One cable core for each PBU and 'wait' lamp.
3. A number of spare cores – usually about 20% of any one cable run.

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11.4 Cable Installation

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All cable systems should be installed as per the requirements of the local electricity authority and the local traffic authority. The following points should be addressed:

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1. Cables should be buried to an adequate depth.
2. All signal hardware and equipment must be sufficiently earthed.
3. All cabling must have adequate separation, isolation, and insulation.
4. Proper termination of cables should be made at controllers, pole tops and junction boxes within mast-arms.

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Main supplies to the controller are to be made by a 32 AMP fuse-unit located on the nearest pole or in a supply pillar close by.

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11.5 Vehicle Loop Detector Cables

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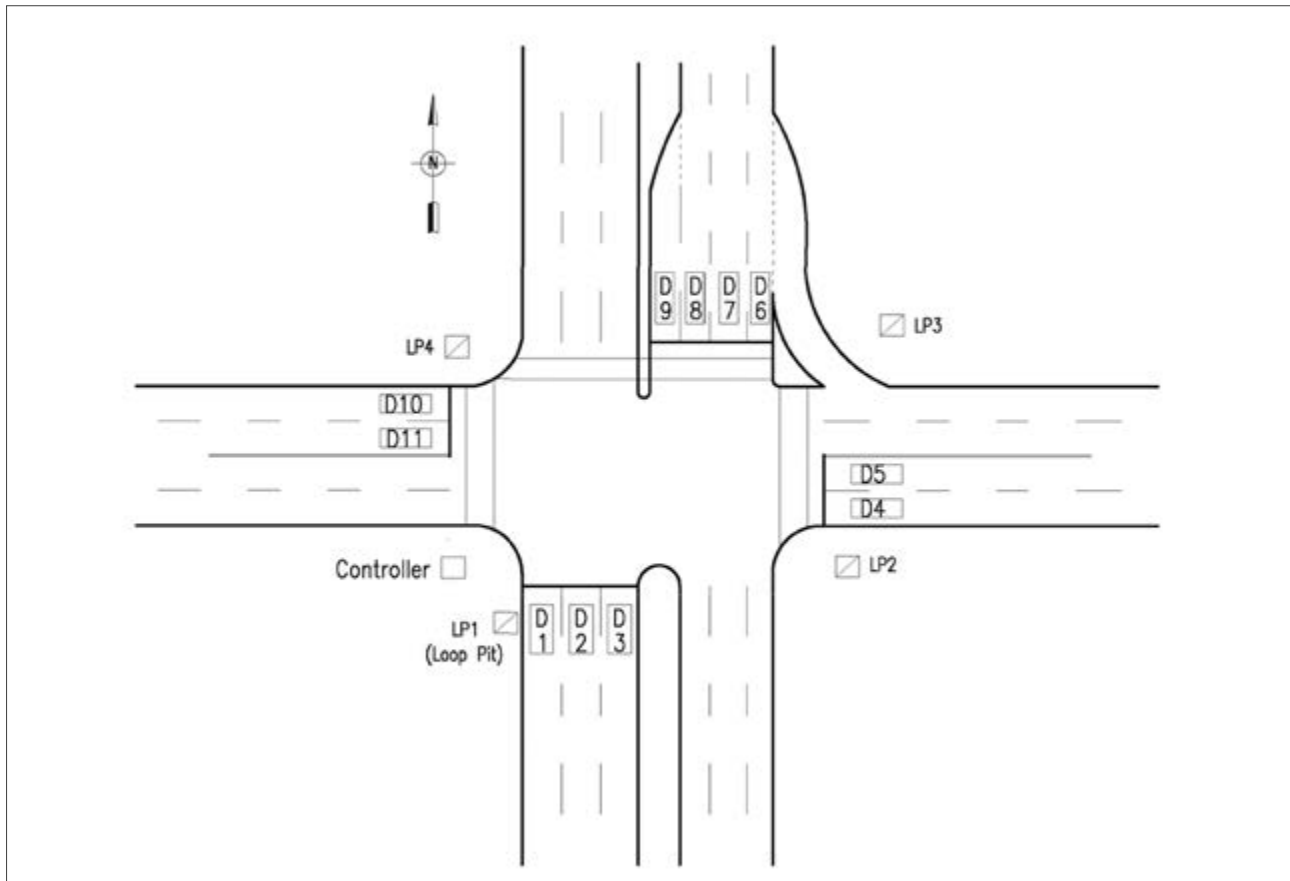
For vehicle loop detector sensors, screened feeder cable should be used between controller and detector road-side access pit. The detector loop cable is to be installed under the road surface by the contractor. A slot is to be cut to a depth of 6 cm by a diamond tip concrete or asphalt cutting blade. The cutting is to be made by a special slot cutting machine. After cutting, the slot channels are to be cleaned with compressed air. Loop cable is to be wound 2 or 3 times into the slot, then the slot is to be backfilled with epoxy resin. The cable is to be connected from the loop to an access pit. The loop tails at the access pit are to be jointed to the feeder cable and continued to the controller loop sensor terminals. **Figure 11.1** shows typical detector location layout at an intersection.

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The sensor to loop distance should be minimised since long feeder cable runs may have adverse effects on detector loop operation.

See Drawing in **Appendix A** for detector loop installation details.

Data link cables can be leased from telecommunication authorities or manufactured and installed to their requirements. Cable connections should be high-speed ADSL, fibre optic, or copper type.

Figure 11.1 Typical Detector Location Layout

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Electrical Design

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Electrical Design

12 Installation Checks And Maintenance

Prior to commissioning, it is necessary to check the installation in order to ensure that it is in accordance with the design and that all equipment are correctly installed and operating properly.

12.1 Commissioning Check List

Prior to commissioning, checks have been carried out according to the manufacturer's instructions and the specifications provided for the project.

12.2 Monitoring and Evaluation of Operation

After the signals have been switched on and operating for a few days, some fine-tuning of signal timings may be required. Thereafter, monitoring and evaluation of the operation shall determine if the design is successful.

Typical problems encountered after installation are variations in traffic flow, parking irregularities affecting traffic lanes.

Monitoring and evaluation should also consider non-peak hours and public holidays in addition to the peak hours.

If deficiencies are noticed, the actions that can be taken are: fine-tuning of signal timings, changes of phase sequences, banning turn movements, extension of kerbside controls - especially for parking or lane blocking.

In some cases, redesign of phasing is required. In some cases additional lane may be required.

12.3 Monitoring Check List

During the monitoring and evaluation a checklist should be used. A sample checklist is provided [Appendix 2](#).

12.4 Maintenance

Traffic signal installations require regular maintenance in order to ensure satisfactory operation in terms of capacity and safety. If a signal site is under central computer control, then problems with operations can be known immediately. Even then, many site faults may go unnoticed, such as dusty aspects, damaged poles, and even missing PBUs. As traffic flow changes at a site, some signal timings may need to be updated.

For signal sites that are not under central control, regular checks should be made for signal timings and operation, and to check for damage to equipment.

Maintenance inspections are to be carried out at regular, defined intervals. The observations should be documented and used for evaluation purpose.

12.5 Maintenance Management

Good signal maintenance management aims to minimise cost while maximising the functional operation of the site. Maintenance site visits are to be scheduled every 3 months, whether or not the site is centrally monitored. The level of maintenance required depends on the type of site signal equipment.

Good maintenance planning must consider the availability of spare parts. Stocks of expendable items should be maintained. Items unique to a signal system or items which have long delivery time should be stocked in sufficient quantity. 10% spares holding is required.

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Installation Checks & Maintenance

12.6 Maintenance Records

In order to adequately schedule a maintenance programme, it is necessary to estimate the Mean Time Between Failure (MTBF) and Mean Time to Repair (MTTR) for various signal components.

Within the Traffic Authority, accurate maintenance records also provide a basis for the review of specifications of equipment tendered for purchase. A good maintenance record should include the following data:

1. Fault advice:
 - a. Date and time.
 - b. Source.
 - c. Location.
 - d. Apparent nature.
2. Service report:
 - a. Date and time of arrival and departure.
 - b. Location.
 - c. Travel involved.
 - d. Personnel identification.
 - e. Action taken to clear fault.
 - f. Details of further work required.
 - g. Materials used.
3. Shop service records:
 - a. Date and time.
 - b. Items returned for service, service actually started, service completed.
 - c. Description and serial numbers of units.
 - d. Repair-hours involved.
 - e. Materials used in repair.

12.7 Urgent Maintenance

Random breakdowns of different natures may occur in a signal system. These must be repaired. Notification of failures can be expected from Police, other agencies or from the public. Fault report telephone numbers are to be placed on roadside signal equipment. Response times are set by the local traffic authority for the maintenance contractor. This time should be limited to around 5 hrs depending on the severity of the fault.

12.8 Accident Damage

Response to reports of accident damage should be given high priority. Damaged equipment should be evaluated promptly in order to ensure that it is electrically safe. It may be necessary to temporarily switch the signals to Flashing Amber, or 'OFF'. Repairs should be executed without delay.

12.9 Cleaning and Painting

Signal hardware should be kept routinely cleaned and painted. Cleaning of lanterns is important in maintaining light output, particularly in dry dusty environments.

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Installation Checks & Maintenance

13 Traffic Control Strategies And Tools

13.1 Overview of Signal Systems

Rapid improvement in microprocessor technology has greatly widened the traffic engineer's options for signal control. This chapter provides various strategies for controlling traffic movement.

13.2 Signal Coordination Methodology

The control mode of traffic signals can be by following:

1. **Fixed-Time Signals:** these have predetermined green periods, cycle times and duration. Time-settings need to be frequently and carefully monitored.
2. **Variable Fixed-Time Signals:** These signals are fixed-time signals with time-settings for different times of the day.
3. **Vehicle Actuated Signals (VA):** These signals respond to traffic demand by using vehicle detectors.
4. **Linked or Coordinated System:** Here signals on several adjacent junctions on the main road are coordinated to provide smooth traffic flow.

13.3 Local Signal Linking Systems

There are numerous options available for signal coordination. These options fall into three basic categories in terms of hardware architecture:

1. Local Linking by Cable.
2. Local Linking by Synchronous Clock (Cable-Less).
3. Linking by Radio.

13.4 Area Wide Traffic Control (ATC) Systems

Wide area traffic control can be done by cable linked system or cable less system.

Cable linked systems are used where junctions are very close together. Information is passed between two or more controllers by buried cable.

On a Cable-Less linked system, co-ordination of two or more junction controllers is achieved by the use of synchronised clocks. The clocks are usually digital microprocessor-based units. Clocks may be synchronised from a radio signal or central control computer.

Radio linked systems are not common, because of unreliability. Radio equipment gets interference from weather and other radio signals. It requires more maintenance and expertise. Bandwidth permits may also be required.

13.4.1 Types of ATC Systems

Traffic signals are linked to the central computer. Signals are controlled by using traffic plans generated by historical data. Plans are normally selected according to the time of day and the day of the week. ATC can be enhanced by using adaptive traffic control system that respond to variations in pedestrian and vehicle volumes. A schematic diagram of ATC components is given in [Figure 13.1](#).

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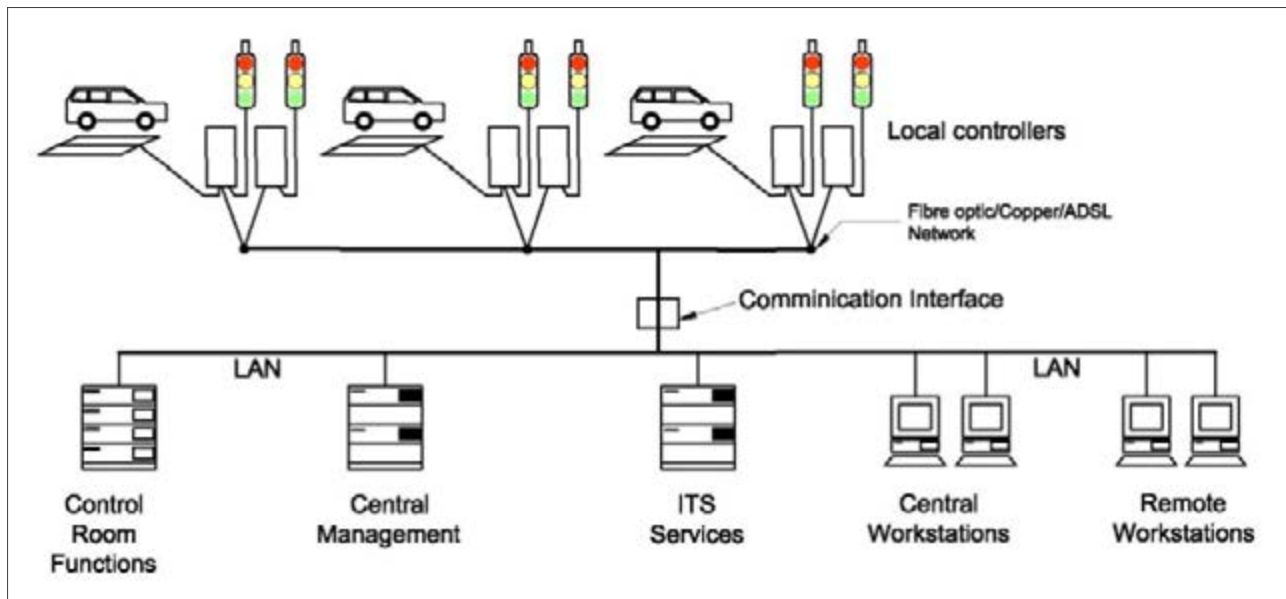
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Figure 13.1 PC Based ATC System Components

13.5 Traffic System Design Tools

There several software packages available to engineers for traffic signal system design. These are of two general categories:

1. Detailed junction design (individual sites or nodes).
2. Coordination design programs (mostly for ATC networks).

Using parameters, such as turning movement traffic volumes, saturation flow, occupancy, and link speeds, it is possible to calculate a set of coordinated signal timings for a network that will minimise delay for those particular traffic conditions.

The features in the engineer's control are:

1. Green split, which is the proportion of green time allocated to an approach.
2. The cycle time (CT), which is the total time required to give green period to all approaches and pedestrian movements.
3. Offset, which is the time from the start of the green at one signal to the start of a selected green at another signal.

Additional factors such as turning radii, lane widths, visibility, signing, environmental considerations and plain common sense must be taken into account in order to achieve good design.

13.6 Time Distance Diagrams

Non-coordinated signals on arterial routes cause drivers to stop frequently at signals because each signal is operated in an isolated manner. When signals are combined and linked, a progression of signals can be provided.

Progression is a time relationship between adjacent signals that allow vehicle platoons to travel in a smooth fashion through a series of intersections.

A platoon is a group of vehicles travelling together with an approximate uniform speed. At isolated signals, platoons begin to disperse and vehicles arrive in random fashion. In a coordinated signal system, a platoon can be maintained by providing green-time at each consecutive intersection. Offsets times are specified to establish time relation between consecutive signals. At each intersection, offset is given to the start of green, based on the travel time from the previous signal. Offsets can be fixed or variable. They are given either in seconds or percentage of cycle length.

Time intervals are drawn on a Time Distance Diagram as two parallel lines. The slope of the line is based on the progression speed. The sequence of green time at the signals is called bandwidth. The bandwidth cannot exceed the least green phase time in the system. Based on the spacing between signals and travel time between them, a short through band can be created. The through band can pass through the beginning, middle or end of the green-time of the individual intersections in the system.

Minimum stopping of vehicles should be considered for providing starting offsets. For the lead vehicles of a platoon, green signals should be available for a smooth flow. Some vehicles in the rear of the platoon may not get green time at some intersection where green time is less. These vehicles will have to wait for the next cycle, experiencing delay.

To minimise delay of the trailing vehicles, finishing offsets can be provided. In that case, the lead vehicles may get red time at signals that have less green time.

A compromise between the two types of offset can be done. Usually finishing offsets are optimised during peak hours and starting offsets are optimised during off-peak hours.

Time-Distance Diagrams provide a graphical tool where vehicle movements can be visualised and timing plans can be prepared. After a timing plan has been implemented, fine tuning at site is necessary to remove unexpected issues. For systems in operation, periodic review is necessary to ensure smooth functioning.

13.6.1 Site Data Collection for Coordination Plans

The data to be collected from site are:

1. Geometric data:
 - a. Intersection spacing (stop-line to stop-line).
 - b. Site layout (widths, lanes, approaches, prohibited movements, parking regulations).
2. Traffic volume data:
 - a. Turning movement volume data at each intersection.
 - b. Free speed of vehicles.

13.6.2 Site Data Analysis for Coordination Plans

Modern analysis tools are capable of producing time distance diagrams automatically from input data. The following steps are recommended for analysis of site data:

1. Select the time period of operation of coordinated plan and establish the required phase times and system cycle length.
2. Determine the capacity for each approach from lane characteristics - number of lanes available, width, gradient, utilisation of shared lanes and the environmental conditions.
3. Select the highest hour of demand during the time period for which the plan is designed.
4. Assess traffic volumes for left turns, right turns, heavy vehicle composition and convert the volumes to PCUs.
5. Specify minimum green-times and phase clearances. Pedestrian walk clearances must be allowed for in the minimum green. Use a software programme to compute signal times for each intersection. Choose the sequence which results in least delay.
6. Compare the cycle lengths derived by the programme and determine which is the highest. This will usually be adopted as the cycle length for the system on this plan.
7. Adjust the phase splits at each intersection to scale it to the system cycle length.

8. Adjustment is to be done by setting the time for fixed cycle length and favoured phase in the software.
9. Determine the green time for the system.
10. Overlap turns allow one direction to continue while halting the other direction. Green interval may differ for the two directions on two-way arterials.
11. Average queue lengths on all approaches at each intersection should be shown in the programme results. This provides an estimate of platoon lengths travelling along the arterial, and indicates the bandwidth required.
12. Queues on side-streets are to be measured. If queues are long, through phases may need to be introduced to clear the queues prior to the progressive movement.

13.6.3 Construction of One Directional Diagrams

A Time-Distance Diagram is arranged with time- on the Y axis and the distance on the X-axis. Every signal site is marked across the X-axis. A vertical line is drawn above each site location to indicate the phases.

The staging for each signal is represented by vertical bars denoting red-time. Green-time is indicated by a Green line or bar or by open spaces. For overlap turning phases, each direction in two-way flow may have different green intervals at the one intersection. Plot the Phase splits previously evaluated for this signal.

Figure 13.2 shows the base graph for time distance diagram. Representation of signal phases in a time distance diagram are shown in Figure 13.3.

Figure 13.2 Time Distance Diagram –Base Graph

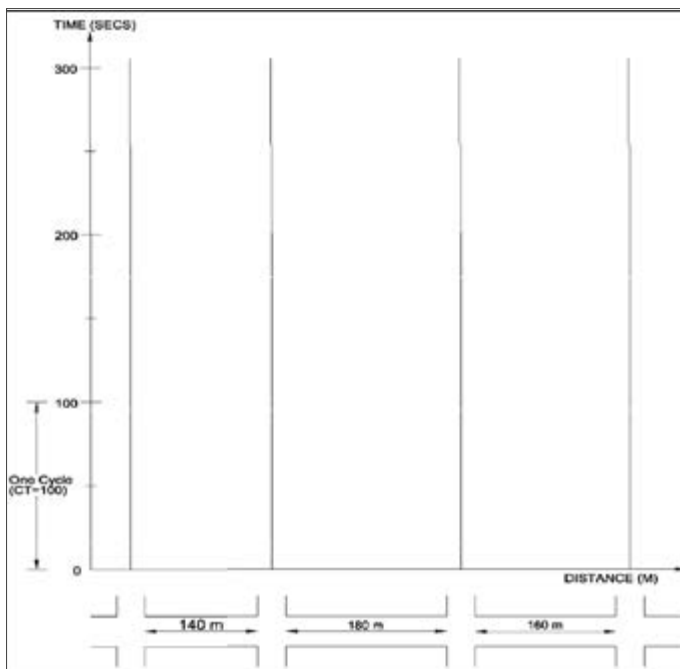
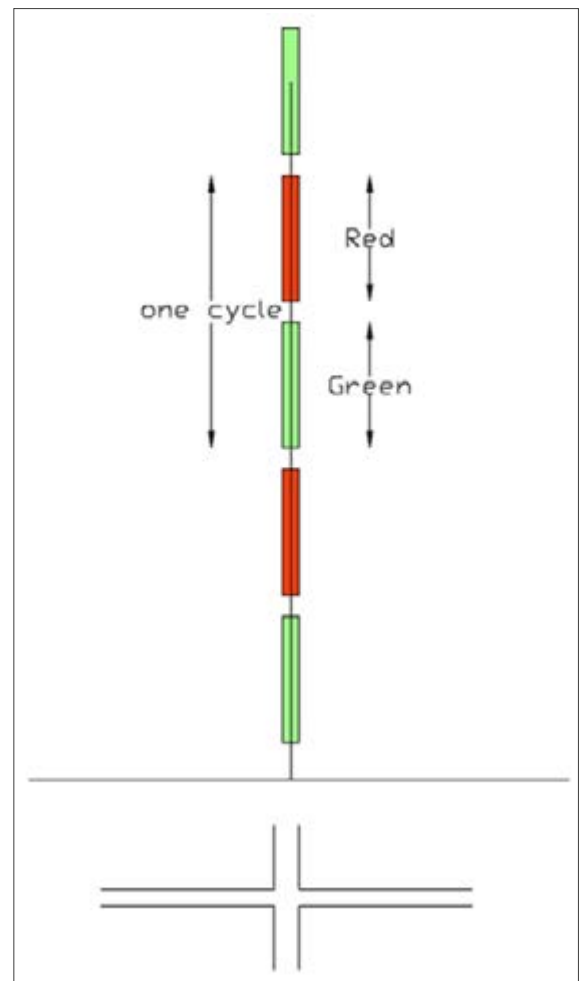


Figure 13.3 Time distance Diagram – Signal Phases



For time distance diagram, the progression speed of platoons is considered. If measured values of free speed are available then those are to be used. Otherwise, estimation is needed. A commonly used value is 44 km/h.

Progression is represented on the time distance diagram by a diagonal line between intersections. The slope of this line is determined by the design progression speed.

For establishing the offsets, two methods can be employed:

1. Linking the start of green time.
2. Linking the end of green time.

To determine the width of the Through Band, a line is drawn parallel to the progression speed line, touching the red interval closest to it. In one-directional coordination, the Bandwidth will equal the shortest Green-time in the system.

Progression line on a time distance diagram is shown in Figure 13.4. Through band is shown in Figure 13.5.

Figure 13.4 Time Distance Diagram – Progression Line

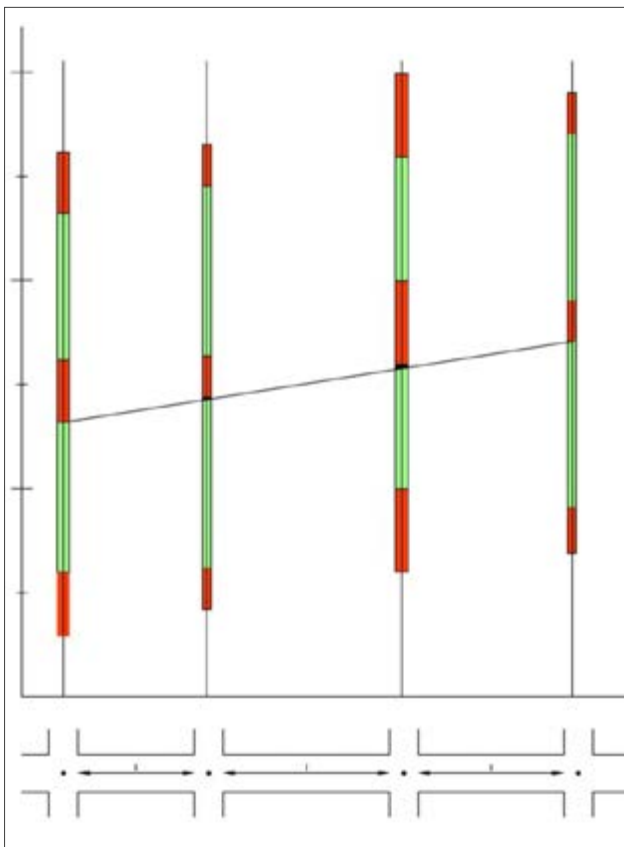
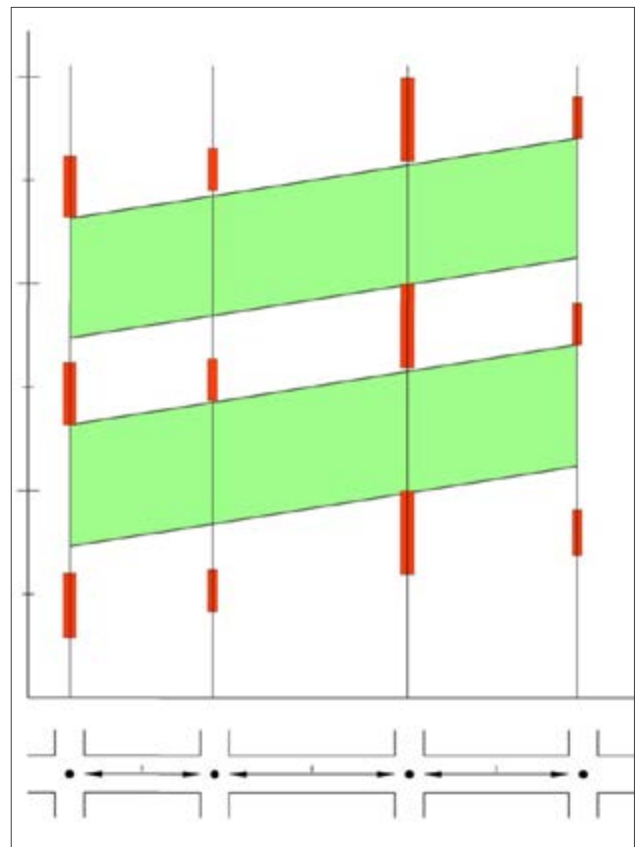


Figure 13.5 Time Distance Diagrams –Through Band



For one-way streets this represents the basic graphical solution to the offset needs of the system. For two-way arterials, the progression in the minor flow direction should be reviewed. It is often possible to shift the offset of under-saturated intersections so that the minor direction bandwidth is improved without adversely affecting the major direction.

13.6.4 Construction of Two Directional Diagrams

This is required for two-way arterial routes with equal demand on both sides.

Multiple trials are needed for creating progression lines. Compromises are needed between the implementation of start-line offsets and finishing offsets.

13.6.5 Compensation for Queue Formation

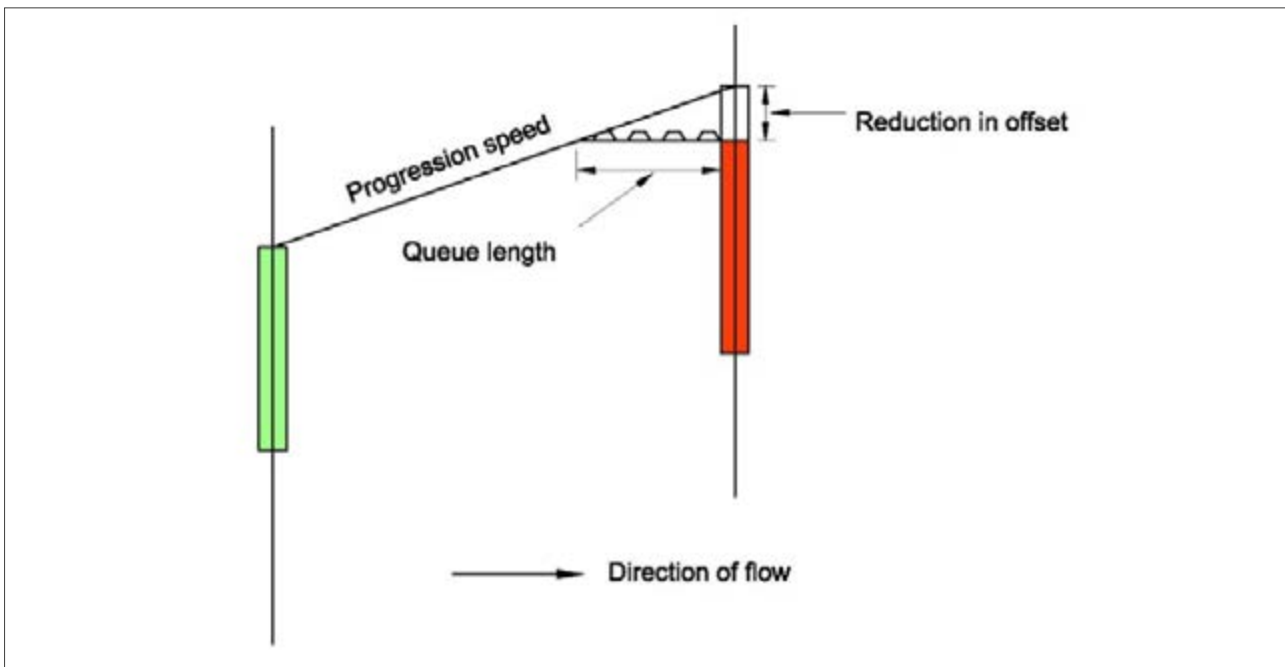
Queues may develop along the minor streets. Compensation for queue can be done by the following process:

1. An estimate of the queue length is obtained from software programme or measurement at site. Assessment of storage lanes available is made.
2. A horizontal line is drawn back from the initial point on the Green-time at the intersection where the queue is waiting.
3. Instead of linking the starts of the through Phases, the progression line is drawn to the end of the queue.

The offset of the downstream site is reduced and the through Phase is made to commence earlier in order to allow the queue to clear.

Compensation in queue formation is shown in Figure Figure 13.6.

Figure 13.6 Time Distance Diagrams –Queue Formation



13.6.6 Offset Strategies

Signal Coordination along a route can be achieved in four:

1. Using starting offsets - start of green periods at all signals are the reference points.
2. Using finishing offsets - end of green periods at all intersections are the reference points.
3. Simultaneous offsets - all green periods start at the same time for a given link.
4. Maximum bandwidth - maximise the amount of green time available to a platoon.

The aim of a starting offset strategy is to ensure that the bulk of cars get through the link without stopping. Vehicles towards the rear of the platoon will get red signal. Those vehicles will form residual queues into the next cycle and may impede the progression. With the finishing Offset system, the disadvantage is that lead vehicles may encounter queues

In order to minimise the negative effects of both starting and finishing offsets, the mid-points of green periods are selected as the coordination points.

The simultaneous starting of green offset is useful for closely spaced signal sites. It is effective in reducing queues in both directions.

13.7 Traffic Signal Coordination

13.7.1 Introduction to Coordination

Signal coordination is achieved by:

1. Operating all signals in the system on the same system cycle time.
2. Maintaining an offset between starting or finishing times of green displays at adjacent signals

The objective of signal coordination plans is to optimise certain selected performance measure. These measures are:

1. Minimise stops.
2. Minimise delay.
3. A combination of both.

The benefits of signal coordination are:

1. Reduction in the number of stops.
2. Reduction in travel time and delay.
3. Improved capacity between adjacent signal sites.
4. Reduction of noise levels, air pollution and energy consumption.
5. Benefits from the increased capacity of the road network which helps to avoid expensive road widening projects.

13.7.2 The Case for Coordination

A signal site can operate in Isolated mode where no obvious advantage can be gained by linking it with adjacent sites.

When signals are at close intervals, vehicle platooning is formed. Benefits from coordination of signals can be achieved for vehicle platoons. For signal spacing less than 1 km, coordination provides benefits.

13.7.3 Local Linking by Cable

These systems usually have a small number of closely spaced signals, where each controller in the link is connected by a cable. One controller is designated as the Master. It contains a number of timing plans.

13.7.4 Local Linking by Synchronous Clock (Cable-Less)

The cable-less linking system or Flexilink System can be applied to a large system of signals, whether or not a central computer system (ATC) is available. Coordination of signals is achieved by using an accurate clock in each signal controller with all clocks maintained at exactly the same time.

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13.7.5 Area Wide Traffic Control Systems

ATC systems involve one or more central computers controlling a large numbers of signals. These systems reside in the traffic control centre and signal operations are constantly monitored by operators throughout the day.

When extraordinary traffic conditions develop, operators can intervene. Operators also monitor traffic incidents and signal equipment malfunctions.

All signal sites in an ATC are connected to the traffic control computers, usually by leased data or telephone lines or high speed dedicated cable systems.

13.8 Coordination Requirements Affecting the Design of Signal Sites

The requirements of signal coordination are:

1. The Controller must have the capability to run synchronous clock Cable-Less or Flexilink facilities, both independently and on ATC.
2. Detector placement and function must be planned according to ATC requirements and for the intended local linking system.
3. Cycle and phase times need to be considered.
4. The phase design needs to be considered in the context of the road or network structure.
5. The design may have to omit certain movements or find other options for providing coordination.
6. The ATC needs to be programmed to actively respond to the presence of public transport vehicle.
7. Bus, BRT, tram, LRT stops may need to be relocated to provide path for smooth flow.

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14 Traffic Signal Timings

14.1 Introduction to Timings

The aim of signal timing is to give appropriate green time for movement of vehicles and pedestrians at an intersection. Safety requirements dictate the time to be given for straight through, right arrow, pedestrian green man, clearance time and inter-green time. Maximum tolerable delay needs to be considered while designing signal timing. While providing time for pedestrians, consideration should be made for crossing time for persons on wheelchair, women with children and pregnant women. This is particularly important at signals near schools, hospitals and public offices.

14.2 Signal Timing Methods

14.2.1 Green Times and Cycle Time (CT)

Software programmes are used to estimate green times based on junction geometrics, traffic count data, actuated controller parameters, and desired signal phasing. The total cycle time is then calculated.

Alternatively, count data are used as input to the software based on a pre-selected cycle time and the software gives the expected degrees of saturation and calculates the expected green times. If levels of saturation are unacceptable, other parameters need to be varied.

The acceptable time range for parameters to estimate suitable green time and cycle time are given in Table 14.1.

14.2.2 Maximum Tolerable Delay

Drivers and pedestrians tend to disobey signals if the waiting time is long. An upper limit needs to be set for maximum green time. The waiting time for vehicles at a red signal depends on:

1. Traffic flows on all approaches.
2. Maximum green time settings.
3. The requirements of signal coordination elsewhere in the network.

Observations suggest that the maximum waiting time is 20 seconds if the intersection is not busy. It is 120 seconds for intersection with heavy traffic.

Increased green time implies increase in red time for a competing movement, and an increase in cycle times. The recommended maximum cycle time for a two-phase signal is a 2 mins. The recommended maximum CT for complex junctions with high traffic demands is 2.5 to 3 mins.

14.2.3 Time Transfer and Stealing

At vehicle-actuated signals, the green times and the cycle times are determined by detectors. The phase sequence can be fixed or variable. A phase can be skipped when there is no demand for it. If a phase is skipped, its time can be transferred to another phase.

If the controller estimates that an upcoming phase will have no demand, the time of that phase can be stolen and used in the current running phase.

Maximum time transfer and maximum time stealing are shown graphically in Figures 14.1 and Figure 14.2.

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Table 14.1 Summary of Signal Controller Time-Settings

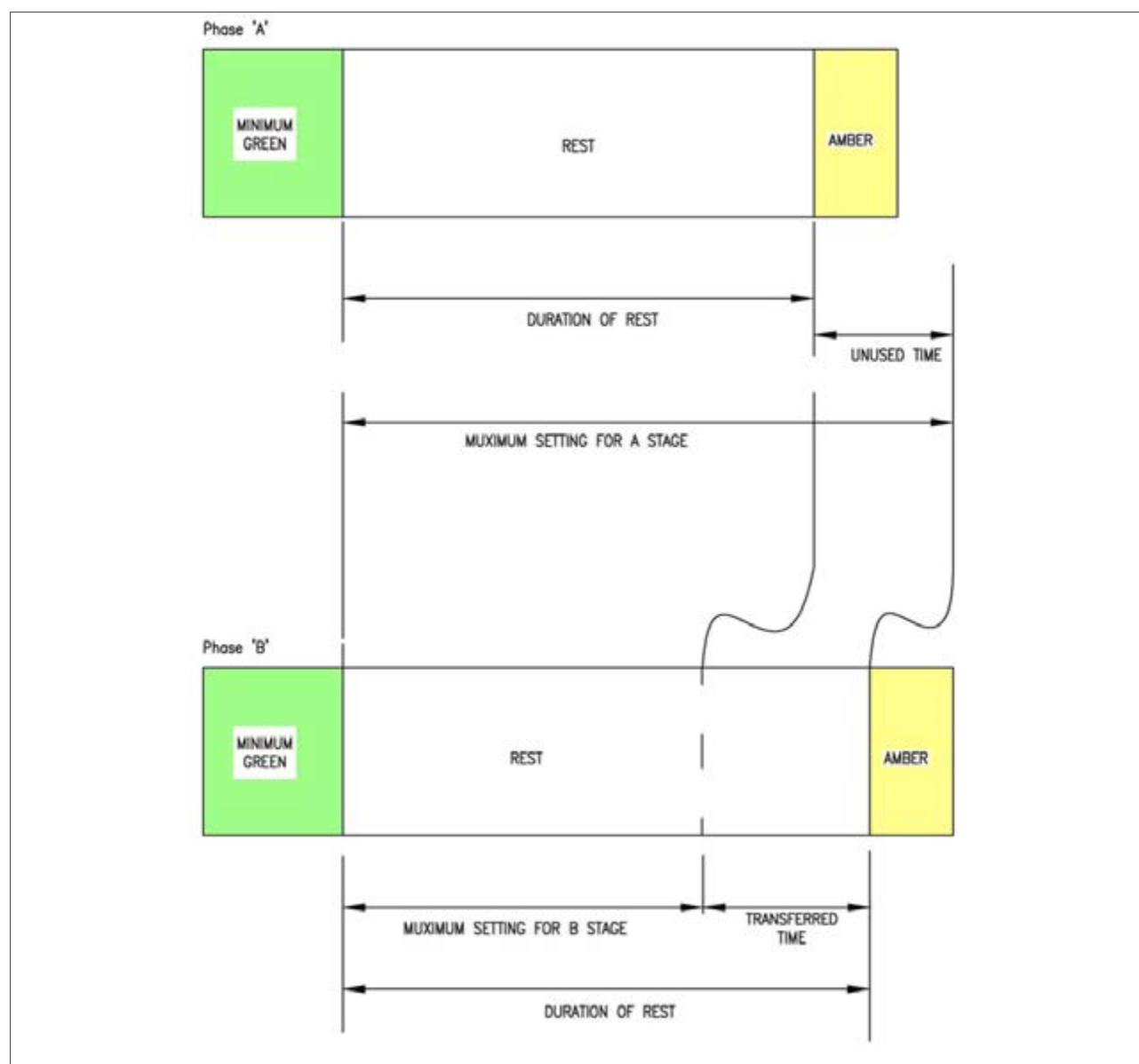
Setting	Purpose	Range
Maximum Cycle Time	Limits the total CT and reduces delays, and queue lengths	100 -120 secs for 2- Phase 150 - 180 secs for 2+ Phases.
Late Start	Allows some signal groups to be delayed at Phase start.	0 - 6 seconds.
Basic Minimum Green	Ensures that the Green signal is displayed for a safe minimum time.	4 - 10 seconds.
Increment (for advance detectors)	Extends the Basic Minimum Green Time so as to clear vehicles stored between detector and the stop line.	0.5 - 2.0 seconds.
Maximum Variable Initial Green - VIG	Limits the VIG period determined by increments – advance detectors	Depends on the distance of advance detectors from stop line
Maximum Extension Green (or Maximum Green)	Controls the maximum extension green time available to each Phase or signal group.	Choose for optimum traffic performance under different conditions
Gap	Sets the time between successive detector actuations before the movement terminates ('gaps out').	1.0 - 4.0 seconds.
Early Cut-Off Green	Allow the termination of some signal groups earlier than others.	0 - 10 seconds.
Amber Time	To provide sufficient warning of the termination of the Phase.	3 – 6 seconds.
All-Red Time	Provide a safe clearance for vehicles that cross the stop line towards the end of the Amber.	1 – 3 seconds.
Presence	Set the period for which a detector must be occupied before a demand is recorded.	0 - 5 seconds.
Headway (Space)	Set the desirable space time between successive detector actuations for efficient traffic flow.	0.3 - 1.5 seconds.
Waste	Set the value to the actual space time at which the phase is terminated.	4 - 12 seconds.
Minimum Red Arrow Time	To allow appropriate red arrow display time considering driver reaction/perception characteristics.	2 - 5 seconds.
Pedestrian Walk Time	Set the duration of the green Walk display.	5 – 16 seconds
Pedestrian Clearance Time	Set the duration of the Flashing Don't Walk display.	6 – 20 seconds
Pedestrian Delay	Provide a delay between the push button actuation and the placement of the pedestrian demand.	0 - 10 seconds.

Source: Guide to Traffic Management, Part 9, Austroads

14.2.4 Vehicle Phase Intervals

Green signal is displayed from the start of minimum green to amber. If a signal group elsewhere within the same phase starts green a few seconds later, the running part of the green phase is from late start to amber. If a signal group terminates to amber a few seconds early, then the running part of green is from late start to early cut-off amber, or ECO amber.

The vehicle phase or signal group intervals are shown in Figure 14.3. It does not include the early cut-off interval.

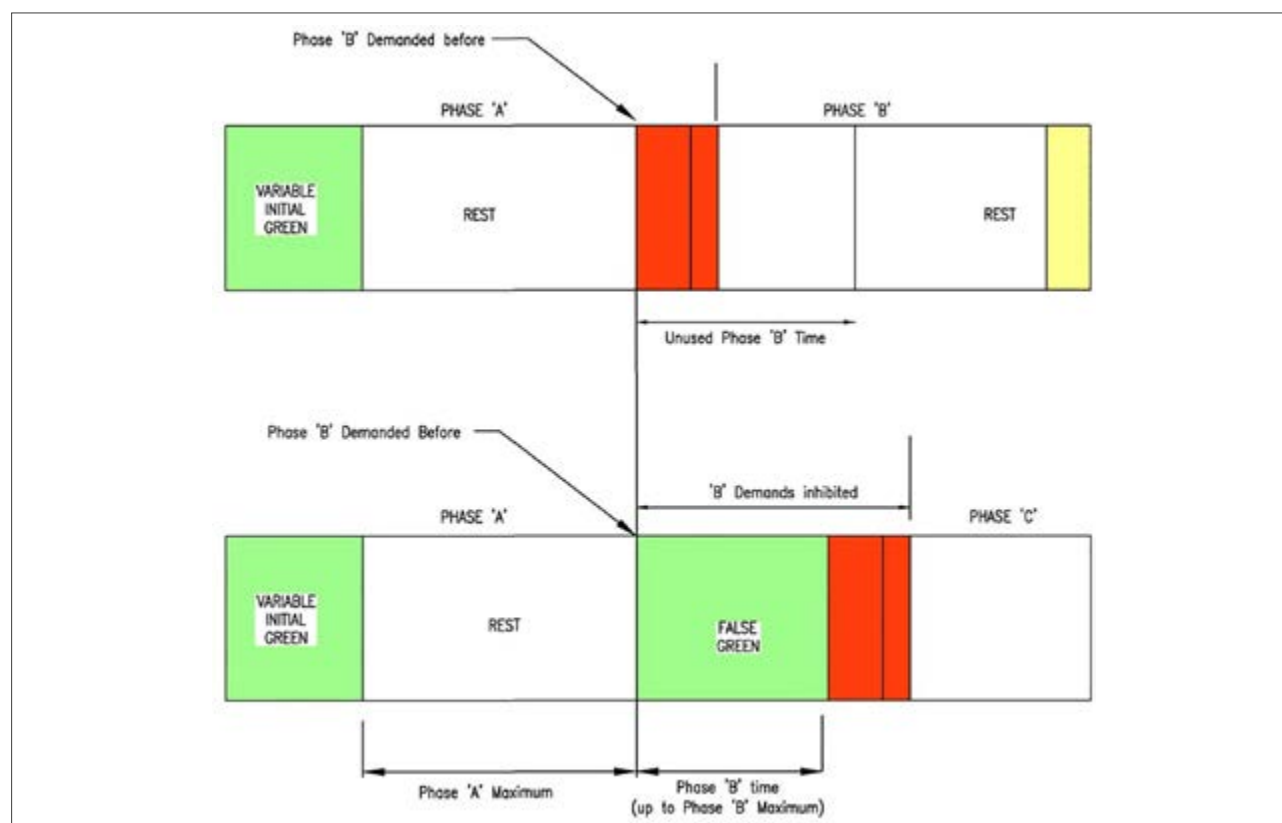
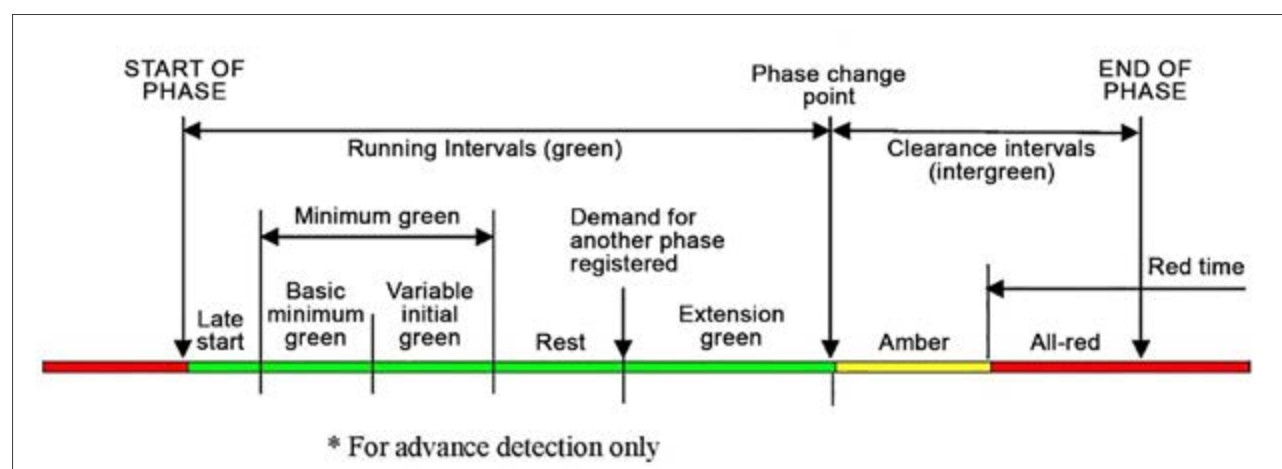
Figure 14.1 Transfer of Unused Maximum Time

14.2.5 Inter-Green

The clearance part of vehicles occurs when the amber and all-red signals are displayed. The clearance period starts at the phase change point and ends at the beginning of the green display. Green time is the duration of the running intervals. Inter-green time is the duration of amber time and all-red time.

14.2.6 Late Start and ECO

Different signal can be displayed for movements in the Late Start and ECO green intervals. The late start interval delays the green signal to some movements. The intervals are set by the late start and ECO green time settings in the controller.

Figure 14.2 Maximum Time 'Stealing'**Figure 14.3** Phase (or Signal Group) Vehicle Intervals

14.2.7 Minimum Green

Two types of minimum green intervals used:

1. Basic minimum green interval.
2. Variable Initial Green interval (VIG).

14.2.8 Variable Initial Green Interval

The Variable Initial Green (VIG) interval provides a time interval according to the number of vehicles queued between the stop-line and the advance detector. The Maximum time for the interval is called Max VIG.

14.2.9 Rest and Extension Green Intervals

Rest interval is an un-timed interval after the minimum green time expires. In this time the controller rests until a demand for another phase is registered. The rest interval is skipped if a demand is registered before the end of the minimum green time.

Extension Green interval is of variable length. In isolated signals, its duration is determined by gap settings, headway, and waste setting. Refer to [section 13.6](#).

14.2.10 Gap Timer

Gap timer operates to terminate green period at the end of the minimum green or the rest interval. Headway and waste timers are additional extension settings.

In addition to a gap timer, a phase can be terminated by a minimum change, or a maximum change. Minimum change occurs when a phase ends at the end of the minimum green period. Maximum change occurs when a gap change or waste change has not occurred during the extension green interval and the total green extension time equals the maximum extension green setting.

In summary, the green period can be terminated by any four methods:

- a. A Minimum change.
- b. A Gap change.
- c. A Waste change.
- d. A Maximum change.

14.3 Pedestrian Intervals

Pedestrian movements are introduced by PBU detection. For parallel crossings at intersections, pedestrian demand needs to be received before the relevant phase starts. Pedestrian movements can also be introduced automatically in areas where heavy pedestrian demands exist.

The pedestrian intervals and it is their relationship with parallel vehicle movement intervals are shown in [Figure 14.4](#). Pedestrian and vehicle phase intervals at a mid-block Pelican crossing are shown in [Figure 14.5](#).

Figure 14.4 Definition of Pedestrian Movement (Phase) Intervals - Parallel Vehicle Movement Intervals

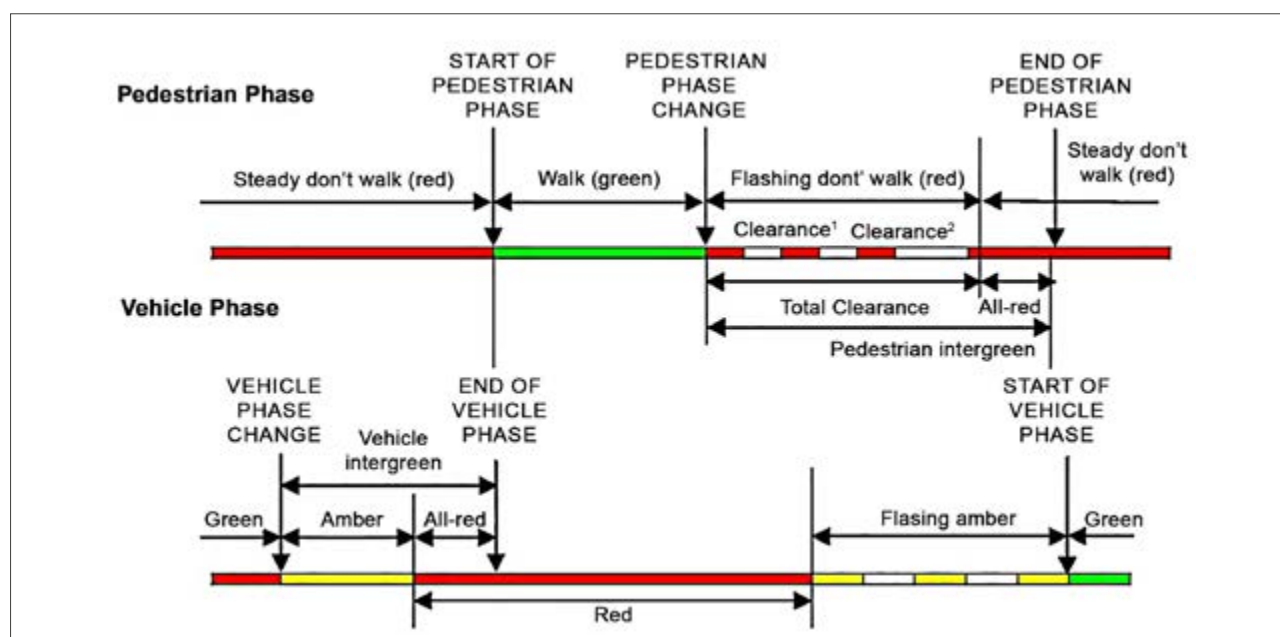
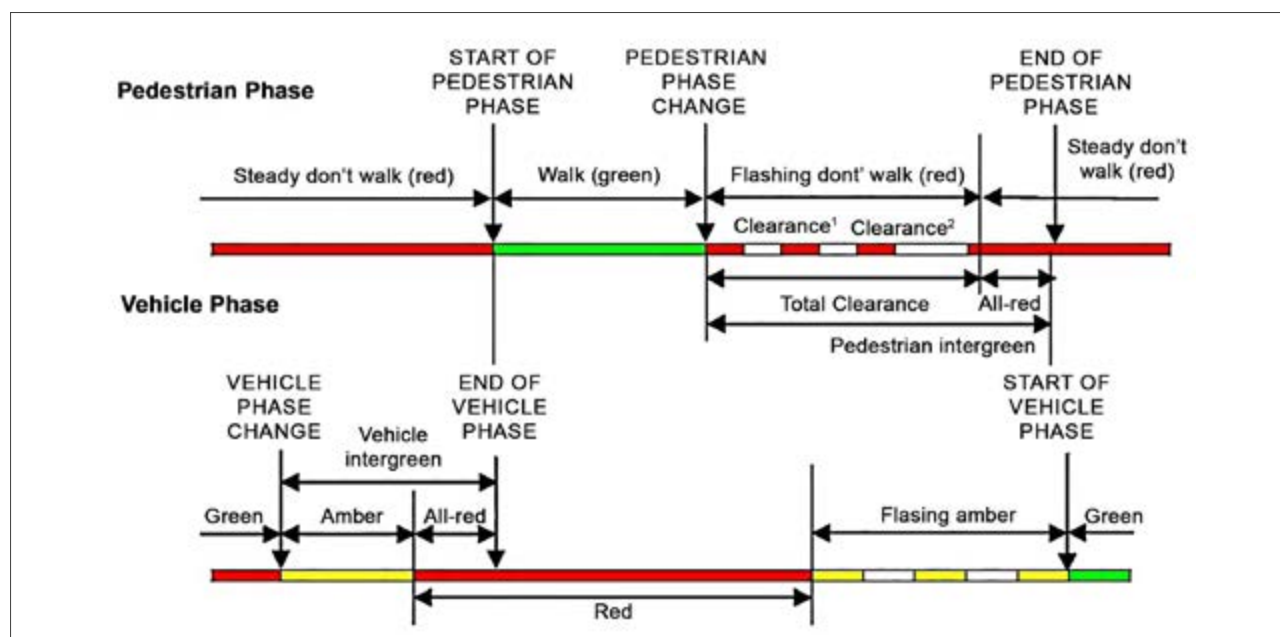


Figure 14.5 Pedestrian & Vehicle Phase Intervals - Pelican Crossing

14.4 Actuated Controller Settings

For determining the values of actuated controller settings, it is necessary to identify the proportion of flow in the critical lanes. Critical lanes have the highest flows and degree of saturation.

Typical signal controller settings for VA loops are given in Table 14.2.

Table 14.2 Typical Signal Controller Settings Used with Stop-Line Detectors

	Through Controlled (and LT) Movements (sec)	Arrow- Controlled RT Movements (sec)	Pedestrian Settings (sec)
Late Start	2 – 5	0 - 6	
Minimum Green	5 – 10	5 – 6	
Max Extension Green	10 – 85	5 – 25	
Gap	2.5 - 4.0	2.0 - 3.0	
Headway	0.3 - 1.5	0.6 - 1.2	
Waste	4 - 11	2 - 8	
Early Cut-Off Green	0 - 8	0	
Amber Time	3.0 - 5.0	3.0 - 6.0	
All-Red Time	1.0 - 3.0	1.0 - 3.0	
Presence Time	0 - 3	0 - 5	
Walk Time			5 - 16
Clearance Time			6 -20

14.5 Vehicle Settings

Vehicle Actuated controller time-settings for Vehicle Intervals are described in the following sections. The values are given in Table 14.2.

14.5.1 Late Start Setting

The purpose of the Late Start Setting is to delay at the start of a phase.

Examples of the use of this setting are:

1. To delay the start of LT by 3 to 6 secs depending on intersection geometry.
2. To delay the start of a filtering RT by 5 secs after a RT Green Arrow display.

14.5.2 Minimum Green Setting

Minimum Green time comprises of the Basic Minimum Green Setting plus a variable time determined by the Max VIG.

The Basic Minimum Green Setting is the time allocated on the basis of a stop-line detector. When using only advance detectors, the minimum Green time is increased to the Maximum VIG to cater for vehicles that may be trapped between the advance detector position and the stop-line. For stop-line detectors, the Maximum VIG is usually set to Zero.

14.5.3 Basic Minimum Green Setting

Basic minimum green time is the minimum time of display of a green signal for safe movement. It is usually considered as 5 secs.

Factors that need consideration when setting the Minimum Green time are:

1. Composition of heavy vehicles.
2. Vertical geometry at the intersection.
3. Number of pedestrians at the intersection.
4. Clearing of opposing turning traffic.

Typical Minimum Green Timings used in practice with stop-line detectors are shown in [Table 14.2](#).

14.5.4 Increment and Maximum Variable Initial Green (VIG)

The increment counter and maximum variable initial green setting is applicable when advance detectors are used without stop-line detectors.

Each vehicle arriving at a red signal adds a small amount of time (equal to the Increment Setting) to the minimum green time. Therefore, the basic minimum green time plus the VIG time should allow sufficient overall green time so that vehicles stored between the advance detector and the stop-line can leave the junction.

Typical increment settings are in the range 0.5 to 2.0 secs.

The Maximum VIG setting limits the additional minimum Green time (Variable Initial Green time) determined by increments. The value of this setting depends on the distance of advance detectors from stop line.

Overall green time, basic minimum green time, VIG time are given by the following equation:

$$\text{Overall green time} = \text{basic minimum green time} + \text{VIG time}$$

Equation 1

$$\text{Incremental delay} = s \times q$$

Equation 2

Where,

s = basic minimum green which is the start time of first vehicle, unit in seconds

q = queue length, unit in number of vehicles

From Equation 1:

$$\text{Overall green} = s + q \times h$$

Equation 3

Where,

h = headway time, unit in seconds

Worked out example:

- Advance detector distance = 60 m
- Average Spacing per vehicle = 6 m
- Start time for first vehicle, s = 2 seconds
- Headway time, h = 2 seconds

Stop-line queue is, obtained by dividing advance distance by average spacing, q = 10 vehicles (60m detector distance / 6m average spacing)

Overall green = 2 seconds + [10 number of vehicles x 2 seconds] = 22 seconds.

If the Basic Minimum Green Setting is taken as 5 seconds, then from Equation 1, the required

Maximum VIG = *Overall green* – *basic min green*, that is 22 seconds – 5 seconds = 17 seconds.

For further details refer to Guide to Traffic Management, Part 10, Austroads.

14.5.5 Maximum Extension Green Setting (Max Green)

The Maximum Extension Green Setting is used to control the Max Green time of each phase when the controller operates in isolated mode. This setting determines the duration of the Extension Green interval. See Figure 14.3.

Max Green time includes the minimum green time.

Typical Max Green Settings used in practice are 10 to 85 seconds for main road Phases, and 5 to 25 seconds for Arrow-controlled right- turn Phases. Side Road or Minor Road Phases are usually set between 12 and 40 seconds.

14.5.6 Gap Setting

The Gap Setting is time when the detector is not occupied. Gap Setting is used to set the maximum allowable time between successive detector actuations before the movement terminates.

For example, if the Gap Setting is 3 secs, then after 3 secs elapses between the departure of a vehicle from the detection zone and the arrival of the next vehicle there, the controller moves to the next phase, by displaying amber.

If the Gap Setting is too short (e.g. 1.5 secs), the phase may terminate before all vehicles pass. If it is too long (e.g. 5.0 secs), the phase extends unnecessarily to its Max Green Time.

Table 14.2 indicates that typical Gap Settings in the range 2.5 to 4.0 seconds for SA and LT movements, and 2.0 to 3.0 seconds for Arrow-controlled RT movements.

14.5.7 Early Cut-Off Green and Early Cut-Off Amber Settings (ECO)

The ECO Green interval allows the termination of some signal groups earlier than others. This arrangement is shown in Figure 14.7.

The area between the 2 stop-lines is sometimes referred to as the reservoir section. The ECO Green Setting depends on intersection geometry and the length of the reservoir section, but normally would not be less than 3 seconds.

The ECO Amber interval is automatically provided with an Amber display for any signal groups that are terminated at the beginning of the ECO Green interval.

The ECO Green interval follows the phase change point, and is considered to be part of the Clearance or Inter-green time.

Figure 14.6 ECO Green and ECO Amber Operation

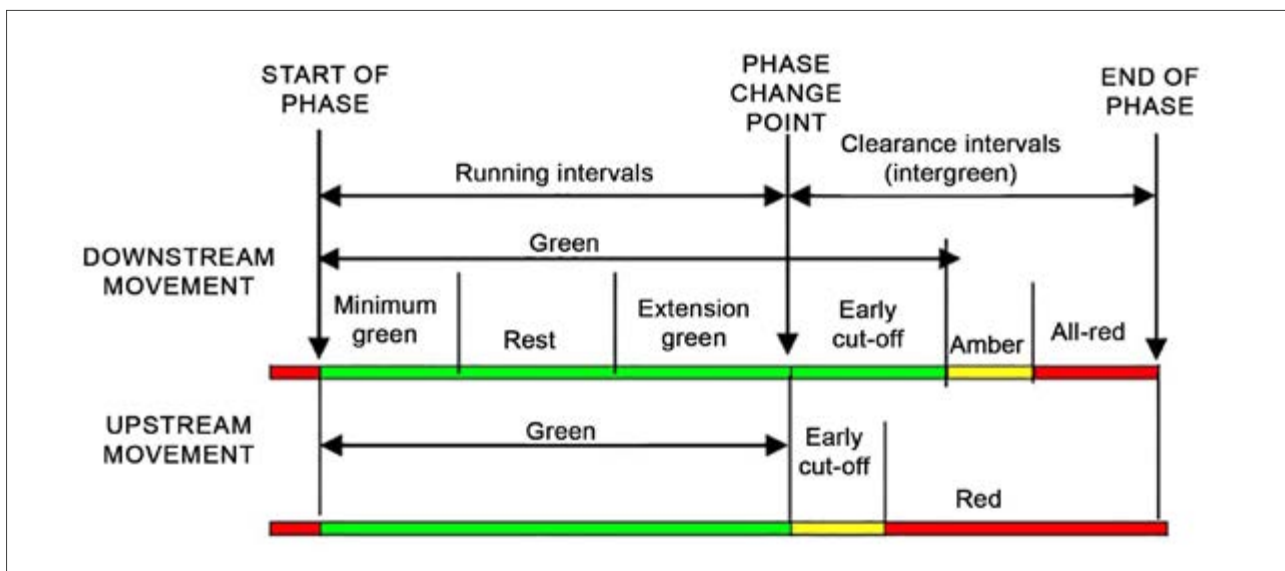
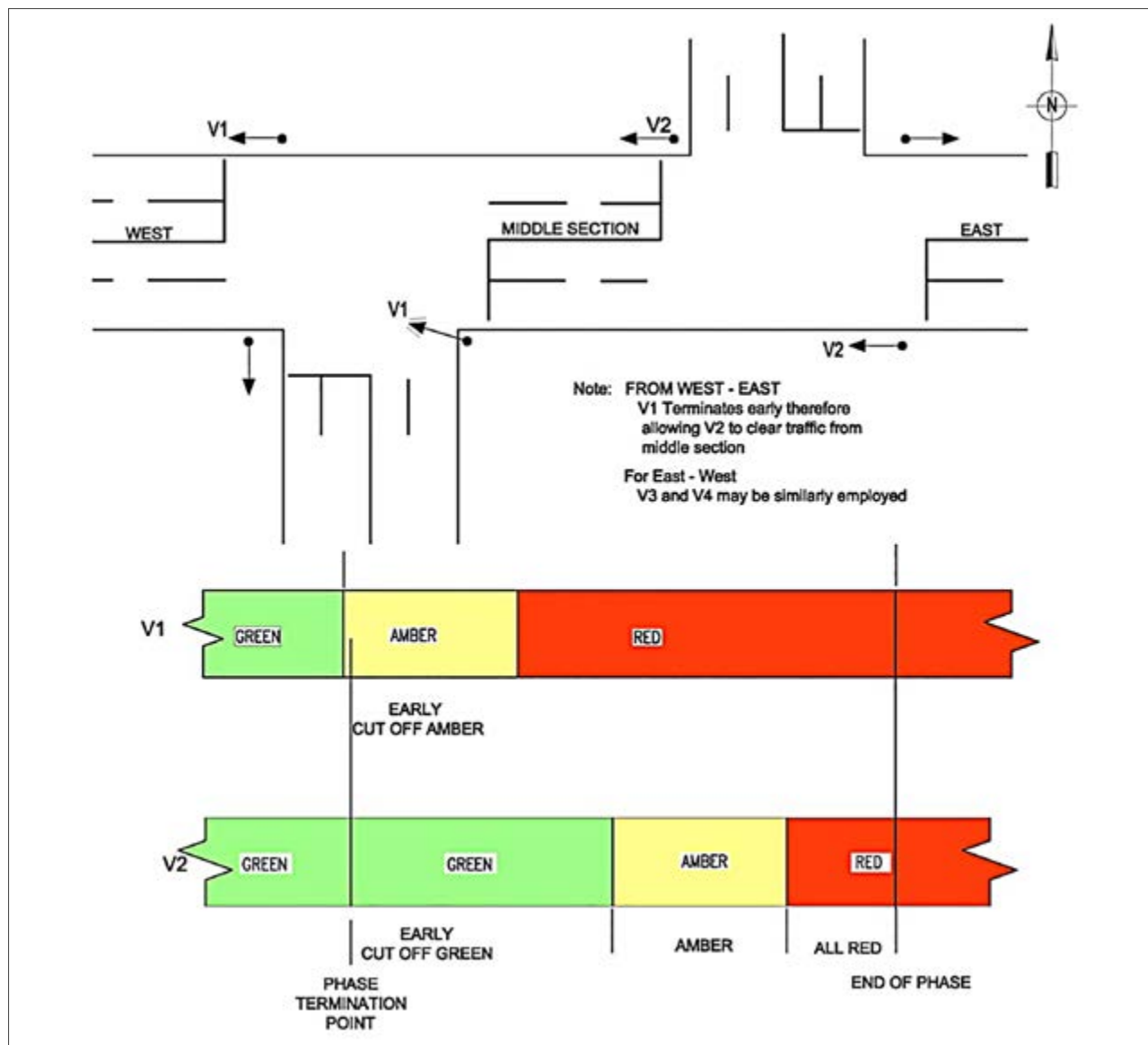


Figure 14.7 Typical Use of Early Cut-Off Periods

14.5.8 Vehicle Clearance (Inter-Green) Settings

The Inter-Green period or vehicle clearance interval is made up of the amber interval that warns drivers that the phase is terminating, and the all-red interval that enables vehicles within the intersection to clear the area.

14.5.9 Amber Time

Amber interval provides warning of the termination of the phase. It should be timed so that approaching vehicles can stop in sufficient time. It is frequently set between 3 and 6 seconds for most urban environments. A driver must stop for an Amber display unless so close to the stop-line that the stopping action would prove hazardous.

The Amber Time is dictated by the acceptable deceleration needs of the majority of the population, usually indicated by the actions of 85 per cent of the driver and vehicle population.

Amber time should be designed such that drivers travelling at a certain speed does not have to abruptly brake seeing an amber display. The amber time should be equal to the sum of the driver's reaction time and the time to reach the stop-line at the design speed, V_d .

The amber times required for different approach speed and gradient are given in [Table 14.3](#).

Table 14.3 Amber Time Values (deceleration 3.0 m / sec)

	Vd = 50 km/h		Vd = 60 km/h		Vd = 70 km/h		Vd = 80 km/h	
Reaction Time $T_r =$	1.0 s	1.5 s	1.0 s	1.5 s	1.0 s	1.5 s	1.0 s	1.5 s
Grade								
-0.08	4.5	5.0	5.0	5.5	5.5	6.0	6.0	6.5
-0.05	4.0	4.5	4.5	5.0	5.0	5.5	5.5	6.0
-0.02	3.5	4.0	4.0	4.5	4.5	5.0	5.0	5.5
0.00	3.5	4.0	4.0	4.5	4.5	5.0	5.0	5.5
0.02	3.0	3.5	3.5	4.0	4.0	4.5	4.5	5.0
0.05	3.0	3.5	3.5	4.0	4.0	4.5	4.5	5.0
0.08	3.0	3.5	3.5	4.0	3.5	4.0	4.0	4.5

14.5.10 All-Red Time

The purpose of the all-red interval is to provide a safe clearance time to vehicles that cross the stop line towards the end of the amber interval.

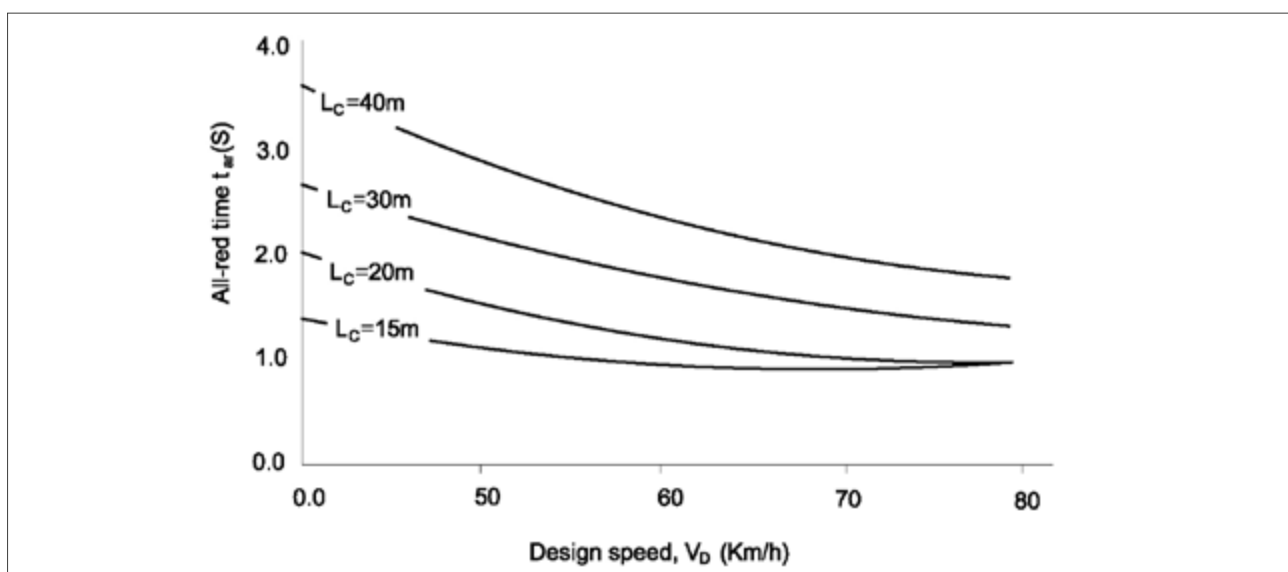
Typical all-red times used are in the range 1.0 to 3.0 seconds, as shown in Table 14.2.

The timing of an all-red interval depends on:

1. The speed of vehicles crossing the stop-line.
2. The distance to the furthest point of conflict.
3. The average length of clearing vehicles.
4. Time for starting traffic to reach the furthest point of conflict.

The all-red time values calculated as a function of speed for various clearance distances are shown in Figure 14.8.

While providing time for pedestrians, consideration should be made for crossing time for women with children and pregnant women. This is particularly important at signals near schools, hospitals and public offices.

Figure 14.8 All-Red Time as a Function of Speed (V_d) and Clearance Distance (L_c)

14.6 Other Vehicle Settings

14.6.1 Timed Presence Time-Setting

The 'Timed' Presence Time-Setting is used to set the length of time for which a detector must be occupied before a demand is recorded. Typical values of the Presence Setting are in the range 0.0 to 5.0 seconds.

Examples of the use of this setting are:

1. Shared lane detector for lagging RT Phase: 2 to 5 seconds.
2. LT detector on overlapped movement: 3 seconds.
3. Queue detectors.

14.6.2 Headway and Waste Time Settings

The headway timer in the traffic controller measures the distance from the end of one vehicle to the start of another.

The purpose of headway setting is to set the desirable time between successive detector actuations, for efficient traffic flow. It is used in along with the waste setting.

Range of headway setting is 0.3 to 1.5 seconds for through movements and 0.6 to 1.2 seconds for arrow-controlled RT movements.

Waste setting is used to set the value at which the phase is terminated. Waste setting time is determined as 10-20 per cent of Maximum Green time. Typical waste settings time is between 4.0 and 11.0 seconds for through movements and 2 to 8 seconds for RT Arrow movements.

14.7 Pedestrian Movement Settings

Pedestrian movement timings are described in the following sections. See [Figure 14.4](#) and [Figure 14.5](#).

14.7.1 Pedestrian Green 'Man' (Walk) Time

The purpose of the Walk time setting is to give pedestrians sufficient time to begin their crossing. This setting determines the duration of the Green Man display. Its value depends on the number and type of pedestrians. Typical pedestrian walk times used are in the range 5 to 16 seconds.

Pedestrian Green Man times can be determined as follows:

1. Use a minimum value of 5 seconds.
2. Add 2 seconds for each additional row of Pedestrians waiting.
3. Allow more time for:
 - a. School.
 - b. Railway stations.
 - c. Elderly, children, disabled or mobility Impaired, women with children and pregnant women.
 - d. Crossing of wide roads in one movement.

While providing time for pedestrians, consideration should be made for crossing time for women with children and pregnant women. This is particularly important at signals near schools, hospitals and public offices.

14.7.2 Pedestrian Clearance Time

The purpose of pedestrian clearance time is to allow pedestrians, who have stepped off the kerb to complete their crossing safely. It comprises of Clearance 1 and Clearance 2 Intervals as shown in Figure 14.4 and Figure 14.5.

The pedestrian clearance interval is implemented using flashing Red Man (Don't Walk) display. As shown in Table 14.2, typical pedestrian clearance times are in the range 6 to 20 seconds.

If there is a median, clearance distance includes both carriageway widths and the median width.

Where a median is wide enough to store pedestrians, a split-movement signal crossing can be used. Crossings are treated separately. The width of appropriate carriageway is used as the clearance distance for each crossing.

Where an exclusive All-Round pedestrian phase is provided at an intersection, the shortest distance between diagonally opposite corner kerb radii for the longest crossing is used as the clearance distance.

14.7.3 Pedestrian Delay Setting

Pedestrian delay setting is used to provide a delay between the PBU actuation and the placement of the pedestrian demand (see Figure 14.4). This helps a number of pedestrians to gather for crossing. Typical pedestrian delay settings are in the range 5.0 to 10.0 seconds.

14.8 Cycle Settings

Two-aspect cycle signal faces can be connected to the two-aspect pedestrian lanterns. The pedestrian Green Man and Clearance times will apply to the cycles as well.

Three-aspect cycle signal faces can be used at main signal sites. The requirements are:

1. For cycle movements parallel to the main road, crossing minor roads, the cycle signal faces can be connected to the adjacent vehicle signal group. Green for cycle can coincide with green for vehicles.
2. For cycle movements across a main road, the cycle signals are driven by a separate signal group. The green time should be designed for cycle speed of 20 km/h.

The following measures can be adopted to allow for slower speed of cyclists compared to vehicle speeds:

1. Adjust the amber time for the cyclist to warn them before other vehicles in the same phase.
2. Allow the cyclists to move off before the vehicle traffic.

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Traffic Signal Timings

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Bibliography

Appendices

Appendix B - Monitoring and Evaluation Checklist

The following checks should be made when monitoring and evaluating the operation of a site:

Sl. No.	Issues	Yes	No	Suggested Remedial Action
1	Do the signs and road markings match the phasing design?			If not, revise the phasing sequence.
2	Are the kerbside controls (e.g. parking) adequate for the smooth flow of traffic?			If not, consult with the Engineer and the traffic police to control kerb side activity.
3	Do current traffic flows match the provided counts?			If not, revise the signal design timings.
4	Is traffic avoiding the 'new' site and creating problems elsewhere?			If yes, investigate the issues that are creating motorist problem, discuss with the Engineer to find solutions.
5	Is the site equipment (controller and detection) operating as intended?			If not, flag the issue with the Engineer. Discuss replacement of the equipment with more suitable ones.
6	Is the amber times provided between phases adequate to clear traffic from the controlled?			If not, revise the phase timing.
7	Is traffic held on a red signal when adequate gaps exist in the oncoming movements?			If yes, re-evaluate the signal timing design.
8	Are delays to vehicles and pedestrians on each approach at satisfactory levels?			If not, re-evaluate the signal timing design.
9	Is there over-queuing or over flowing at traffic bays?			If yes, re-evaluate the signal timing design.
10	Is there blocking of the site by any movements?			If yes, re-evaluate the signal timing design. Discuss with the Engineer about minor geometric modification.
11	Is there is good progression as planned under coordination or central computer control?			If not, re-evaluate the signal timing design.
12	Are vehicle detectors operating as intended in both the passage and presence mode?			If not, re-evaluate the detector design and revise signal timing design.
13	Is the waiting traffic as anticipated in the design?			If not, re-evaluate the signal timing design. Discuss with the Engineer about minor geometric modification.
14	Are any vehicles being missed by detectors, or activating detectors for which they are not intended?			If yes, re-evaluate the detector design and revise signal timing design.

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Appendices